It's Note Time

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1 Helpful Relations, Units, Constants, and Identities

1.1 Units and Prefixes

The fundamental units are the metre (m), second (s), kilogram (kg), kelvin (K), mole (mol), candela (cd), and Ampere (A). All other units can be expressed in terms of these units as follows:

Radian:	$\mathrm{rad} = \mathrm{m} \cdot \mathrm{m}^{-1}$	(Plane Angle)
Steradian:	$\mathrm{sr} = \mathrm{m}^2 \cdot \mathrm{m}^{-2}$	(Solid Angle)
Hertz:	$\mathrm{Hz} = \mathrm{s}^{-1}$	(Frequency)
Newton:	$\mathrm{N} = \mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$	(Force)
Pascal:	$\mathrm{Pa} = \mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2} = \mathrm{N} \cdot \mathrm{m}^{-2}$	(Pressure)
Joule:	$\mathrm{J} = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-2} = \mathrm{N} \cdot \mathrm{m}$	(Energy)
Watt:	$\mathrm{W} = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-3} = \mathrm{J} \cdot \mathrm{s}^{-1}$	(Power, Radiant Flux)
Coulomb:	$\mathrm{C} = \mathrm{A} \cdot \mathrm{s}$	(Electric Charge)
Volt:	$\mathrm{V} = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-3} \cdot \mathrm{A}^{-1} = \mathrm{J} \cdot \mathrm{C}^{-1}$	(Voltage, emf)
Farad:	$\mathrm{F} = \mathrm{kg}^{-1} \cdot \mathrm{m}^{-2} \cdot \mathrm{s}^4 \cdot \mathrm{A}^2 = \mathrm{C} \cdot \mathrm{V}^{-1}$	(Capacitance)
Ohm:	$\Omega = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-3} \cdot \mathrm{A}^{-2} = \mathrm{V} \cdot \mathrm{A}^{-1}$	(Resistance)
Siemens:	$\mathrm{S} = \mathrm{kg}^{-1} \cdot \mathrm{m}^{-2} \cdot \mathrm{s}^3 \cdot \mathrm{A}^2 = \Omega^{-1}$	(Electrical Conductance)
Weber:	$\mathrm{Wb} = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-2} \cdot \mathrm{A}^{-1} = \mathrm{V} \cdot \mathrm{s}$	(Magnetic Flux)
Tesla:	$\mathrm{T} = \mathrm{kg} \cdot \mathrm{s}^{-2} \cdot \mathrm{A}^{-1} = \mathrm{Wb} \cdot \mathrm{m}^{-2}$	(Magnetic Flux Density)
Henry:	$\mathrm{H} = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-2} \cdot \mathrm{A}^{-2} = \mathrm{Wb} \cdot \mathrm{A}^{-1}$	(Inductance)
Degree Celcius:	$^\circ\mathrm{C} = \mathrm{K}$	(Temperature Relative to 273.15 K)
Lumen:	$\mathrm{lm} = \mathrm{cd} \cdot \mathrm{sr}$	(Luminous Flux)
Lux:	$\mathrm{lx} = \mathrm{m}^{-2} \cdot \mathrm{cd} = \mathrm{lm} \cdot \mathrm{m}^{-2}$	(Illuminance)
Becquerel:	$\mathrm{Bq} = N_{\mathrm{Decay}} \cdot \mathrm{s}^{-1}$	(Radioactivity)
Gray:	$\mathrm{Gy} = \mathrm{m}^2 \cdot \mathrm{s}^{-2} = \mathrm{J} \cdot \mathrm{kg}^{-1}$	(Absorbed Dose)
Sievert:	$\mathrm{Sv} = \mathrm{m}^2 \cdot \mathrm{s}^{-2} = \mathrm{J} \cdot \mathrm{kg}^{-1}$	(Equivalent Dose)
Katal:	$\mathrm{kat} = \mathrm{mol} \cdot \mathrm{s}^{-1}$	(Catalytic Activity)

The established SI order of magnitude prefixes are:

Yotta (Y) – 10^{24}	Yocto (y) – 10^{-24}
Zetta (Z) – 10^{21}	Zepto (z) – 10^{-21}
Exa (E) – 10^{18}	Atto (a) – 10^{-18}
Peta (P) – 10^{15}	Femto (f) – 10^{-15}
Tera (T) – 10^{12}	Pico (p) – 10^{-12}
Giga (G) – 10^{9}	Nano (n) – 10^{-9}

Mega (M) – 10^6	Micro (μ) – 10^{-6}
Kilo (k) – 10^3	Milli (m) – 10^{-3}
Hecto (h) – 10^2	Centi (c) – 10^{-2}
Deca (da) – 10^1	Deci (d) – 10^{-1}

Although not SI, the prefix for the order of magnitude 10^{-10} is often given the name Angstroms, which is denoted by the symbol Å.

1.2 Fundamental Constants

Planck Constant:	$h = 6.62607015 \times 10^{-34}\ \mathrm{J \cdot s}$
Planck Reduced Constant:	$\hbar = h/2\pi \approx 1.05457182 \times 10^{-34}\ \mathrm{J} \cdot s$
Speed of Light in Vacuum:	$c = 2.99792458 \times 10^{8}\ \mathrm{m \cdot s^{-1}}$
Elementary Charge:	$e = 1.60217663 \times 10^{-19}\ \mathrm{C}$
Electron Rest Mass:	$m_e = 9.10938370 \times 10^{-31}\ \mathrm{kg} = 0.51099895\ \mathrm{MeV} \cdot c^{-2}$
Vacuum Electric Permittivity:	$\varepsilon_0 = 8.85418781 \times 10^{-12}\ \mathrm{C^2 \cdot J^{-1} \cdot m^{-1}}$
Vacuum Magnetic Permeability:	$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{J \cdot s \cdot C^{-2}}$
Avagadro Constant:	$N_A = 6.02214076 \times 10^{23}\ \mathrm{mol^{-1}}$
Molar Gas Constant:	$R = N_A k_B = 8.31446261\ \mathrm{J \cdot mol^{-1} \cdot K^{-1}}$
Boltzmann Constant:	$k_B = 1.380649 \times 10^{-23}\ \mathrm{J \cdot K^{-1}}$
Gravitational Constant:	$G = 6.6743015 \times 10^{-11}\ \mathrm{N \cdot m^2 \cdot kg^{-2}}$
Fine-Structure Constant:	$\alpha = \dfrac{e^2}{4\pi\varepsilon_0\hbar c} = 0.00729735 \approx 1/137$
Rydberg Constant:	$R_\infty = \dfrac{m_e e^4}{8\,\varepsilon_0^2 c h^3} = 10973731.56816021\ \mathrm{m^{-1}}$
Characteristic Impedance of Vacuum:	$Z_0 = 376.73031367\ \Omega$

1.3 Frequently Used Constants

1.3.1 Subatomic Physics

Proton Rest Mass:	$m_p = 1.67262192 \times 10^{-27}\ \mathrm{kg} = 938.27208816\ \mathrm{MeV} \cdot c^{-2}$
Neutron Rest Mass:	$m_n = 1.67492749 \times 10^{-27}\ \mathrm{kg} = 939.56542052\ \mathrm{MeV} \cdot c^{-2}$
Tau Rest Mass:	$m_\tau = 3.1675421 \times 10^{-27}\ \mathrm{kg} = 1776.85895968\ \mathrm{MeV} \cdot c^{-2}$
Muon Rest Mass:	$m_\mu = 1.88353163 \times 10^{-28}\ \mathrm{kg} = 105.65837552\ \mathrm{MeV} \cdot c^{-2}$
Deuteron Rest Mass:	$m_d = 3.34358377 \times 10^{-27}\ \mathrm{kg} = 1875.61294257\ \mathrm{MeV} \cdot c^{-2}$

Alpha Particle Rest Mass:	$m_\alpha = 6.64465734 \times 10^{-27}\ \mathrm{kg} = 3727.37940661\ \mathrm{MeV} \cdot c^{-2}$
Nuclear Magneton:	$\mu_N = \dfrac{e\hbar}{2m_p c} = 5.05078375 \times 10^{-27}\ \mathrm{J} \cdot \mathrm{T}^{-1} = 3.15245126 \times 10^{-8}\ \mathrm{eV} \cdot \mathrm{T}^{-1}$
Fermi Coupling Constant:	$G_F^0 = \dfrac{G_F}{(\hbar c)^3} = 4.5437957 \times 10^{14}\ \mathrm{J}^{-2} = 1.16637876 \times 10^{-5}\ \mathrm{GeV}^{-2}$

1.3.2 Astrophysics

Solar Mass:	$M_\odot = 1.98847 \times 10^{30}$ kg	(The mass of our Sun)
Earth Mass:	$M_\oplus = 5.9722 \times 10^{24}$ kg	
Astronomical Unit:	au = 149597870700 m	(Approximate Earth-Sun distance)
Light Year:	ly = 9.46×10^{12} km	(Distance travelled by light in a year)
Parsec:	pc = $\dfrac{648000}{\pi}$ au ≈ 3.26156 ly	(The distance at which 1 au subtends an angle of 1 arc second)
Hubble Constant:	$H_0 = 69.3\ \mathrm{km} \cdot \mathrm{s}^{-1} \cdot \mathrm{Mpc}^{-1}$	
	$= 2.25 \times 10^{-18}\ \mathrm{s}^{-1}$	(The current value of the Hubble parameter)

1.3.3 Atomic Physics

Electron Volt:	eV = $1.60217663 \times 10^{-19}$ J	(The energy gained by an electron moved across an electric potential difference of 1 Volt)
Unified Atomic Mass Unit:	u = $1.66053904 \times 10^{-27}$ kg	
	$= 931.49410242\ \mathrm{MeV} \cdot c^{-2}$	(1/12 of the mass of an unbound neutral atom of Carbon-12 in its nuclear and electronic ground state, and at rest)
Bohr Radius:	$a_0 = \dfrac{4\pi\varepsilon_0\hbar^2}{m_e e^2} = \dfrac{\hbar}{m_e c\,\alpha}$	
	$= 5.29177211 \times 10^{-11}$ m	
	= 0.529177211 Å	
Compton Wavelength:	$\lambda_c = \dfrac{h}{mc}$	
Electron g-Factor:	$g_e = -2.002319304362$	
Bohr Magneton:	$\mu_B = \dfrac{e\hbar}{2m_e c}$	
	$= 9.274009 \times 10^{-24}\ \mathrm{J} \cdot \mathrm{T}^{-1}$	
Hyperfine Transition Frequency of Cs-133:	$\Delta\nu_{\mathrm{Cs}} = 9192631770$ Hz	
Hartree Energy:	$E_h = 4.35974472 \times 10^{-18}$ J	

$= 27.21138631 \text{ eV}$

1.3.4 Mechanics and Thermodynamics

Standard Atmosphere: $\text{atm} = 101325 \text{ Pa} \approx 763.43 \text{ mm Hg}$ (The pressure at sea level on Earth)

Standard Gravity: $g = 9.80665 \text{ m} \cdot \text{s}^{-2}$ (The acceleration due to gravity on Earth)

Stefan-Boltzmann Constant: $\sigma = \dfrac{2\pi^5 k_B^4}{15\, h^3 c^2} \approx 5.67037442 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$

1.3.5 Electromagnetism

Faraday Constant: $F = 96485.33212 \text{ C} \cdot \text{mol}^{-1}$

Classical Electron Radius: $r_e = \dfrac{e^2}{4\pi\varepsilon_0 m_e c^2} \approx 2.81794033 \times 10^{-15} \text{ m}$

Thomson Scattering Cross-Section of Electron: $\sigma_e = \dfrac{8\pi}{3}\left(\dfrac{e^2}{4\pi\varepsilon_0 m_e c^2}\right)^2 = 6.65245873 \times 10^{-29} \text{ m}^2$

Conductance Quantum: $G_0 = \dfrac{2e^2}{h} = 7.74809173 \times 10^{-5} \text{ S}$

Magnetic Flux Quantum: $\Phi_0 = \dfrac{h}{2e} \approx 2.06783385 \times 10^{-15} \text{ Wb}$

1.4 Mathematical Constants

Euler's Number: $e = \lim_{n\to\infty}\left(1 + \dfrac{1}{n}\right)^n = 2.718281828\ldots$

Pi: $\pi = \lim_{n\to\infty} \dfrac{2^{4n}(n!)^4}{n\,((2n)!)^2} = 3.14159265\ldots$

Imaginary Number: $i = \sqrt{-1}$

Euler-Mascheroni Number: $\gamma = \lim_{n\to\infty}\left[1 + \dfrac{1}{2} + \dfrac{1}{3} + \ldots + \dfrac{1}{n} - ln(n+1)\right]$
$= 0.577215661901532\ldots$

Radians to Degrees: $1 \text{ rad} = \dfrac{180}{\pi}^{\circ} \approx 57.29577951^{\circ}$

Arc Seconds to Degrees: $1 \text{ as} = \dfrac{1}{3600}^{\circ} \approx 0.00027778^{\circ}$

1.5 Specialized Unit Systems

The following special unit systems are often employed in physical fields such as General Relativity, Electromagnetism, Atomic Physics, and Quantum Gravity (to name a few) so that equations may either be scaled or simplified. It is important to note that this list is not exhaustive.

1.5.1 Planck Units

Planck units are a system of natural units obtained through combination of the reduced Planck constant ($\hbar$), speed of light in vacuum (c), gravitational constant (G), Boltzmann constant (k_B), and vacuum electric permittivity constant (ε_0). In this system, these 5 constants are set equal to 1 and must satisfy the equations:

$$
\begin{aligned}
l_P &= c\, t_P \\
F_P &= \frac{m_P\, l_P}{t_P^2} = G\frac{m_P^2}{l_P^2} \\
E_P &= \frac{m_P\, l_P^2}{t_P^2} = \hbar\frac{1}{t_P} \\
F_P &= \frac{m_P\, l_P}{t_P^2} = \frac{1}{4\pi\varepsilon_0}\frac{q_P^2}{l_P^2} \\
E_P &= \frac{m_P\, l_P^2}{t_P^2} = k_B T_P
\end{aligned}
$$

The Planck units expressed in SI units are then:

Planck Length:	$l_P = \sqrt{\frac{\hbar G}{c^3}}$	$\approx 1.616 \times 10^{-35}$ m
Planck Time:	$t_P = \sqrt{\frac{\hbar G}{c^5}}$	$\approx 5.391 \times 10^{-44}$ s
Planck Mass:	$m_P = \sqrt{\frac{\hbar c}{G}}$	$\approx 2.176 \times 10^{-8}$ kg
Planck Temperature:	$T_P = \sqrt{\frac{\hbar c^5}{G k_B^2}}$	$\approx 1.417 \times 10^{32}$ K
Planck Charge:	$q_P = \sqrt{4\pi\hbar c\varepsilon_0}$	$\approx 1.876 \times 10^{-18}$ C

These 5 units constitute the Planck scale, which is used as an indicator for when quantum gravitational effects can no longer be ignored in calculations (mainly in the standard model of particle physics).

1.5.2 Geometrized Units

Geometrized units are a system of natural units typically used in general relativity and can be obtained by setting the speed of light in vacuum (c) and the gravitational constant (G) equal to 1. There is also additional freedom to set the Boltzmann constant (k_B), elementary charge (e), and Coulomb constant ($1/4\pi\varepsilon_0$) equal to 1. This freedom does not extend to the reduced Planck constant, which is the main difference from Planck units.

The conversion factors to switch between different SI units, with the geometrized unit system are:

	m	kg	s	C	K
m	1	$G^{-1}c^2$	c^{-1}	$\sqrt{4\pi\varepsilon_0 c^4 G^{-1}}$	$c^4(Gk_B)^{-1}$
kg	Gc^{-2}	1	Gc^{-3}	$\sqrt{4\pi\varepsilon_0 G}$	$c^2 k_B^{-1}$
s	c	$c^3 G^{-1}$	1	$\sqrt{4\pi\varepsilon_0 c^6 G^{-1}}$	$c^5(Gk_B)^{-1}$
C	$\sqrt{G(4\pi\varepsilon_0 c^4)^{-1}}$	$(\sqrt{4\pi G\varepsilon_0})^{-1}$	$\sqrt{G(4\pi\varepsilon_0 c^6)^{-1}}$	1	$\sqrt{c^4(4\pi\varepsilon_0 G k_B^2)^{-1}}$
K	$Gk_B c^{-4}$	$k_B c^{-2}$	$Gk_B c^{-5}$	$\sqrt{4\pi\varepsilon_0 G k_B^2 c^{-4}}$	1

1.5.3 Natural Units (Particle Physics and Cosmology)

The name natural units is technically an umbrella term for all the unit systems described in this section, however, in particle physics and cosmology it generally means setting the reduced Planck constant ($\hbar$), the speed of light in vacuum (c), and the Boltzmann constant (k_B) equal to 1. Additionally, the electron volt (eV) is utilized so that the system of units can describe any physical quantity. In using the eV we can further choose Gaussian or Lorentz-Heaviside units to represent the elementary charge.

The natural units expressed in SI units are then:

1 eV^{-1} of Length: $$\frac{\hbar c}{1\,\text{eV}} \approx 1.97327 \times 10^{-7}\ \text{m}$$

1 eV^{-1} of Time: $$\frac{\hbar}{1\,\text{eV}} \approx 6.58212 \times 10^{-16}\ \text{s}$$

1 eV of Mass: $$\frac{1\,\text{eV}}{c^2} \approx 1.78266 \times 10^{-36}\ \text{kg}$$

1 eV of Temperature: $$\frac{1\,\text{eV}}{k_B} \approx 1.16045 \times 10^{4}\ \text{K}$$

1 Unit of Charge (L-H): $$\frac{e}{\sqrt{4\pi\alpha}} \approx 5.29012 \times 10^{-19}\ \text{C}$$

1 Unit of Charge (G): $$\frac{e}{\sqrt{\alpha}} \approx 1.87530 \times 10^{-18}\ \text{C}$$

1.5.4 Hartree Atomic Units

Hartree atomic units are a system of natural units typically employed in atomic physics and can be obtained by setting the electron mass (m_e), elementary charge (e), reduced Planck constant ($\hbar$), and inverse Coulomb constant ($4\pi\varepsilon_0$) equal to 1. A further derived unit needed to express all physical quantities of interest is the Hartree energy (E_h).

The Hartree atomic units and some derived units expressed in SI units are then:

1 Bohr (Length): $$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} \approx 5.29177211 \times 10^{-11}\ \text{m}$$

Time: $$\frac{\hbar}{E_h} \approx 2.41888433 \times 10^{-17}\ \text{s}$$

Mass:	$m_e = 9.10938370 \times 10^{-31}$ kg
1 Hartree (Energy):	$E_h = \dfrac{\hbar^2}{m_e a_0^2} \approx 4.35974472 \times 10^{-18}$ J
Charge:	$e = 1.60217663 \times 10^{-19}$ C
Action:	$\hbar = 1.05457182 \times 10^{-34}$ J $\cdot$ s
Velocity:	$v = \dfrac{e^2}{4\pi\varepsilon_0 c\hbar} \approx 2.18769126 \times 10^6$ m $\cdot$ s^{-1}

Some physical constants expressed in Hartree atomic units are:

Bohr Radius:	$a_0 = \dfrac{4\pi\varepsilon_0\hbar^2}{m_e e^2} = 1$ a.u.
Speed of Light in Vacuum:	$c = \dfrac{1}{\alpha}$ a.u.
Reduced Compton Wavelength of Electron:	$\lambda\!\!\!^{-}_e = \dfrac{\hbar}{m_e c} = \alpha$ a.u.
Classical Electron Radius:	$r_e = \dfrac{e^2}{4\pi\varepsilon_0 m_e c^2} = \alpha^2$ a.u.
Velocity:	$v = \dfrac{e^2}{4\pi\varepsilon_0 c\hbar} = 1$ a.u.
Rydberg Constant:	$R_\infty = \dfrac{m_e e^4}{8\,\varepsilon_0^2 c h^3} = \dfrac{1}{2}$ a.u.
Bohr Magneton:	$\mu_B = \dfrac{e\hbar}{2m_e} = \dfrac{1}{2}$ a.u.
Electric Dipole Moment:	$\mu_E = e\,a_0 = 1$ a.u.

1.6 Properties of Exponents and Algebraic Identities

Exponentiation, symbolically expressed as b^n (for $b \neq 0$), is a mathematical operation involving a base b, that is multiplied by itself n times This definition makes intuitive sense if the exponent n, is a positive integer, however, it applies for any complex exponent $z \in \mathbb{C}$. (Note that 0^0 is undefined)

The Following is a list of exponent properties (provided the base is non-zero):

Zero Exponent:

$$b^0 = 1$$

Negative Exponent:

$$b^{-n} = \frac{1}{b^n}$$

Addition of Exponents:

$$b^{m+n} = b^m \cdot b^n$$

Subtraction of Exponents:

$$b^{m-n} = \frac{b^m}{b^n}$$

Multiplication of Exponents:

$$(b^m)^n = b^{m \cdot n}$$

Distributivity of Exponents:

$$(b \cdot c)^n = b^n \cdot c^n$$

The Exponential Function:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = \lim_{n\to\infty} \left(1 + \frac{z}{n}\right)^n$$

Euler's Identity:

$$e^{i\theta + 2n\pi i} = e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

Alternate Exponential Form:

$$b^z = e^{z\,\ln(b)}$$

1.6.1 Binomial Theorem

The Binomial theorem allows for many algebraic identities to be expressed. It is encapsulated by the expression:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} y^k x^{n-k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} y^k x^{n-k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k y^{n-k} \,. \tag{1}$$

Some useful identities for computing binomial coefficients are:

$$\binom{n}{k} = \binom{n}{n-k}, \qquad n > 0\,;\ k \le n \qquad\qquad \binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}, \qquad n > 0\,;\ k \le n-1$$

$$\binom{n}{k} = \frac{n-k+1}{k}\binom{n}{k-1}, \qquad n > 0\,;\ k \le n+1 \qquad\qquad \binom{n}{k} = \frac{n}{n-k}\binom{n-1}{k}, \qquad n > 0\,;\ k \le n-1$$

$$\binom{n-1}{k} = \frac{k+1}{n}\binom{n}{k+1}, \qquad n > 0\,;\ k \le n-1 \qquad\qquad \binom{m+n}{k} = \sum_{r=0}^{k} \binom{m}{r}\binom{n}{k-r}, \qquad r \le m\,;\ k \le m+n$$

This can be further generalized to include complex exponents $r \in \mathbb{C}$ using the gamma function

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, \mathrm{d}t \,,$$

which extends the factorial beyond positive integers to all $z \in \mathbb{C}\backslash\{n \in \mathbb{Z}^-\}$ yielding:

$$(x+y)^r = \sum_{k=0}^{n} \frac{\Gamma(r+1)}{\Gamma(r-k+1)\,k!} y^k x^{r-k} \,, \tag{2}$$

where $x, y \in \mathbb{R}$ and $|x| > |y|$. If we wish to extend this to the case where x and y are complex numbers, we again make the assumption that $|x| > |y|$ but now we choose a holomorphic branch of the logarithm defined on an open disk of radius $|x|$ centred at x.

For sums of more than two terms we have the multinomial theorem, which is given by the expression:

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{k_1+k_2+\cdots+k_m=n} \binom{n}{k_1,\, k_2,\, \ldots,\, k_m} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m} \,, \tag{3}$$

where the summation is taken over all sequences of non-negative integer indices k_1 through k_m such that the sum of all k_i is n, and the multinomial coefficients are:

$$\binom{n}{k_1,\, k_2,\, \ldots,\, k_m} = \frac{n!}{k_1! \cdot k_2! \cdots k_m!} \,.$$

For products of binomials we have the multi-binomial theorem, which is given by the expression:

$$(x_1+y_1)^{n_1} \cdots (x_d+y_d)^{n_d} = \sum_{k_1=0}^{n_1} \cdots \sum_{k_d=0}^{n_d} \binom{n_1}{k_1} x_1^{k_1} y_1^{n_1-k_1} \cdots \binom{n_d}{k_d} x_d^{k_d} y_d^{n_d-k_d} \,. \tag{4}$$

The following is a list of algebraic and absolute value identities:

Product Formulae:

$$(ax \pm by)^2 = (ax)^2 \pm 2abxy + (by)^2$$

$$(ax \pm by)^3 = (ax)^3 \pm 3(ax)^2 by + 3ax(by)^2 \pm (by)^3$$

$$(ax \pm by)^4 = (ax)^4 \pm 4(ax)^3 by + 6(ax)^2(by)^2 \pm 4ax(by)^3 + (by)^4$$

Factoring Formulae:

$$x^2 - y^2 = (x+y)(x-y)$$

$$x^3 - y^3 = (x-y)(x^2 + xy + y^2)$$

$$x^3 + y^3 = (x+y)(x^2 - xy + y^2)$$

$$x^4 - y^4 = (x^2 + y^2)(x^2 - y^2)$$

Absolute Value Definition:

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Triangle Inequality:

$$|x \pm y| \leq |x| + |y|$$

Multiplication of Absolute Values:

$$|xy| = |x|\,|y|$$

Division of Absolute Values:

$$\left|\frac{x}{y}\right| = \frac{|x|}{|y|}, \quad y \neq 0$$

1.7 Properties of Logarithms

The logarithm is the inverse operation of exponentiation, meaning that the logarithm of a number z is the exponent to which a base b must be raised to in order to get the number z:

$$y = \log_b(z) \quad \Rightarrow \quad b^y = z$$

The Following is a list of logarithmic properties (provided the base is non-zero):

Zero Logarithm: $\log_b(1) = 0$

Logarithm of Products: $\log_b(xy) = \log_b(x) + \log_b(y)$

Logarithm of Quotients: $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$

Logarithm of Sums/Differences: $\log_b\left(\sum_{i=0}^{N} a_i\right) = \log_b(a_0) + \log_b\left(1 + \sum_{i=1}^{N} \frac{a_i}{a_0}\right)$

Powers: $\log_b(x^p) = p\log_b(x)$

Roots: $\log_b(\sqrt[p]{x}) = \frac{\log_b(x)}{p}$

An Exponent Identity: $x^{\frac{\log(a)}{\log(x)}} = a$

Change of Base: $\log_b(x) = \frac{\log_k(x)}{\log_k(b)}$

Swapping Variables: $x^{\log_b(y)} = y^{\log_b(x)}$

Reciprocal Bases: $-\log_b(a) = \log_{\frac{1}{b}}(a)$

Product Bases: $\log_{xy}(a) = \frac{\log_y(a)\log_x(a)}{\log_y(a) + \log_x(a)}$

The Logarithmic Function: $$\mathrm{Log}(z) = \ln(|z|) + i\arg(z) = \sum_{n=0}^{\infty} \frac{2}{2n+1}\left(\frac{z-1}{z+1}\right)^{2n+1}, \quad z > 0$$
$$= \lim_{n\to\infty} n(z^{1/n} - 1)$$

Some common bases of the logarithm are:

Binary Logarithm:	$\mathrm{lb}(z) = \log_2(z) = y$	$\Rightarrow$	$2^y = z$
Natural Logarithm:	$\ln(z) = \log_e(z) = y$	$\Rightarrow$	$e^y = z$
Common Logarithm:	$\lg(z) = \log_{10}(z) = y$	$\Rightarrow$	$10^y = z$

1.8 Properties of Sequences and Limits

Sequences are an enumerated collection of objects with an ordering, where repetitions are permitted. Sequences can be denoted by:

$$\{a_0, a_1, a_2, \ldots, a_{n-2}, a_{n-1}, a_n\} = \{a_k\}_{k=0}^{n} = (a_k)_{k=0}^{n} = (a_k)_{k\in S\subseteq\mathbb{N}},$$

and if the sequence is infinite or bi-infinite then the upper indices should be replaced with infinity. More formally, sequences are a function whose domain is an interval of integers. The length of a sequence is defined as the number of terms in the sequence. A sequence of a finite length n is also called an n-tuple. Finite sequences include the empty sequence ( ) that has no elements.

A sequence is said to be monotonically increasing, if each term is greater than or equal to the one before it. For example, the sequence $(a_n)_{n=1}^{\infty}$ is monotonically increasing if and only if $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$. If each consecutive term is strictly greater than the previous term then the sequence is called strictly monotonically increasing. A sequence is monotonically decreasing, if each consecutive term is less than or equal to the previous one, and strictly monotonically decreasing, if each is strictly less than the previous. If a sequence is either increasing or decreasing it is called a monotone sequence. This is a special case of the more general notion of a monotonic function. (The terms nondecreasing and nonincreasing are often used in place of increasing and decreasing in order to avoid any possible confusion with strictly increasing and strictly decreasing, respectively.)

If the sequence of real numbers (a_n) is such that all the terms are less than some real number M, then the sequence is said to be bounded from above. In other words, this means that there exists M such that for all n, $a_n \leq M$. Any such M is called an upper bound. Likewise, if, for some real m, $a_n \geq m$ for all n greater than some N, then the sequence is bounded from below and any such m is called a lower bound. If a sequence is both bounded from above and bounded from below, then the sequence is said to be bounded.

A subsequence of a given sequence is a sequence formed from the given sequence by deleting some of the elements without disturbing the relative positions of the remaining elements. For instance, the sequence of positive even integers $(2, 4, 6, \ldots)$ is a subsequence of the positive integers $(1, 2, 3, \ldots)$. The positions of some elements change when other elements are deleted. However, the relative positions are preserved. Formally, a subsequence of the sequence $(a_n)_{n\in\mathbb{N}}$ is any sequence of the form $(a_{n_k})_{k\in\mathbb{N}}$, where $(n_k)_{k\in\mathbb{N}}$ is a strictly increasing sequence of positive integers.

1.8.1 Limits of Sequences

The sequence $\{a_n\}$ has limit L if a_n is as close as we like to L for all sufficiently large integers n. if the limit exists then we say the sequence converges, otherwise we say the sequence diverges. Formally, we call L the

limit of the sequence $\{a_k\}$ if, for each real number $\epsilon > 0$, there exists a natural number N such that, for every natural number $n \geq N$, we have $|a_n - L| < \epsilon$. The sequence $\{a_n\}$ is said to converge to or tend to the limit L, written $a_n \to L$ or $\lim\limits_{n\to\infty} a_n = L$ (Note that the limit of a given sequence is unique).

Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences with limits L and M respectively. Then the following general properties hold (provided the limits on both sides exist):

(i) $\lim\limits_{n\to\infty} (a_n \pm b_n) = \lim\limits_{n\to\infty} a_n \pm \lim\limits_{n\to\infty} b_n = L \pm M$

(ii) $\lim\limits_{n\to\infty} c\, a_n = c \cdot \lim\limits_{n\to\infty} a_n = c\, L$

(iii) $\lim\limits_{n\to\infty} |a_n| = |L|$

(iv) $\lim\limits_{n\to\infty} (a_n \cdot b_n) = \left(\lim\limits_{n\to\infty} a_n\right) \cdot \left(\lim\limits_{n\to\infty} b_n\right) = L\, M$

(v) $\lim\limits_{n\to\infty} \left(\dfrac{a_n}{b_n}\right) = \dfrac{\lim\limits_{n\to\infty} a_n}{\lim\limits_{n\to\infty} b_n} = \dfrac{L}{M}$ provided $\lim\limits_{n\to\infty} b_n \neq 0$

(vi) $\lim\limits_{n\to\infty} a_n^p = \left[\lim\limits_{n\to\infty} a_n\right]^p$

If $a_n \leq b_n$ for all n greater than some N, then $\lim\limits_{n\to\infty} a_n \leq \lim\limits_{n\to\infty} b_n$.

If $a_n \leq c_n \leq b_n$ for all $n > N$, and $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = L$, then $\lim\limits_{n\to\infty} c_n = L$ (Squeeze Theorem).

Some other important properties of limits of real sequences include the following:

1. If a sequence is bounded and monotonic then it is convergent.
2. A sequence is convergent if and only if every subsequence is convergent.
3. If every subsequence of a sequence has its own subsequence which converges to the same point, then the original sequence converges to that point.

The limit superior and limit inferior of a sequence are defined as:

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \left(\sup_{m\geq n} a_m\right) \quad \text{and} \quad \liminf_{n\to\infty} a_n = \lim_{n\to\infty} \left(\inf_{m\geq n} a_m\right).$$

Sequences can be given in the form above, and if possible, can be given by an explicit expression or defined recursively. A recurrence relation is an expression for the nth term in a sequence, which has the form:

$$u_n = \varphi(n, u_{n-1}, u_{n-2}, \ldots, u_{n-k}), \quad n \geq k \quad \text{where} \quad \varphi : \mathbb{N} \times X^k \to X.$$

The factorial is an example of a recurrence relation, it is given by:

$$n! = \begin{cases} n(n-1)! & n > 0 \\ 1 & n = 0 \end{cases}$$

1.8.2 Limits of Functions

A viable informal definition of the limit of a function is that a function f approaches the limit L near c (symbolically, $\lim_{x\to c} f(x) = L$) if we can make $f(x)$ as close as we like to L by requiring that x be sufficiently close to, but unequal to, c. The formal (ε, δ) definition of the limit of a function is as follows:

> Let f be a real-valued function defined on a subset D of the real numbers. Let c be a limit point of D and let L be a real number. We say that $\lim_{x\to c} f(x) = L$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that, for all $x \in D$, if $0 < |x - c| < \delta$, then $|f(x) - L| < \varepsilon$.

In the above definitions it is implicitly true that the right- and left-hand limits of f must be equal for the limit to exist. Symbolically, this is:

$$\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = \lim_{x\to c} f(x) .$$

Continuity:

A function f is said to be continuous at c if it is both defined at c and its value at c equals the limit of f as x approaches c: $\lim_{x\to c} f(x) = f(c)$. The formal (ε, δ) definition for a continuous function can be obtained from the definition of a limit by replacing $0 < |x - c| < \delta$ with $|x - c| < \delta$ to ensure that f is defined at c and equals the limit. f is said to be continuous on an interval I if it is continuous at every point c of I.

An equivalent definition of continuity in terms of sequences is that for any sequence $(x_n)_{n\in\mathbb{N}}$ of points in the domain D which converges to c (i.e. $\lim_{n\to\infty} x_n = c$), the corresponding sequence $(f(x_n))_{n\in\mathbb{N}}$ converges to $f(c)$ (i.e. $\lim_{n\to\infty} f(x_n) = f(c)$).

If f and g are continuous at c, then each of the following functions are also continuous at c:

(i) $|f|$
(ii) $c\,f$, for each $c \in \mathbb{R}$
(iii) $f \pm g$
(iv) $f \cdot g$
(v) $\dfrac{f}{g}$, provided that $g(c) \neq 0$
(vi) $f \circ g$, provided f is continuous at $g(c)$

If f is a continuous function on an interval $[a, b]$ then f is bounded on $[a, b]$ and there exist constants $c, d \in [a, b]$, such that $f(c) \leq f(x) \leq f(d)$ for all $x \in [a, b]$.

Some types of discontinuities include:

1. **Removable:** Both one-sided limits of the function f exist and equal each other at the given point c, however, the value of f at c is not equal to the limit value L. Symbolically,

   $$\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = \lim_{x\to c} f(x) = L \neq f(c) .$$

   If f is undefined at c then we call this a removable singularity.

2. **Jump:** The one-sided limits of the function f exist and are finite but are not equal to each other at the point c. Symbolically, $\lim_{x\to c^-} f(x) = L^- \neq L^+ = \lim_{x\to c^+} f(x)$. f can take any value at c.

3. **Essential:** One or both of the one-side limits of the function f are either infinite, or undefined at the point c.

Uniform and Pointwise Convergence:

Pointwise Convergence:

> Let $\{f_n\}$ be a sequence of functions on a set X. Let f be a function on X. We say that $\{f_n\}$ converges pointwise to f on X if $\lim\limits_{n\to\infty} f_n(x) = f(x)$ for every $x \in X$.

Uniform Convergence:

> Let $\{f_n\}$ be a sequence of functions on a set X. Let f be a function on X. We say that $\{f_n\}$converges uniformly to f on X if for every $\varepsilon > 0$, there exists a positive integer N such that if $n \geq N$, then $|f_n(x) - f(x)| < \varepsilon$ for every $x \in X$. (Note that N depends on ε but not on x)

Let $f(x)$ and $g(x)$ be continuous functions with limits L and M respectively. Then the following general properties hold (provided the limits on both sides exist):

(i) $\lim\limits_{x\to c} (f(x) \pm a) = L \pm a$

(ii) $\lim\limits_{x\to c} (f(x) \pm g(x)) = \lim\limits_{x\to c} f(x) \pm \lim\limits_{x\to c} g(x) = L \pm M$

(iii) $\lim\limits_{x\to c} a\, f(x) = a \cdot \lim\limits_{x\to c} f(x) = a\, L$

(iv) $\lim\limits_{x\to c} |f(x)| = |L|$

(v) $\lim\limits_{x\to c} (f(x) \cdot g(x)) = \left(\lim\limits_{x\to c} f(x)\right) \cdot \left(\lim\limits_{x\to c} g(x)\right) = L\, M$

(vi) $\lim\limits_{x\to c} \left(\dfrac{f(x)}{g(x)}\right) = \dfrac{\lim\limits_{x\to c} f(x)}{\lim\limits_{x\to c} g(x)} = \dfrac{L}{M}$ provided $\lim\limits_{x\to c} g(x) \neq 0$

(vii) $\lim\limits_{x\to c} [f(x)]^n = \left[\lim\limits_{x\to c} f(x)\right]^n = L^n, \quad n \in \mathbb{Z}^+$

(viii) $\lim\limits_{x\to c} a^{f(x)} = a^{\left[\lim\limits_{x\to c} f(x)\right]} = a^L$

If $g(x)$ is continuous at L and $\lim\limits_{x\to c} f(x) = L$ then $\lim\limits_{x\to c} g(f(x)) = g(L)$

If $f(x) \leq g(x)$ for all x in an interval that contains c, except possibly c itself, and the limit of $f(x)$ and $g(x)$ both exist at c, then $\lim\limits_{x\to c} f(x) \leq \lim\limits_{x\to c} g(x)$

If $f(x) \leq g(x) \leq h(x)$ for all x in an open interval that contains c, except possibly c itself, and $\lim\limits_{x\to c} f(x) = \lim\limits_{x\to c} h(x) = L$, then $\lim\limits_{x\to c} g(x) = L$ (Squeeze Theorem) .

Differentiation:

The notion of a derivative (of one variable) is formally defined through limits of functions as:

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x},$$

where h is the infinitesimal change in the variable x. Some common rules of differentiation can also be cast in this form (assuming f and g are both differentiable at x) as:

$$\lim_{h\to 0} \frac{f\circ g(x+h) - f\circ g(x)}{h} = f'[g(x)]\, g'(x) \qquad \text{(Chain Rule)}$$

$$\lim_{h\to 0} \frac{f(x+h)\, g(x+h) - f(x)\, g(x)}{h} = f'(x)\, g(x) + f(x)\, g'(x) \qquad \text{(Product Rule)}$$

$$\lim_{h\to 0} \left(\frac{f(x+h)}{f(x)}\right)^{\frac{1}{h}} = \exp\left(\frac{f'(x)}{f(x)}\right)$$

If $f(x)$ and $g(x)$ are differentiable on an open interval containing c, except possibly c itself, and $\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0$ or $\pm\infty$, L'Hopital's rule can be used:

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}\,.$$

Limiting Behaviour:

Asymptotic equivalences, $f(x) \sim g(x)$, are true if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1$. Therefore, they can also be reframed as limits. Some notable asymptotic equivalences include:

$\lim_{x\to\infty} \frac{x/\ln(x)}{\pi(x)} = 1$, due to the prime number theorem, $\pi(x) \sim \frac{x}{\ln(x)}$, where $\pi(x)$ is the prime counting function.

$\lim_{n\to\infty} \frac{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n}{n!} = 1$, due to Stirling's approximation, $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$.

The behaviour of functions described by Big O notation can also be described by limits. For example:

$$f(x) \in O(g(x)) \quad \text{if} \quad \limsup_{x\to\infty} \frac{|f(x)|}{g(x)} < \infty\,.$$

Limits of Exponential Functions:

$$\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

Limits of Logarithmic Functions:

$$\lim_{n\to\infty} n(x^{1/n} - 1) = \log(x)$$

$$\lim_{x\to c} \log_a(x) = \log_a(c) \qquad a > 1\,;\ c > 0$$

$$\lim_{x\to 0^+} \log_a(x) = -\infty \qquad a > 1$$

$$\lim_{x\to 0^+} \log_a(x) = \infty \qquad 0 < a < 1$$

$$\lim_{x\to\infty} \log_a(x) = \infty \qquad a > 1$$

$$\lim_{x\to\infty} \log_a(x) = -\infty \qquad 0 < a < 1$$

$$\lim_{x\to 0^+} x^b \log_a(x) = 0 \qquad b > 0$$

$$\lim_{x\to\infty} \frac{\log_a(x)}{x^b} = 0 \qquad b > 0$$

Limits of Trigonometric Functions:

$$\lim_{x\to a} \sin(x) = \sin(a)$$

$$\lim_{x\to a} \cos(x) = \cos(a)$$

$$\lim_{x\to 0} \frac{\sin(ax)}{ax} = 1 \quad a \neq 0$$

$$\lim_{x\to 0} \frac{\tan(ax)}{ax} = 1$$

$$\lim_{x\to 0} \frac{\cos(ax) - 1}{ax} = 0$$

$$\lim_{n\to\infty} \sum_{k=-n}^{n} \frac{1}{\pi\left(\frac{1}{2} - k\right) - z} = \tan(z)\,, \qquad \frac{z}{\pi} - \frac{1}{2} \notin \mathbb{Z}$$

$$\lim_{n\to\infty} \sum_{k=-n}^{n} \frac{1}{z - \pi k} = \cot(z)\,, \qquad \frac{z}{\pi} \notin \mathbb{Z}$$

$$\lim_{n\to\infty} \sum_{k=-n}^{n} \frac{(-1)^k}{z - \pi k} = \csc(z)\,, \qquad \frac{z}{\pi} \notin \mathbb{Z}$$

$$\lim_{n\to\infty} \sum_{k=-n}^{n} \frac{(-1)^k}{\pi\left(\frac{1}{2} + k\right) - z} = \tan(z)\,, \qquad \frac{z}{\pi} - \frac{1}{2} \notin \mathbb{Z}$$

Limits of Special Functions:

Generalized Hypergeometric Function:

$${}_{p-1}F_q(a_2, \ldots, a_p\,; b_1, \ldots, b_q\,; z) = \lim_{a\to\infty} {}_pF_q\left(a, a_2, \ldots, a_p\,; b_1, \ldots, b_q\,; \frac{z}{a}\right)$$

$${}_pF_{q-1}(a_1, \ldots, a_p\,; b_2, \ldots, b_q\,; z) = \lim_{b\to\infty} {}_pF_q\left(a_1, \ldots, a_p\,; b, b_2, \ldots, b_q\,; bz\right)$$

$$\theta(z) = \lim_{n\to\infty} \frac{1}{e^{-nz} + 1} = \lim_{n\to\infty} \frac{1}{2}\left(\tanh(nz) + 1\right) \qquad \text{(Heaviside Step Function)}$$

$$\delta(z) = \lim_{n\to 0} \frac{1}{2\sqrt{\pi n}} e^{-\frac{z^2}{4n}} = \lim_{n\to\infty} \frac{n}{\pi(1 + n^2 z^2)} = \lim_{n\to\infty} \frac{\sin(nz)}{\pi z} \qquad \text{(Dirac Delta Function)}$$

$$\zeta(z) = \lim_{n\to\infty} \left(\sum_{k=1}^{n} k^{-z} - \frac{n^{1-z} - 1}{1 - z}\right) - \frac{1}{1 - z}\,, \quad \mathrm{Re}(z) > 0 \qquad \text{(Riemann Zeta Function)}$$

$$\gamma = \lim_{z\to 1} \left(\zeta(z) - \frac{1}{z - 1}\right) \qquad \text{(Euler-Mascheroni Constant)}$$

$$1 = \lim_{n\to\infty} \frac{\Gamma(a + n)\, n^{b-a}}{\Gamma(b + n)}$$

$$\mathrm{Si}(z) = \frac{\pi}{2} - \lim_{n\to\infty} \left(\sum_{k=n+1}^{\infty} \frac{1}{k} \sin\left(\frac{kz}{n}\right)\right)\,, \quad z \in \mathbb{R}\,;\, z > 0 \qquad \text{(Sine Integral)}$$

$$\mathrm{Ci}(z) = -\lim_{n\to\infty} \left(\sum_{k=n+1}^{\infty} \frac{1}{k} \cos\left(\frac{kz}{n}\right)\right)\,, \quad z \in \mathbb{R}\,;\, z > 0 \qquad \text{(Cosine Integral)}$$

$$J_{-\mu}(z) = \lim_{n\to\infty} n^{-\mu} P_n^{\mu}\left(\cos\left(\frac{z}{n}\right)\right) \qquad \text{(Bessel Function of the 1st Kind)}$$

(Where $P_n^{\mu}(z)$ is the associated Legendre polynomial of the 1st kind)

$$J_{\lambda}(z) = \lim_{n\to\infty} \left(\frac{z}{2n}\right)^{\lambda} L_n^{\lambda}\left(\frac{z^2}{4n}\right)$$

(Where $L_n^{\lambda}(z)$ is the associated Laguerre polynomial)

$$Y_\nu(z) = \lim_{\lambda\to\infty} -\frac{2}{\pi}\lambda^\nu Q_\lambda^{-\nu}\left(\cos\left(\frac{z}{\lambda}\right)\right) \qquad \text{(Bessel Function of the 2nd Kind)}$$

(Where $Q_\lambda^\nu(z)$ is the associated Legendre function of the 2nd kind)

$$I_\lambda(z) = \lim_{n\to\infty} \left(\frac{z}{2n}\right)^\lambda L_n^\lambda\left(-\frac{z^2}{4n}\right) \qquad \text{(Modified Bessel Function of the 1st Kind)}$$

(Where $L_n^\lambda(z)$ is the associated Laguerre polynomial)

$$I_\nu(z) = \lim_{\lambda\to\infty} (\lambda)^\nu P_\lambda^{-\nu}\left(\cosh\left(\frac{z}{\lambda}\right)\right)$$

(Where $P_\lambda^\nu(z)$ is the associated Legendre polynomial of the 1st kind)

$$K_\nu(z) = \lim_{\lambda\to\infty} \left(\lambda e^{i\pi}\right)^{-\nu} Q_\lambda^\nu\left(\cosh\left(\frac{z}{\lambda}\right)\right) \qquad \text{(Modified Bessel Function of the 2nd Kind)}$$

(Where $Q_\lambda^\nu(z)$ is the associated Legendre polynomial of the 2nd kind)

$$H_n(z) = \lim_{\lambda\to\infty} n!\left(\frac{2}{\lambda}\right)^{n/2} L_n^\lambda\left(\lambda - \sqrt{2\lambda}\, z\right) \qquad \text{(Hermite Polynomial)}$$

(Where $L_n^\lambda(z)$ is the associated Laguerre polynomial)

$$\frac{2\sin(z)}{\sqrt{\pi}} = \lim_{n\to\infty} \frac{(-1)^n}{4^n n!} H_{2n+1}\left(\frac{z}{2\sqrt{n}}\right)$$

$$\frac{\cos(z)}{\sqrt{\pi}} = \lim_{n\to\infty} \frac{(-1)^n\sqrt{n}}{4^n n!} H_{2n}\left(\frac{z}{2\sqrt{n}}\right)$$

$$j_\nu(z) = \lim_{\lambda\to\infty} \sqrt{\frac{\lambda\pi}{2z}}\lambda^\nu P_\lambda^{-(\nu+\frac{1}{2})}\left(\cos\left(\frac{z}{\lambda}\right)\right) \qquad \text{(Spherical Bessel Function of the 1st Kind)}$$

(Where $P_\lambda^\nu(z)$ is the associated Legendre function of the 1st kind)

$$j_\nu(z) = \lim_{n\to\infty} \sqrt{\frac{\pi}{4n}}\left(\frac{z}{2n}\right)^\nu L_n^{\nu+\frac{1}{2}}\left(\frac{z^2}{4n}\right)$$

(Where $L_n^\nu(z)$ is the associated Laguerre polynomial)

1.9 Properties of Sums and Series

The sum of a sequence $a_1, a_2, \ldots$ defined over an arbitrary field F is given by:

$$S_n = \sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_{n-1} + a_n \tag{5}$$

and in the limit $n \to \infty$, the sum is called a series and is given by:

$$\lim_{n\to\infty} \sum_{k=m}^{n} a_k = \sum_{k=m}^{\infty} a_k = a_m + a_{m+1} + a_{m+2} + \ldots \tag{6}$$

(The original sum in this case can now be seen as a partial sum of the first n terms of the infinite sequence)

A series is said to be convergent if it converges to the limit L:

$$L = \sum_{k=0}^{\infty} a_k \ ,$$

otherwise it is said to be divergent. More formally, a series is convergent if there exists a number L such that for every arbitrarily small positive number ε, there is a sufficiently large number $N \in \mathbb{Z}$ such that for all $n \geq N$, $|S_n - L| \leq \varepsilon$.

There are different types of convergence, namely absolute convergence and conditional convergence. A series $\sum_{k=0}^{\infty} a_k$ is said to be absolutely convergent if the series of absolute values $\sum_{k=0}^{\infty} |a_k|$ converges. A series which is convergent but not absolutely convergent is said to be conditionally convergent. The Riemann series theorem says that if a series is conditionally convergent, then its terms can be permuted to form a new series which either converges to an arbitrary number or diverges.

Some tests to determine the convergence of a series are discussed below.

n-th Term Test:

The nth term test says that if $\lim_{n\to\infty} a_n \neq 0$ then the series must diverge. The test is inconclusive if the limit evaluates to zero. (This is the first test one should employ when testing for convergence of a series)

Ratio Test:

Suppose there exists r such that

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = r \ .$$

If $r < 1$ then the series is absolutely convergent, if $r > 1$ then the series diverges, and if $r = 1$ then the test is inconclusive and another test must be used to determine convergence.

Root Test:

Suppose there exists r such that

$$r = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} \ .$$

If $r < 1$ then the series is absolutely convergent, if $r > 1$ then the series diverges, and if $r = 1$ then the test is inconclusive and another test must be used to determine convergence. (note that whenever the ratio test determines convergence, the root test does as well, but not conversely)

Integral Test:

Let $f : [1, \infty) \to \mathbb{R}^+$ be a nonnegative and monotonically decreasing function such that $f(n) = a_n$, then if

$$\int_1^{\infty} f(x)\,\mathrm{d}x = \lim_{t\to\infty} \int_1^{t} f(x)\,\mathrm{d}x < \infty \,, \tag{7}$$

the series will converge. If the integral diverges then the series does so as well.

An application of this test to the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ shows that it converges for $p > 1$ and diverges for $p \leq 1$ (this is known as the p-series).

Direct Comparison Test:

If the series $\sum_{n=1}^{\infty} b_n$ is absolutely convergent and $|a_n| \leq |b_n|$ for sufficiently large n, then the series $\sum_{n=1}^{\infty} a_n$ is also absolutely convergent.

Limit Comparison Test:

for two strictly positive sequences $\{a_n\}$ and $\{b_n\}$, if $\lim_{n\to\infty} \frac{a_n}{b_n}$ exists, is finite, and non-zero, then $\sum_{n=1}^{\infty} a_n$ diverges if and only if $\sum_{n=1}^{\infty} b_n$ diverges.

Cauchy Condensation Test:

Let $\{a_n\}$ define a positive non-increasing sequence, then the sum $A = \sum_{n=1}^{\infty} a_n$ converges if and only if the sum $A = \sum_{n=0}^{\infty} 2^n a_{2^n}$ converges. (if they converge then $A \leq A \leq 2A$ holds)

Abel's Test:

Suppose $\sum a_n$ is a convergent series and $\{b_n\}$, a monotonic and bounded sequence, then the series $\sum b_n a_n$ is also convergent.

Alternating Series Test:

Let $\{a_n\}$ be a sequence and suppose that for every n, $a_{n+1} \leq a_n$, and that $\lim_{n\to\infty} a_n = 0$. Then $\sum_{n=k}^{\infty} (-1)^n a_n$ and $\sum_{n=k}^{\infty} (-1)^{n+1} a_n$ are convergent series.

Dirichlet's Test:

Let $\{a_n\}$ be a sequence of real numbers and $\{b_n\}$ be a sequence of complex numbers such that $a_{n+1} \leq a_n$, $\lim_{n\to\infty} a_n = 0$, and $\left|\sum_{n=1}^{N} b_n\right| \leq M$ for every $N \in \mathbb{Z}^+$, where M is a constant, then the series $\sum_{n=1}^{\infty} b_n a_n$ converges.

Raabe-Duhamel's Test:

Let $\sum a_n$ be a series of real numbers for which $a_n > 0$ and define $b_n = n\left(\frac{a_n}{a_{n+1}} - 1\right)$. If $b > 1$ and $K \in \mathbb{N}$ exist such that $\left|\frac{a_{n+1}}{a_n}\right| \leq 1 - \frac{b}{n}$ for all $n > K$ then $\sum a_n$ is convergent.

Bertrand's Test:

Let $\{a_n\}$ be a sequence of positive numbers and define $b_n = \ln\left(n^2\left(\frac{a_n}{a_{n+1}} - 1\right) - n\right)$. If the limit of b_n exists and equals L then for $L > 1$ the series converges, for $L < 1$ the series diverges, and for $L = 1$ the test is inconclusive.

Gauss' Test:

Let $\{a_n\}$ be a sequence of positive numbers. If $\dfrac{a_n}{a_{n+1}} = 1 + \dfrac{\alpha}{n} + O\left(\dfrac{1}{n^\beta}\right)$ for some $\beta > 1$, then $\sum a_n$ converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.

Some general sum identities include:

Distributivity:

$$\sum_{n=s}^{t} C \cdot f(n) = C \cdot \sum_{n=s}^{t} f(n)$$

$$\left(\sum_{i=0}^{n} a_i\right)\left(\sum_{k=0}^{n} b_k\right) = \sum_{i=0}^{n}\sum_{k=0}^{n} a_i\, b_k$$

$$\sum_{i=s}^{m}\sum_{j=t}^{n} a_i\, c_j = \left(\sum_{i=s}^{m} a_i\right)\left(\sum_{j=t}^{n} c_j\right)$$

Commutativity and Associativity:

$$\sum_{n=s}^{t} f(n) \pm \sum_{n=s}^{t} g(n) = \sum_{n=s}^{t}(f(n) \pm g(n))$$

$$\sum_{i=k_0}^{k_1}\sum_{j=l_0}^{l_1} a_{i,j} = \sum_{j=l_0}^{l_1}\sum_{i=k_0}^{k_1} a_{i,j}$$

$$\sum_{k\leq j\leq i\leq n} a_{i,j} = \sum_{i=k}^{n}\sum_{j=k}^{i} a_{i,j} = \sum_{j=k}^{n}\sum_{i=j}^{n} a_{i,j} = \sum_{j=0}^{n-k}\sum_{i=k}^{n-j} a_{i+j,i}$$

Index Shift:

$$\sum_{n=s}^{t} f(n) = \sum_{n=s+p}^{t+p} f(n-p)$$

Generalized Index Shift:

$$\sum_{n\in B} f(n) = \sum_{m\in A} f(\sigma(m))$$

(for a bijection σ from a finite set A onto a set B)

Index Splitting:

$$\sum_{n=s}^{t} f(n) = \sum_{n=s}^{j} f(n) + \sum_{n=j+1}^{t} f(n)$$

$$\sum_{n=a}^{b} f(n) = \sum_{n=0}^{b} f(n) - \sum_{n=0}^{a-1} f(n)$$

$$\sum_{n=2s}^{2t+1} f(n) = \sum_{n=s}^{t} f(2n) + \sum_{n=s}^{t} f(2n+1) \qquad \text{(Even Indices)}$$

$$\sum_{n=2s+1}^{2t} f(n) = \sum_{n=s+1}^{t} f(2n) + \sum_{n=s+1}^{t} f(2n-1) \qquad \text{(Odd Indices)}$$

Reverse Order Sum Equivalence:

$$\sum_{n=s}^{t} f(n) = \sum_{n=0}^{t-s} f(t-n)$$

$$\sum_{n=0}^{t} f(n) = \sum_{n=0}^{t} f(t-n)$$

Sum to Product Log Property:

$$\sum_{n=s}^{t} \log_b(f(n)) = \log_b\left(\prod_{n=s}^{t} f(n)\right)$$

Sum to Product Exp Property:

$$C^{\sum_{n=s}^{t} f(n)} = \prod_{n=s}^{t} C^{f(n)}$$

An analogue to integration by parts also exists for sums of products of sequences. Given the sequences $\{f_k\}$ and $\{g_k\}$, summation by parts is defined as:

$$\sum_{k=m}^{n} f_k(g_{k+1} - g_k) = (f_n\, g_{n+1} - f_m\, g_m) - \sum_{k=m+1}^{n} g_k(f_k - f_{k-1})\,. \tag{8}$$

Summation by Parts and the Cauchy Product:

As well, we can define the Cauchy product of two infinite series using a discrete convolution:

$$\left(\sum_{k=0}^{\infty} a_k\right) \cdot \left(\sum_{i=0}^{\infty} b_i\right) = \sum_{j=0}^{\infty} c_j = \sum_{j=0}^{\infty} \left[\sum_{l=0}^{j} a_l\, b_{j-l}\right] , \tag{9}$$

and in general for $n \in \mathbb{Z}^+$ absolutely convergent (except for the nth one, which only converges) infinite series we have:

$$\sum_{k_1=0}^{\infty} \sum_{k_2=0}^{k_1} \cdots \sum_{k_n=0}^{k_{n-1}} a_{1,k_n}\, a_{2,k_{n-1}-k_n} \cdots a_{n,k_1-k_2} = \prod_{j=1}^{n} \left(\sum_{j=0}^{\infty} a_{j,k_j}\right) . \tag{10}$$

Method of Differences:

A telescoping sum is a sum of the difference in consecutive terms of a sequence, which cancel each other when summed, leaving only the first and final terms. Symbolically, this is given by:

$$\sum_{n=1}^{N} (a_n - a_{n-1}) = a_N - a_0 ,$$

and in the limit $N \to \infty$, provided $a_n \to 0$:

$$\sum_{n=1}^{\infty} (a_n - a_{n-1}) = -a_0 .$$

(For quotients, a partial fraction decomposition may reveal this type of sum/series)

1.9.1 Euler and Bernoulli Numbers

The Bernoulli ($B_n^{\pm}$) and Euler ($E_n^{\pm}$) numbers are sequences that appear frequently in representations of functions, they are defined as:

$$B_n^- = \sum_{k=0}^{n} \sum_{v=0}^{k} (-1)^v \binom{k}{v} \frac{v^n}{k+1} , \qquad B_n^+ = \sum_{k=0}^{n} \sum_{v=0}^{k} (-1)^v \binom{k}{v} \frac{(v+1)^n}{k+1} ,$$

$$E_{2n} = \sum_{k=1}^{2n} (-1)^k \frac{1}{2^k} \sum_{l=0}^{2k} (-1)^l \binom{2k}{l} (k-l)^{2n}$$

(Only the even Euler numbers are given since the odd Euler numbers are all zero).

The Bernoulli numbers can also be defined recursively as:

$$B_n^- = \delta_{n,0} - \sum_{k=0}^{n-1} \binom{n}{k} \frac{B_k^-}{n-k+1} \quad \text{and} \quad B_n^+ = 1 - \sum_{k=0}^{n-1} \binom{n}{k} \frac{B_k^+}{n-k+1}$$

The Bernoulli numbers are related to the Euler numbers through the expressions:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} \frac{n}{4^n - 2^n} E_k \quad \text{and} \quad E_n = \sum_{k=1}^{n} \binom{n}{k-1} \frac{2^k - 4^k}{k} B_k$$

for $n = 2, 4, 6, \ldots$.

The Euler and Bernoulli numbers can also be used to define the Bernoulli $B_n(z)$ and Euler $E_m(z)$ polynomials, given by:

$$B_n(z) = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}\, z^k \quad \text{and} \quad E_m(z) = \sum_{k=0}^{m} \frac{E_k}{2^k} \binom{m}{k} \left(z - \frac{1}{2}\right)^{m-k} ,$$

for $n \geq 0$, or in terms of their generating functions as:

$$\frac{t\, e^{zt}}{e^t - 1} = \sum_{n=0}^{\infty} B_n(z)\, \frac{t^n}{n!} \quad \text{and} \quad \frac{2e^{zt}}{e^t + 1} = \sum_{n=0}^{\infty} E_n(z)\, \frac{t^n}{n!} .$$

1.9.2 Harmonic Numbers

The generalized harmonic numbers H_n^k are given by the expression:

$$H_n^m = \sum_{k=1}^{n} \frac{1}{k^m} = \sum_{k=1}^{n-1} \frac{H_k^{m-1}}{k\,(k+1)} + \frac{H_n^{m-1}}{n} ,$$

and have the additional relation:

$$\sum_{k=1}^{n} H_k^m = (n+1)\, H_n^m - H_n^{m-1} ,$$

for $m \geq 0$. The harmonic numbers H_n are recovered for $m = 1$ with an integral representation given by:

$$H_n = \int_0^1 \frac{1 - x^n}{1 - x}\, \mathrm{d}x$$

1.9.3 Summation Formulae

Summation methods are methods employed to assign a value to divergent series. More specifically, a summation method is a partial function from the set of series to values. Some desirable properties for these summation methods include:

1. **Regularity:** whenever the sequence s converges to x, then $\boldsymbol{A}(s) = x$.
2. **Linearity:** $\boldsymbol{A}$ is a linear functional on the sequences where it is defined, meaning that $\boldsymbol{A}(kr + s) = k\boldsymbol{A}(r) + \boldsymbol{A}(s)$ for sequences r, s and real or complex scalar k.
3. **Stability:** If s is a sequence starting from s_0 and s' is the sequence obtained by omitting the first value and subtracting it from the rest, so that $s'_n = s_{n+1} - s_0$, then $\boldsymbol{A}(s)$ is defined if and only if $\boldsymbol{A}(s')$ is defined, and $\boldsymbol{A}(s) = s_0 + \boldsymbol{A}(s')$.

Between distinct summation methods $\boldsymbol{A}$ and $\boldsymbol{B}$, a desirable property to have is consistency, which is to say that for every sequence s, both methods assign a value $\boldsymbol{A}(s) = \boldsymbol{B}(s)$. (Note that some methods do not posses the above properties)

Some standard summation methods are discussed below.

<u>Norlund Means</u>

For a sequence of positive terms $\{p_n\}$ such that

$$\frac{p_n}{\sum_{k=0}^{n} p_k} \to 0 \,,$$

the transformation of a sequence s using p to give weighted means is:

$$t_m = \frac{\sum_{k=0}^{m} p_{m-k}\, s_k}{\sum_{k=0}^{m} p_k} \,,$$

and in the limit of t_n as n goes to infinity, this quantity is called the Norlund mean $N_p(s)$. The Norlund mean is regular, linear, stable, and consistent with any other Norlund mean.

Cesaro Summation:

Define the infinite sequence $\{a_n\}^{\infty}$, and let its kth partial sum be given by:

$$s_k = a_1 + \cdots a_k = \sum_{n=1}^{k} a_n \,.$$

The above infinite sequence is said to be Cesaro summable, with Cesaro sum $A \in \mathbb{R}$, if, as n tends to infinity, the arithmetic mean of its first n partial sums $s_1, s_2, \ldots, s_n$ tends to A:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} s_k = A \,.$$

This method can be generalized to a family of methods called (C, α), for $\alpha \in \mathbb{R}\backslash\mathbb{Z}^{-}$. Given a series $\sum a_n$, we start by defining the quantity A_n^{α} implicitly as

$$\sum_{n=0}^{\infty} A_n^{\alpha}\, x^n = \frac{1}{(1-x)^{1+\alpha}} \sum_{n=0}^{\infty} a_n\, x^n$$

(Note that the upper indices are not exponents) and define E_n^{α} to be the A_n^{α} for the series $1+0+0+0+\ldots$ (In particular, E_n^{α} are the binomial coefficients of power $-(1+\alpha)$). Then the (C, α) sum of $\sum a_n$ is denoted by (C, α)-$\sum a_n$ and has the value:

$$(C, \alpha) - \sum_{j=0}^{\infty} a_j = \lim_{n\to\infty} \frac{A_n^{\alpha}}{E_n^{\alpha}}$$

if it exists. The $(C, 0)$ method is just ordinary summation, while the $(C, 1)$ method is the method described above.

We can also describe the Cesaro summability of an integral: let $\alpha \geq 0$, then the integral $\int_0^{\infty} f(x)\,\mathrm{d}x$ is (C, α) summable if

$$\lim_{\lambda\to\infty} \int_0^{\lambda} \left(1 - \frac{x}{\lambda}\right)^{\alpha} f(x)\,\mathrm{d}x$$

exists and is finite. The value of this limit, should it exist, is the (C, α) sum of the integral.

Cesaro sums (C, k), for $k \geq 0$, are Norlund means $(C, k) - \sum s = N_{p^k}(s)$, where the sequence p^k is given by

$$p_n^k = \binom{n+k-1}{k-1} \,.$$

Abelian Means

Suppose $\lambda = \{\lambda_0, \lambda_1, \lambda_2, \ldots\}$ is a strictly increasing sequence tending towards infinity, and that $\lambda_0 \geq 0$. Furthermore, suppose

$$f(x) = \sum_{n=0}^{\infty} a_n \, e^{-\lambda_n x}$$

converges for all real numbers $x > 0$. Then the Abelian mean A_λ is defined as

$$A_\lambda(s) = \lim_{x \to 0^+} f(x) \, .$$

More generally, if the series for f only converges for large x but can be analytically continued to all positive real x, then one can still define the sum of the divergent series by the limit above.

Abelian means are regular and linear, but not stable and not always consistent between different choices of λ.

Abel Summation:

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real or complex numbers. Define the partial sum function A as:

$$A(t) = \sum_{0 \leq n \leq t} a_n \, ,$$

for any real number t. Fix real numbers $x < y$, and let ϕ be a continuously differentiable function on $[x, y]$. Then:

$$\sum_{x < n \leq y} a_n \, \phi(n) = A(y) \, \phi(y) - A(x) \, \phi(x) - \int_x^y A(u) \, \phi'(u) \, \mathrm{d}u \, .$$

The formula is derived by applying integration by parts for a Riemann–Stieltjes integral to the functions A and ϕ.

Some variations of the above formula are discussed below.

Taking the left endpoint to be -1 gives the formula:

$$\sum_{0 \leq n \leq x} a_n \, \phi(n) = A(x) \, \phi(x) - \int_0^x A(u) \, \phi'(u) \, \mathrm{d}u \, .$$

If the sequence (a_n) is indexed starting at $n = 1$, then we may formally define $a_0 = 0$. The previous formula then becomes:

$$\sum_{1 \leq n \leq x} a_n \, \phi(n) = A(x) \, \phi(x) - \int_1^x A(u) \, \phi'(u) \, \mathrm{d}u \, .$$

A common way to apply Abel's summation formula is to take the limit of one of these formulas as $x \to \infty$. The resulting formulae are:

$$\sum_{n=0}^{\infty} a_n \, \phi(n) = \lim_{x \to \infty} \left(A(x) \, \phi(x) \right) - \int_0^{\infty} A(u) \, \phi'(u) \, \mathrm{d}u,$$

$$\sum_{n=1}^{\infty} a_n \, \phi(n) = \lim_{x \to \infty} \left(A(x) \, \phi(x) \right) - \int_1^{\infty} A(u) \, \phi'(u) \, \mathrm{d}u \, .$$

These equations hold whenever both limits on the right-hand side exist and are finite.

A particularly useful case is the sequence $a_n = 1$ for all $n \geq 0$. In this case, $A(x) = \lfloor x+1 \rfloor$. For this sequence, Abel's summation formula simplifies to:

$$\sum_{0 \leq n \leq x} \phi(n) = \lfloor x+1 \rfloor \phi(x) - \int_0^x \lfloor u+1 \rfloor \phi'(u)\, \mathrm{d}u \,.$$

Similarly, for the sequence $a_0 = 0$ and $a_n = 1$ for all $n \geq 1$, the formula becomes:

$$\sum_{1 \leq n \leq x} \phi(n) = \lfloor x \rfloor \phi(x) - \int_1^x \lfloor u \rfloor \phi'(u)\, \mathrm{d}u \,.$$

Upon taking the limit as $x \to \infty$, we find:

$$\sum_{n=0}^{\infty} \phi(n) = \lim_{x \to \infty} \left(\lfloor x+1 \rfloor \phi(x) \right) - \int_0^{\infty} \lfloor u+1 \rfloor \phi'(u)\, \mathrm{d}u \,,$$

$$\sum_{n=1}^{\infty} \phi(n) = \lim_{x \to \infty} \left(\lfloor x \rfloor \phi(x) \right) - \int_1^{\infty} \lfloor u \rfloor \phi'(u)\, \mathrm{d}u \,,$$

assuming that both terms on the right-hand side exist and are finite.

Abel's summation formula can be generalized to the case where ϕ is only assumed to be continuous if the integral is interpreted as a Riemann–Stieltjes integral:

$$\sum_{x < n \leq y} a_n \phi(n) = A(y)\phi(y) - A(x)\phi(x) - \int_x^y A(u)\, \mathrm{d}\phi(u) \,.$$

By taking ϕ to be the partial sum function associated to some sequence, this leads to the summation by parts formula.

Perron's Formula:

Let $a(n)$ be an arithmetic function, and let $g(s) = \sum_{n=1}^{\infty} \frac{a(n)}{n^s}$ be the corresponding uniformly convergent Dirichlet series for $\mathrm{Re}(s) > \sigma$. Perron's formula is then given by:

$$A(x) = \sum_{n \leq x} a(n) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} g(z) \frac{x^z}{z}\, \mathrm{d}z \,,$$

where it should be understood that the last term of the sum is to be multiplied by $1/2$ when $x \in \mathbb{Z}$, and that the arbitrary constant c should satisfy $c > 0$, $c > \sigma$, and $x \in \mathbb{R}^+$.

Perron's formula can be seen as a special case of the Mellin discrete convolution:

$$\sum_{n=1}^{\infty} a(n)\, f\left(\frac{n}{x}\right) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)\, G(s)\, x^s\, \mathrm{d}s \,,$$

where

$$G(s) = \sum_{n=1}^{\infty} \frac{a(n)}{n^s} \quad \text{and} \quad F(s) = \int_0^{\infty} f(x)\, x^{s-1}\, \mathrm{d}x \,,$$

using the test function $f\left(\frac{n}{x}\right) = \theta(x-1)$, where $\theta(x)$ is the Heaviside step function.

Abel-Plana Formula:

The Abel-Plana formula is given by:

$$\sum_{n=0}^{\infty} f(n) = \int_0^{\infty} f(x)\,\mathrm{d}x + \frac{1}{2} f(0) + i \int_0^{\infty} \frac{f(it) - f(-it)}{e^{2\pi t} - 1}\,\mathrm{d}t\,,$$

for functions f that are holomorphic in the region $\mathrm{Re}(z) \geq 0$ and which satisfy a suitable growth condition in this region (For example it is enough to assume that $|f|$ is bounded by $C\,|z|^{-(1+\varepsilon)}$ in this region for some constants $C, \varepsilon > 0$).

A formula for alternating sums is given by:

$$\sum_{n=0}^{\infty} (-1)^n f(n) = \frac{1}{2} f(0) + i \int_0^{\infty} \frac{f(it) - f(-it)}{2\sinh(\pi t)}\,\mathrm{d}t\,.$$

Lindelof Summation:

If $\lambda_n = n\,\log(n)$, then (indexing from one) we have:

$$f(x) = a_1 + a_2 2^{-2x} + a_3 3^{-3x} + \cdots\,.$$

Then $L(s)$, the Lindelof sum, is the limit of $f(x)$ as x goes to positive zero. The Lindelof sum is a powerful method when applied to power series among other applications, summing power series in the Mittag-Leffler star.

If $g(z)$ is analytic in a disk around zero, and hence has a Maclaurin series $G(z)$ with a positive radius of convergence, then $L(G(z)) = g(z)$ in the Mittag-Leffler star. Moreover, convergence to $g(z)$ is uniform on compact subsets of the star.

Analytic Continuation

Analytic Continuation of Power Series:

If $\sum a_n x^n$ converges for small complex x and can be analytically continued along some path from $x = 0$ to the point $x = 1$, then the sum of the series can be defined to be the value at $x = 1$. This value may depend on the choice of path.

Euler Summation:

Given a series $\sum a_n$, if its Euler transform converges to a sum, then that sum is called the Euler sum of the original series. As well as being used to define values for divergent series, Euler summation can be used to speed the convergence of series.

Euler summation can be generalized into a family of methods denoted (E, q), where $q \geq 0$. The $(E, 1)$ sum is the ordinary Euler sum. All of these methods are strictly weaker than Borel summation; for $q > 0$ they are incomparable with Abel summation.

For some value y we may define the Euler sum (if it converges for that value of y) corresponding to a particular formal summation as:

$${}_{E_y} \sum_{j=0}^{\infty} a_j := \sum_{i=0}^{\infty} \frac{1}{(1+y)^{i+1}} \sum_{j=0}^{i} \binom{i}{j} y^{j+1} a_j\,.$$

If the formal sum actually converges, an Euler sum will equal it. But Euler summation is particularly used to accelerate the convergence of alternating series and sometimes it can give a useful meaning to divergent sums.

To justify the approach notice that for interchanged sum, Euler's summation reduces to the initial series, because

$$y^{j+1} \sum_{i=j}^{\infty} \binom{i}{j} \frac{1}{(1+y)^{i+1}} = 1 \, .$$

This method itself cannot be improved by iterated application, as

$$E_{y_1} E_{y_2} \sum = E_{\frac{y_1 y_2}{1+y_1+y_2}} \sum \, .$$

Analytic Continuation of Dirichlet Series:

This method defines the sum of a series to be the value of the analytic continuation of the Dirichlet series:

$$f(s) = \frac{a_1}{1^s} + \frac{a_2}{2^s} + \frac{a_3}{3^s} + \cdots$$

at $s = 0$, if this exists and is unique. This method is sometimes confused with zeta function regularization. If $s = 0$ is an isolated singularity, the sum is defined by the constant term of the Laurent series expansion.

Zeta Function Regularization:

If the series

$$f(s) = \frac{1}{a_1^s} + \frac{1}{a_2^s} + \frac{1}{a_3^s} + \cdots$$

(for positive values of the a_n) converges for large real s and can be analytically continued along the real line to $s = -1$, then its value at $s = -1$ is called the zeta regularized sum of the series $a_1 + a_2 + \ldots$. Zeta function regularization is nonlinear.

In applications, the numbers a_i are sometimes the eigenvalues of a self-adjoint operator A with compact resolvent, and $f(s)$ is then the trace of A^{-s}. For example, if A has eigenvalues $1, 2, 3, \ldots$ then $f(s)$ is the Riemann zeta function, $\zeta(s)$, whose value at $s = -1$ is $-1/12$, assigning a value to the divergent series $1 + 2 + 3 + 4 + \ldots$. Other values of s can also be used to assign values for the divergent sums $\zeta(0) = 1 + 1 + 1 + \ldots = -1/12$, $\zeta(-2) = 1 + 4 + 9 + \ldots = 0$ and in general:

$$\zeta(-s) = \sum_{n=1}^{\infty} n^s = 1^s + 2^s + 3^s + \cdots = -\frac{B_{s+1}}{s+1} \, ,$$

where B_k is a Bernoulli number.

<u>Integral Function Means</u>

If $J(x) = \sum p_n x^n$ is an integral function, then the J sum of the series $\sum_{n=0}^{\infty} a_n$ is defined to be:

$$\lim_{x \to \infty} \frac{\sum_n p_n (a_0 + \cdots + a_n) x^n}{\sum_n p_n x^n} \, ,$$

if this limit exists.

There is a variation of this method where the series for J has a finite radius of convergence r and diverges at $x = r$. In this case one defines the sum as above, except taking the limit as x tends to r rather than infinity.

Valiron's Method:

Valiron's method is a generalization of Borel summation to certain more general integral functions J. Valiron showed that under certain conditions it is equivalent to defining the sum of a series as:

$$\lim_{n\to\infty} \sqrt{\frac{H(n)}{2\pi}} \sum_{h\in\mathbb{Z}} e^{-\frac{1}{2}h^2 H(n)} (a_0 + \cdots + a_h) ,$$

where H is the second derivative of G and $c(n) = e^{-G(n)}$, and $a_0 + \ldots + a_h$ is to be interpreted as 0 when $h < 0$.

<u>Moment Methods</u>

Suppose that $\mathrm{d}\mu$ is a measure on the real line such that all the moments

$$\mu_n = \int x^n \, \mathrm{d}\mu$$

are finite. If $\sum_{n=0}^{\infty} a_n$ is a series such that

$$a(x) = \frac{a_0 x^0}{\mu_0} + \frac{a_1 x^1}{\mu_1} + \cdots$$

converges for all x in the support of μ, then the $(\mathrm{d}\mu)$ sum of the series is defined to be the value of the integral

$$\int a(x) \, \mathrm{d}\mu$$

if it is defined. (Note that if the numbers μ_n increase too rapidly then they do not uniquely determine the measure μ.)

Borel Summation:

There are (at least) three slightly different methods called Borel summation. They differ in which series they can sum, but are consistent, meaning that if two of the methods sum the same series they give the same answer.

Throughout let $A(z)$ denote a formal power series:

$$A(z) = \sum_{k=0}^{\infty} a_k z^k ,$$

and define the Borel transform of A to be its equivalent exponential series:

$$\mathcal{B}A(t) \equiv \sum_{k=0}^{\infty} \frac{a_k}{k!} t^k .$$

1. Borel's exponential summation method:

Let $A_n(z)$ denote the partial sum:

$$A_n(z) = \sum_{k=0}^{n} a_k z^k .$$

A weak form of Borel's summation method defines the Borel sum of A to be:

$$\lim_{t\to\infty} e^{-t} \sum_{n=0}^{\infty} \frac{t^n}{n!} A_n(z) \ .$$

If this converges at $z \in \mathbb{C}$ to some $a(z)$, we say that the weak Borel sum of A converges at z, and write $\sum a_k z^k = a(z)\ (\boldsymbol{wB})$.

2. Borel's integral summation method:

Suppose that the Borel transform converges for all positive real numbers to a function growing sufficiently slowly that the following integral is well defined (as an improper integral), the Borel sum of A is given by:

$$\int_0^{\infty} e^{-t} \mathcal{B}A(tz)\, \mathrm{d}t \ .$$

This is the case when the measure $\mathrm{d}\mu = e^{-t}\mathrm{d}t$ for positive x and 0 for negative x. If the integral converges at $z \in \mathbb{C}$ to some $a(z)$, we say that the Borel sum of A converges at z, and write $\sum a_k z^k = a(z)\ (\boldsymbol{B})$.

There is a generalization of this depending on a variable α, called the (B', α) sum, where the sum of a series $\sum_{k=0}^{\infty} a_k$ is defined to be:

$$\int_0^{\infty} e^{-t} \sum \frac{a_n t^{n\alpha}}{\Gamma(n\alpha + 1)}\, \mathrm{d}t$$

if this integral exists.

3. Borel's integral summation method with analytic continuation:

This is similar to Borel's integral summation method, except that the Borel transform need not converge for all t, but converges to an analytic function of t near 0 that can be analytically continued along the positive real axis.

The methods $(\boldsymbol{B})$ and $(\boldsymbol{wB})$ are both regular summation methods, meaning that whenever $A(z)$ converges (in the standard sense), then the Borel sum and weak Borel sum also converge, and do so to the same value. i.e.

$$\sum_{k=0}^{\infty} a_k z^k = A(z) < \infty \quad \Rightarrow \quad \sum a_k z^k = A(z)\ (\boldsymbol{B},\ \boldsymbol{wB}) \ .$$

Regularity of $(\boldsymbol{B})$ is easily seen by a change in order of integration, which is valid due to absolute convergence: if $A(z)$ is convergent at z, then

$$A(z) = \sum_{k=0}^{\infty} a_k z^k = \sum_{k=0}^{\infty} a_k \left(\int_0^{\infty} e^{-t} t^k \mathrm{d}t \right) \frac{z^k}{k!} = \int_0^{\infty} e^{-t} \sum_{k=0}^{\infty} a_k \frac{(tz)^k}{k!} \mathrm{d}t \ ,$$

where the rightmost expression is exactly the Borel sum at z.

Regularity of $(\boldsymbol{B})$ and $(\boldsymbol{wB})$ imply that these methods provide analytic extensions to $A(z)$.

Generalized Borel Transform:

Bounding may be defined for other functions besides the exponential function. In general, a function $\Psi(t)$ is a comparison function if it has a series:

$$\Psi(t) = \sum_{n=0}^{\infty} \Psi_n t^n$$

with $\Psi_n > 0$ for all n, and

$$\lim_{n\to\infty} \frac{\Psi_{n+1}}{\Psi_n} = 0 \, .$$

Comparison functions are necessarily entire, which follows from the ratio test. If $\Psi(t)$ is such a comparison function, one then says that f is of Ψ-type if there exist constants M and τ such that:

$$\left| f\left(re^{i\theta}\right) \right| \leq M\Psi(\tau r)$$

as $r \to \infty$. If τ is the infimum of all such τ one says that f is of Ψ-type τ.

Nachbin's theorem states that a function $f(z)$ with the series:

$$f(z) = \sum_{n=0}^{\infty} f_n z^n$$

is of Ψ-type τ if and only if:

$$\limsup_{n\to\infty} \left| \frac{f_n}{\Psi_n} \right|^{1/n} = \tau \, .$$

The generalized Borel transform is given by:

$$F(w) = \sum_{n=0}^{\infty} \frac{f_n}{\Psi_n w^{n+1}} \, .$$

If f is of Ψ-type τ, then the exterior of the domain of convergence of $F(w)$, and all of its singular points, are contained within the disk $|w| \leq \tau$. Furthermore, one has:

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma} \Psi(zw) \, F(w) \, \mathrm{d}w \, ,$$

where the contour of integration γ encircles the disk $|w| \leq \tau$. This generalizes the usual Borel transform for exponential type, where $\Psi(t) = e^t$. The integral form for the generalized Borel transform follows as well. Let $\alpha(t)$ be a function whose first derivative is bounded on the interval $[0, \infty)$, so that:

$$\frac{1}{\Psi_n} = \int_0^{\infty} t^n \, \mathrm{d}\alpha(t) \, ,$$

where $\mathrm{d}\alpha(t) = \alpha'(t) \, \mathrm{d}t$. Then the integral form of the generalized Borel transform is:

$$F(w) = \frac{1}{w} \int_0^{\infty} f\left(\frac{t}{w}\right) \, \mathrm{d}\alpha(t) \, .$$

The ordinary Borel transform is regained by setting $\alpha(t) = e^{-t}$. Note that the integral form of the Borel transform is just the Laplace transform.

The generalized Borel transform can be used to sum divergent series that escape to the usual Borel summation or even to solve (asymptotically) integral equations of the form:

$$g(s) = s \int_0^\infty K(st) f(t)\, \mathrm{d}t \,,$$

where $f(t)$ may or may not be of exponential growth and the kernel $K(u)$ has a Mellin transform. The solution can be obtained as

$$f(x) = \sum_{n=0}^{\infty} \frac{a_n}{M(n+1)} x^n \,,$$

with

$$g(s) = \sum_{n=0}^{\infty} a_n s^{-n} \,,$$

and $M(n)$ is the Mellin transform of $K(u)$.

In some cases as an extra condition we require $\int_0^\infty K(t)\, t^n$ to be finite for $n = 0, 1, 2, 3, \ldots$, and different from 0.

<u>**Miscellaneous Methods**</u>

Holder Summation:

Given a series $\sum_{i=1}^{\infty} a_i$, define the partial sums:

$$H_n^0 = \sum_{i=1}^{n} a_i$$

$$H_n^{k+1} = \frac{1}{n} \sum_{i=1}^{n} H_i^k \,.$$

If the limit $\lim_{n\to\infty} H_n^k$ exists for some k, then this value is called the Holder sum of the series, denoted (H, k).

Ingham Summability:

The series $\sum a_n$ is called Ingham summable to s if:

$$\lim_{x\to\infty} \sum_{1\le n\le x} a_n \frac{n}{x} \left[\frac{x}{n}\right] = s \,.$$

If δ is any positive number then $(C, -\delta)$ summability implies Ingham summability, and Ingham summability implies (C, δ) summability.

Lambert Summability:

A series $\sum a_n$ is Lambert summable to A, denoted $\sum a_n = A(L)$, if:

$$\lim_{r\to 1^-} (1-r) \sum_{n=1}^{\infty} \frac{n\, a_n r^n}{1-r^n} = A \,.$$

If a series is convergent to A then it is Lambert summable to A. From the above criterion, we can also define the Lambert series as a series of the form:

$$S(q) = \sum_{n=1}^{\infty} a_n \frac{q^n}{1-q^n} \ .$$

Le Roy Summation:

The series $\sum a_n$ is called Le Roy summable to s if:

$$\lim_{\zeta \to 1^-} \sum_n \frac{\Gamma(1+\zeta n)}{\Gamma(1+n)} a_n = s \ .$$

Mittag-Leffler Summation:

The series $\sum a_n$ is called Mittag-Leffler summable to s if:

$$\lim_{\delta \to 0} \sum_n \frac{a_n}{\Gamma(1+\delta n)} = s \ .$$

Riemann Summability:

The series $\sum a_n$ is called (R, k) or Riemann summable to s if:

$$\lim_{h \to 0} \sum_n a_n \left(\frac{\sin(nh)}{nh} \right)^k = s \ .$$

The series $\sum a_n$ is called R_2 summable to s if:

$$\lim_{h \to 0} \frac{2}{\pi} \sum_n \frac{\sin^2(nh)}{n^2 h} (a_1 + \cdots + a_n) = s \ .$$

Riesz Means:

Given a series $\sum s_n$, the Riesz mean of the series is defined by:

$$s^\delta(\lambda) = \sum_{n \le \lambda} \left(1 - \frac{n}{\lambda}\right)^\delta s_n$$

Sometimes, a generalized Riesz mean is defined as:

$$R_n = \frac{1}{\lambda_n} \sum_{k=0}^{n} (\lambda_k - \lambda_{k-1})^\delta s_k$$

Here, the λ_n are a sequence with $\lambda_n \to \infty$ and with $\lim_{n \to \infty} \frac{\lambda_{n+1}}{\lambda_n} = 1$. Other than this, the λ_n are taken as arbitrary.

Stated another way;

If λ_n form an increasing sequence of real numbers and

$$A_\lambda(x) = a_0 + \cdots + a_n \quad \text{for} \quad \lambda_n < x \leq \lambda_{n+1}$$

then the Riesz (R, λ, κ) sum of the series $\sum a_n$ is defined to be

$$\lim_{\omega\to\infty} \frac{\kappa}{\omega^\kappa} \int_0^\omega A_\lambda(x)(\omega - x)^{\kappa-1}\, \mathrm{d}x \,.$$

Riesz means are often used to explore the summability of sequences; typical summability theorems discuss the case of $s_n = \sum_{k=0}^{n} a_k$ for some sequence $\{a_k\}$. Typically, a sequence is summable when the limit $\lim_{n\to\infty} R_n$ exists, or the limit $\lim_{\delta\to 1, \lambda\to\infty} s^\delta(\lambda)$ exists, although the precise summability theorems in question often impose additional conditions.

Vallee-Poussin Summability:

The series $\sum_{n=1}^{\infty} a_n$ is called Vallee-Poussin summable to s if:

$$\lim_{m\to\infty} \sum_{k=0}^{m} \frac{[\Gamma(m+1)]^2}{\Gamma(m-k+1)\,\Gamma(m+k+1)}\, a_k = \lim_{m\to\infty} \left[a_0 + \frac{m}{m+1} a_1 + \frac{m(m-1)}{(m+1)(m+2)} a_2 + \cdots\right] = s \,.$$

1.9.4 Approximation by Series

Darboux's Formula:

Darboux's formula provides a method to sum infinite series using integration or to evaluate integrals using infinite series, it is given by:

$$\sum_{m=0}^{n} (-1)^m (z-a)^m \left[\varphi^{(n-m)}(1)\, f^{(m)}(z) - \varphi^{(n-m)}(0)\, f^{(m)}(a)\right] = (-1)^n (z-a)^{n+1} \int_0^1 \varphi(t)\, f^{(n+1)}(a+t(z-a))\, \mathrm{d}t, \tag{11}$$

where $\varphi(t)$ is a polynomial of degree n and f an analytic function. Important special cases of the Darboux formula include the Taylor series, for $\varphi(t) = (t-1)^n$ and $\lim_{n\to\infty}$, and the Euler-Maclaurin formula, for $\varphi(t) = B_n(t)$ (Bernoulli polynomials).

Euler-Maclaurin Formula:

The Euler-Maclaurin formula provides a method to approximate definite integrals with finite sums, it is given by:

$$\sum_{i=m+1}^{n} f(i) - \int_m^n f(x)\, \mathrm{d}x = \sum_{k=1}^{p} \frac{B_k}{k!} \left(f^{(k-1)}(n) - f^{(k-1)}(m)\right) + R_p \,, \tag{12}$$

where $p \in \mathbb{Z}^+$, f is p-times continuously differentiable on the interval $[m, n]$, B_k is the k-th Bernoulli number, and R_p is an error term to account for the fact that the integral is not perfectly approximated by the sum. The error term can be given by the expression:

$$R_p = (-1)^{p+1} \int_m^n f^{(p)}(x)\, \frac{B_p(x - \lfloor x \rfloor)}{p!}\, \mathrm{d}x \,, \tag{13}$$

which has an upper bound given by the inequality:

$$\left|R_p\right| \leq \frac{2\,\zeta(p)}{(2\pi)^p} \int_m^n \left|f^{(p)}(x)\right| \mathrm{d}x\,, \tag{14}$$

where $\zeta(s)$ is the Riemann zeta function.

Ramanujan Summation:

If we take the limit as p goes to infinity of the Euler-Maclaurin formula for $m = 0$, then we arrive at the expression:

$$\sum_{k=1}^{x} f(k) = C + \int_0^x f(t)\,dt + \frac{1}{2}f(x) + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} f^{(2k-1)}(x)\,,$$

where C is a constant specific to the series and its analytic continuation. Comparing both formulae and assuming that R tends to 0 as x tends to infinity, we see that, in a general case, for functions $f(x)$ with no divergence at $x = 0$:

$$C(a) = \int_0^a f(t)\,\mathrm{d}t - \frac{1}{2}f(0) - \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} f^{(2k-1)}(0)\,,$$

where Ramanujan assumed $a = 0$. By taking $a = \infty$ we normally recover the usual summation for convergent series. For functions $f(x)$ with no divergence at $x = 1$, we obtain:

$$C(a) = \int_1^a f(t)\,\mathrm{d}t + \frac{1}{2}f(1) - \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} f^{(2k-1)}(1)\,,$$

$C(0)$ was then proposed to use as the sum of the divergent sequence. It is like a bridge between summation and integration. If we have a functional given by an integral we can also use this method, consider for example:

$$\int_a^\infty x^{m-s}\,\mathrm{d}x = \frac{m-s}{2}\int_a^\infty x^{m-1-s}\,\mathrm{d}x + \zeta(s-m) - \sum_{i=1}^{a} i^{m-s} + a^{m-s}$$

$$- \sum_{r=1}^{\infty} \frac{B_{2r}\theta(m-s+1)}{(2r)!\Gamma(m-2r+2-s)}(m-2r+1-s)\int_a^\infty x^{m-2r-s}\,\mathrm{d}x\,,$$

which is the natural extension to integrals of the Zeta regularization algorithm.

This recurrence equation is finite, since for $m - 2r < -1$,

$$\int_a^\infty \mathrm{d}x\, x^{m-2r} = -\frac{a^{m-2r+1}}{m-2r+1}\,.$$

Note that this involves

$$I(n,\Lambda) = \int_0^\Lambda \mathrm{d}x\, x^n\,.$$

With $\Lambda \to \infty$, the application of this Ramanujan resummation lends to finite results in the renormalization of quantum field theories.

Euler-Boole Summation:

Euler-Boole summation provides a method for summing alternating series based on the Euler polynomials and periodic Euler functions. The periodic Euler functions are given by:

$$\tilde{E}_n(x+1) = -\tilde{E}_n(x) \qquad \text{and} \qquad \tilde{E}_n(x) = E_n(x)\,, \quad 0 < x < 1\,.$$

The Euler-Boole summation formula is then given by:

$$\sum_{j=a}^{n-1}(-1)^j\, f(j+h) = \frac{1}{2}\sum_{k=0}^{m-1}\frac{E_k(h)}{k!}\left((-1)^{n-1}\, f^{(k)}(n) + (-1)^a\, f^{(k)}(a)\right) + \frac{1}{2(m-1)!}\int_a^n f^{(m)}(x)\,\tilde{E}_{m-1}(h-x)\,\mathrm{d}x\,, \tag{15}$$

where $a, m, n \in \mathbb{N}$, $a < n, h \in [0,1]$ and $f^{(k)}$ is the k-th derivative.

Many approximations to functions are often given by the family of power series defined below. Taylor series expansions in particular, are very useful in physics.

1.9.5 Generalized Power Series

A power series of one variable is an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c)^1 + a_2(x-c)^2 + \cdots\,, \tag{16}$$

where the coefficient a_n is the n-th term, which is also independent of x, and c is a constant. Power series are related to Taylor series for infinitely differentiable functions, in fact, Borel's theorem implies that every power series is the Taylor series of some smooth function.

The Taylor series of a function f can be expressed for a specific choice of a_n, namely:

$$\sum_{n=0}^{\infty}\frac{f^{(n)}(c)}{n!}(x-c)^n\,, \tag{17}$$

where $f^{(0)}(c) = f(c)$.

A power series will converge for some values of $x-c$ but diverge for others. the values of $x-c$ for which the power series converges is called the radius of convergence and is denoted by the positive real number r. The power series converges absolutely for $|x-c| < r$ and diverges for $|x-c| > r$. r can be expressed as:

$$r = \liminf_{n\to\infty}\ |a_n|^{-\frac{1}{n}}\,, \tag{18}$$

or equivalently as:

$$r^{-1} = \limsup_{n\to\infty}\ |a_n|^{\frac{1}{n}}\,. \tag{19}$$

If the limit exists, we can also state the relation:

$$r^{-1} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|\,. \tag{20}$$

If $|x-c| = r$, we cannot make any general statements about convergence or divergence and must come to a conclusion using some other method or result (i.e. Abel's theorem for power series)

Some general properties of power series include:

Addition and Subtraction: $$\sum_{n=0}^{\infty} a_n(x-c)^n \pm \sum_{n=0}^{\infty} b_n(x-c)^n = \sum_{n=0}^{\infty}(a_n \pm b_n)\,(x-c)^n$$

(Note that this property may not hold for the respective radii of convergence)

Multiplication: $$\left(\sum_{n=0}^{\infty} a_n\,(x-c)^n\right)\left(\sum_{n=0}^{\infty} b_n\,(x-c)^n\right) = \sum_{n=0}^{\infty}\left(\sum_{i=0}^{n} a_i\,b_{n-i}\right)(x-c)^n$$

Division: $$\frac{\sum_{n=0}^{\infty} a_n\,(x-c)^n}{\sum_{n=0}^{\infty} b_n\,(x-c)^n} = \sum_{n=0}^{\infty} d_n\,(x-c)^n$$

(The coefficients d_n can be solved for recursively by comparing coefficients)

Differentiation: $$\frac{\mathrm{d}}{\mathrm{d}x}\sum_{n=1}^{\infty} a_n\,(x-c)^n = \sum_{n=1}^{\infty} n\,a_n\,(x-c)^{n-1} = \sum_{n=0}^{\infty} a_{n+1}\,(n+1)\,(x-c)^n$$

Integration: $$\int \sum_{n=1}^{\infty} a_n\,(x-c)^n\,\mathrm{d}x = \sum_{n=0}^{\infty}\frac{a_n\,(x-c)^{n+1}}{n+1} + k = \sum_{n=1}^{\infty}\frac{a_{n-1}\,(x-c)^n}{n} + k$$

(The resulting series after integration and differentiation still have the same radius of convergence)

A power series of more than one variable can be given in the form:

$$\sum_{j_1,\ldots,j_n=0}^{\infty} a_{j_1,\ldots,j_n}\prod_{k=1}^{\infty}(x_k-c_k)^{j_k}\,, \tag{21}$$

where $j_1,\ \ldots\ ,j_n$ are a vector of natural numbers.

The preceding discussion was valid for indices $n \in \mathbb{N}$. Generalizations to negative and fractional n, as well as complex x exist. We will outline some of them below.

Laurent Series:

The Laurent series for a complex function $f(z)$ about a point c is given by:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-c)^n\,, \tag{22}$$

where the constants a_n are defined as:

$$a_n = \frac{1}{2\pi i}\oint_{\gamma}\frac{f(z)}{(z-c)^{n+1}}\,dz\,. \tag{23}$$

The integration path γ is counter-clockwise around a Jordan curve enclosing c and lying in an annulus A in which $f(z)$ is holomorphic. The Laurent series extends the power series to include negative indices and complex argument z.

Puiseux Series:

The Puiseux series, with indeterminate T, for a function f is given by:

$$f = \sum_{k=k_0}^{\infty} c_k T^{k/n}\,, \tag{24}$$

where $n \in \mathbb{Z}^+$, k_0 is an arbitrary integer, and c_k are coefficients defined over a field K. It is understood that the exponent $k/n \in \mathbb{Q}$ is the smallest rational such that the coefficient c_k of the term with exponent

k/n is non-zero. The Puiseux series also admits negative indices as long as they are bounded below (in this case the lower bound is k_0). The Puiseux series extends the power series to include negative and fractional indices, and generalizes the argument to an arbitrary field.

There exists a further generalization of the Puiseux series called the Hahn series, whose exponents and indices are defined over an ordered group, and does not require a lower bound for the indices.

1.9.6 General Dirichlet Series

A general Dirichlet series is the infinite series of the form:

$$\sum_{n=1}^{\infty} a_n \, e^{-\lambda_n s} \,, \tag{25}$$

where $a_n, s = \sigma + ti \in \mathbb{C}$ and $\{\lambda_n\}$ is a strictly increasing sequence of nonnegative real numbers that tends to infinity.

For suitable substitutions of λ_n we can obtain some other important series. If $\lambda_n = \ln(n)$, then the ordinary Dirichlet series is recovered:

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} \,,$$

(for $a_n = 1$ this is the Riemann zeta function) and if $\lambda_n = n$, then a power series is recovered:

$$\sum_{n=1}^{\infty} a_n \, (e^{-s})^n \,.$$

The general Dirichlet series has an abscissa of convergence defined by:

$$\sigma_c = \inf \left\{ \sigma \in \mathbb{R} : \sum_{n=1}^{\infty} a_n \, e^{-\lambda_n s} \text{ converges for every } s \text{ for which } \operatorname{Re}(s) > \sigma \right\} .$$

Thus, $\sigma = \sigma_c$ is called the line of convergence and $\mathbb{C}_{\sigma_c} = \{s \in \mathbb{C} : \operatorname{Re}(s) > \sigma_c\}$ is called the half-plane of convergence.

An important theorem states that if a Dirichlet series is convergent at $s_0 = \sigma_0 + t_0 i$, then it is uniformly convergent in the domain $|\arg(s - s_0)| \leq \theta < \frac{\pi}{2}$, and convergent for any $s = \sigma + ti$ where $\sigma > \sigma_0$.

Some common series are:

$$\sum_{k=1}^{n} c = nc \qquad \sum_{k=1}^{n} k = \frac{n\,(n+1)}{2} \qquad \sum_{k=1}^{n} k^p = \frac{2n+p+1}{2(p+1)} \, n^p + \sum_{k=2}^{p} \binom{p+1}{k} B_k \, n^{p-k+1}$$

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n \qquad \sum_{k=1}^{n} k^2 = \frac{n\,(n+1)\,(2n+1)}{6} \qquad = \sum_{k=0}^{p} \frac{(-1)^{p-k}}{k+1} \binom{p}{k} B_{p-k} \, n^{k+1}$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0 \qquad \sum_{k=1}^{n} k^3 = \left[\frac{n\,(n+1)}{2}\right]^2 \qquad = \frac{1}{p+1} \left(B_{p+1}(n+1) - (-1)^{p+1} \, B_{p+1} \right)$$

$$\sum_{k=0}^{n-1} a^k = \frac{1-a^n}{1-a}$$

$$\sum_{k=0}^{n-1} (b+kd)a^k = b\left(\frac{1-a^n}{1-a}\right) + d\left(\frac{a-na^n+(n-1)a^{n+1}}{(1-a)^2}\right)$$

$$\sum_{k=0}^{n-1} ((k+1)^b - k^b) = n^b$$

$$\sum_{k=1}^{n} \log(k) = \log(n!)$$

$$\sum_{k=1}^{\infty} \frac{4(-1)^{k+1}}{2k-1} = \pi$$

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \ln(2)$$

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e$$

1.9.7 Rational and Irrational Functions

$$\sum_{k=0}^{n} z^k = \frac{1-z^{n+1}}{1-z}$$

$$\sum_{k=1}^{n} k^m z^k = \left(z\frac{\mathrm{d}}{\mathrm{d}z}\right)^m \frac{1-z^{n+1}}{1-z}$$

$$\sum_{k=0}^{\infty} \binom{\alpha}{k} z^k = (1+z)^\alpha\,, \quad |z|<1$$

$$\sum_{k=0}^{\infty} \binom{\alpha+k-1}{k} z^k = \frac{1}{(1-z)^\alpha}\,, \quad |z|<1$$

1.9.8 Trigonometric and Inverse Trigonometric Functions

A trigonometric series is a series of the form:

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} \left(A_n \cos(nz) + B_n \sin(nz)\right)\ .$$

If the coefficients A_n and B_n are given by:

$$A_n = \frac{1}{\pi}\int_0^{2\pi} f(z)\cos(nz)\,\mathrm{d}z \quad n = 0, 1, 2, 3, \ldots\ ,$$

$$B_n = \frac{1}{\pi}\int_0^{2\pi} f(z)\sin(nz)\,\mathrm{d}z \quad n = 1, 2, 3, \ldots\ ,$$

where f is an integrable function, then the series is called a Fourier series (see section 23 for further details).

Some series representations of trigonometric functions are given below:

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$\cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}$$

$$\csc(z) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2(2^{2n-1}-1)B_{2n} z^{2n-1}}{(2n)!}\,, \quad 0<|z|<\pi$$

$$\sec(z) = \sum_{n=0}^{\infty} \frac{(-1)^n E_{2n} z^{2n}}{(2n)!}\,, \quad |z|<\frac{\pi}{2}$$

$$\tan(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} 2^{2n} (2^{2n}-1) B_{2n} z^{2n-1}}{(2n)!}, \quad |z| < \frac{\pi}{2}$$

$$\cot(z) = \sum_{n=0}^{\infty} \frac{(-1)^{n} 2^{2n} B_{2n} z^{2n-1}}{(2n)!}, \quad 0 < |z| < \pi$$

$$\arcsin(z) = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{z^{2n+1}}{(2n+1)}, \quad |z| \le 1$$

$$\operatorname{arccsc}(z) = \sum_{n=0}^{\infty} \left(\frac{\Gamma(n+1/2)}{\sqrt{\pi}}\right) \frac{z^{-(2n+1)}}{(2n+1)\, n!}, \quad |z| > 1$$

$$\arccos(z) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \left(\frac{\Gamma(n+1/2)}{\sqrt{\pi}}\right) \frac{z^{2n+1}}{(2n+1)\, n!}, \quad |z| < 1$$

$$\operatorname{arcsec}(z) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \left(\frac{\Gamma(n+1/2)}{\sqrt{\pi}}\right) \frac{z^{-(2n+1)}}{(2n+1)\, n!}, \quad |z| < 1$$

$$\arctan(z) = \sum_{n=0}^{\infty} \frac{(-1)^{n} z^{(2n+1)}}{(2n+1)}, \quad |z| \le 1$$

$$\operatorname{arccot}(z) = \sum_{n=0}^{\infty} \frac{(-1)^{n} z^{-(2n+1)}}{(2n+1)}, \quad |z| > 1$$

where E_n and B_n are the n-th Euler and Bernoulli numbers respectively.

1.9.9 Hyperbolic and Inverse Hyperbolic Functions

$$\sinh(z) = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!}$$

$$\operatorname{csch}(z) = z^{-1} + \sum_{n=1}^{\infty} \frac{2(1-2^{2n-1}) B_{2n} z^{2n-1}}{(2n)!}, \quad 0 < |z| < \pi$$

$$\cosh(z) = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}$$

$$\operatorname{sech}(z) = \sum_{n=0}^{\infty} \frac{E_{2n} z^{2n}}{(2n)!}, \quad |z| < \frac{\pi}{2}$$

$$\tanh(z) = \sum_{n=1}^{\infty} \frac{2^{2n}(2^{2n}-1) B_{2n} z^{2n-1}}{(2n)!}, \quad |z| < \frac{\pi}{2}$$

$$\coth(z) = z^{-1} + \sum_{n=1}^{\infty} \frac{2^{2n} B_{2n} z^{2n-1}}{(2n)!}, \quad 0 < |z| < \pi$$

$$\operatorname{arsinh}(z) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n (2n)!}{2^{2n}(n!)^2}\right) \frac{z^{2n+1}}{(2n+1)}, \quad |z| < 1$$

$$\operatorname{arsech}(z) = \ln\left(\frac{2}{z}\right) - \sum_{n=1}^{\infty} \left(\frac{(2n)!}{2^{2n}(n!)^2}\right) \frac{z^{2n}}{(2n)}, \quad 0 < z \le 1$$

$$\operatorname{arcsch}(z) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n (2n)!}{2^{2n}(n!)^2}\right) \frac{z^{-(2n+1)}}{(2n+1)}, \quad |z| > 1$$

$$\operatorname{artanh}(z) = \sum_{n=0}^{\infty} \frac{z^{(2n+1)}}{(2n+1)}, \quad |z| < 1$$

$$\operatorname{arcosh}(z) = \ln(2z) - \sum_{n=1}^{\infty} \left(\frac{(2n)!}{2^{2n}(n!)^2}\right) \frac{z^{-2n}}{(2n)}, \quad |z| > 1$$

$$\operatorname{arcoth}(z) = \sum_{n=0}^{\infty} \frac{z^{-(2n+1)}}{(2n+1)}, \quad |z| > 1$$

where E_n and B_n are the n-th Euler and Bernoulli numbers respectively.

1.9.10 Exponential and Logarithmic Functions

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$\ln(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(z-1)^n}{n}, \quad |z-1| < 1$$

$$\mathrm{Li}_\nu(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^\nu}\,, \quad |z| < 1 \quad \text{(Polylogarithm Function)}$$

1.9.11 Special Functions

Series Representation of Special Functions:

Generalized Hypergeometric Function:

$$_pF_q(a_1,\ldots,a_p\,;b_1,\ldots,b_q\,;z) = \sum_{k=0}^{\infty} \frac{\prod_{j=1}^{p} \frac{\Gamma(k+a_j)}{\Gamma(a_j)}}{\prod_{j=1}^{q} \frac{\Gamma(k+b_j)}{\Gamma(b_j)}} \frac{z^k}{k!}\,, \quad b_1,\ldots,b_p \notin \mathbb{Z}^- \text{ except when } -b_k < a_j$$

The generalized hypergeometric function converges for any finite z if $p < q+1$ and diverges for any finite non-zero z if $p > q+1$. If $p = q+1$, then it will converge for all $|z| < 1$ and diverge for all $|z| > 1$.

Dirac Delta Function:

$$\delta(z) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} e^{ikz}\,, \quad -\pi < z < \pi$$

Heaviside Step Function:

$$\theta(z) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{\sin(z(2k+1))}{2k+1}\,, \quad -\pi < z < \pi$$

Riemann Zeta Function:

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}\,, \quad \mathrm{Re}(s) > 1$$

Polygamma Function:

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{m+1}}\,, \quad m > 0$$

Error Function:

$$\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n\, z^{2n+1}}{n!\,(2n+1)}$$

Logarithmic Integral:

$$\mathrm{li}(z) = \sum_{k=1}^{\infty} \frac{(\ln(z))^k}{k\,k!} + \gamma + \ln(\ln(z))$$

Beta Function:

$$B(a,b) = \sum_{k=0}^{\infty} \frac{\Gamma(1-b+k)}{\Gamma(1-b)\,(a+k)\,k!}\,, \quad \mathrm{Re}(a,b) > 0$$

Exponential Integral:

$$\mathrm{Ei}(z) = \sum_{k=1}^{\infty} \frac{z^k}{k\,k!} + \gamma + \ln(z)$$

Incomplete Beta Function:

$$B_z(a,b) = \sum_{k=0}^{\infty} \frac{\Gamma(1-b+k)\, z^{k+a}}{\Gamma(1-b)\,(a+k)\,k!}\,, \quad -a \notin \mathbb{N}$$

Poly Exponential Integral:

$$E_\nu(z) = z^{\nu-1}\,\Gamma(1-\nu) - \frac{1}{(1-\nu)^2} \sum_{k=0}^{\infty} \frac{(-z)^k}{k!\,(k+1-\nu)}$$

Sine Integral:

$$\mathrm{Si}(z) = \sum_{k=0}^{\infty} \frac{(-1)^k\, z^{2k+1}}{\Gamma(2k+3)}$$

Hyperbolic Sine Integral:

$$\mathrm{Shi}(z) = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{\Gamma(2k+3)}$$

Cosine Integral:

$$\mathrm{Ci}(z) = \ln(z) + \gamma + \sum_{k=1}^{\infty} \frac{(-1)^k z^{2k}}{\Gamma(2k+2)}$$

Hyperbolic Cosine Integral:

$$\mathrm{Chi}(z) = \ln(z) + \gamma + \sum_{k=1}^{\infty} \frac{z^{2k}}{\Gamma(2k+2)}$$

Complete Elliptic Integral of the 1st Kind:

$$K(k^2) = \sum_{n=0}^{\infty} \frac{k^{2n}(\Gamma(n+1/2))^2}{2(n!)^2}, \quad k^2 < 1$$

Complete Elliptic Integral of the 2nd Kind:

$$E(k^2) = -\sum_{n=0}^{\infty} \frac{k^{2n}\Gamma(n+1/2)\,\Gamma(n-1/2)}{4(n!)^2}, \quad k^2 < 1$$

Incomplete Elliptic Integral of the 1st Kind:

$$F(z|k^2) = \frac{2zK(k^2)}{\pi} + \sum_{n=1}^{\infty}\sum_{m=0}^{\infty} \frac{(-1)^n k^{2(n+m)} \sin(2nz)\ [\Gamma(m+n+1/2)]^2\ \Gamma(2n+1)}{4^n\, m!\,(n+1)!\,\sqrt{\pi}\,\Gamma(m+1+2n)\,\Gamma(n+1/2)}$$

Incomplete Elliptic Integral of the 2nd Kind:

$$E(z|k^2) = \frac{2zE(k^2)}{\pi} - \sum_{n=1}^{\infty}\sum_{m=0}^{\infty} \frac{(-1)^n k^{2(n+m)} \sin(2nz)\,[\Gamma(m+n-1/2)]^2 \left(m+n-\frac{1}{2}\right)\Gamma(2n+1)}{4^n(n+1)!\,m!\,2\sqrt{\pi}\,\Gamma(m+1+2n)\,\Gamma(n-1/2)\left(n-\frac{1}{2}\right)}$$

Bessel Function of the 1st Kind:

$$J_\nu(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}$$

Bessel Function of the 2nd Kind:

$$Y_\nu(z) = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{k!\,2^{2k}} \left(\frac{2^{-\nu}z^{\nu}\cot(\pi\nu)}{\Gamma(k+\nu+1)} - \frac{2^{\nu}z^{-\nu}\csc(\pi\nu)}{\Gamma(k-\nu+1)}\right), \quad \nu \notin \mathbb{Z}$$

Modified Bessel Function of the 1st Kind:

$$I_\nu(z) = \sum_{k=0}^{\infty} \frac{1}{k!\,\Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}$$

Modified Bessel Function of the 2nd Kind:

$$K_\nu(z) = \sum_{k=0}^{\infty} \frac{\csc(\pi\nu)}{k!\,2^{2k+1}} \left(\frac{\pi 2^{\nu} z^{2k-\nu}}{\Gamma(k-\nu+1)} - \frac{\pi 2^{-\nu} z^{2k+\nu}}{\Gamma(k+\nu+1)}\right), \quad \nu \notin \mathbb{Z}$$

Spherical Bessel Function of the 1st Kind:

$$j_\nu(z) = \sqrt{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k 2^{-(\nu+1+2k)} z^{2k+\nu}}{k!\,\Gamma(k+\nu+3/2)}$$

Spherical Bessel Function of the 2nd Kind:

$$y_\nu(z) = \sqrt{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{k!\,2^{2k}} \left(\frac{-2^{-\nu}z^{\nu}\tan(\pi\nu)}{2\,\Gamma(k+\nu+3/2)} - \frac{2^{\nu}z^{-\nu}\sec(\pi\nu)}{z\Gamma(k-\nu+1/2)}\right)$$

Hermite Polynomials:

$$H_n(z) = n!\sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^k (2z)^{n-2k}}{k!\,(n-2k)!}, \quad n \in \mathbb{N}$$

Generalized Hermite Polynomials:

$$\gamma_n^m(z) = \sum_{k=0}^{\lfloor \frac{n}{m} \rfloor} \frac{(-1)^k (mz)^{n-mk}}{k!\,(n-mk)!}, \quad n, m \in \mathbb{N}$$

Laguerre Polynomials:

$$L_n(z) = \sum_{k=0}^{n} \frac{\Gamma(k-n)\,z^k}{(k!)^2\,\Gamma(-n)}, \quad n \in \mathbb{N}$$

Associated Laguerre Polynomials:

$$L_n^\lambda(z) = \frac{\Gamma(\lambda+n+1)}{n!\,\Gamma(-n)} \sum_{k=0}^{n} \frac{\Gamma(k-n)\,z^k}{k!\,\Gamma(\lambda+k+1)}, \quad n \in \mathbb{N}$$

Legendre Polynomials of the 1st Kind:

$$P_n(z) = \left(\frac{z-1}{2}\right)^n \sum_{k=0}^{n} \binom{n}{k}^2 \left(\frac{z+1}{z-1}\right)^k, \quad n \in \mathbb{N}$$

Associated Legendre Polynomials of the 1st Kind:

$$P_n^\mu(z) = \left(\frac{z+1}{1-z}\right)^{\mu/2} \sum_{k=0}^{n} \frac{\Gamma(k-n)\,\Gamma(k+n+1)}{k!\,\Gamma(k-\mu+1)\,\Gamma(-n)\,\Gamma(n+1)} \left(\frac{1-z}{2}\right)^k,$$

Legendre Polynomials of the 2nd Kind:

$$Q_n(z) = \frac{1}{2}\ln\left(\frac{1+z}{1-z}\right)\left(\frac{z-1}{2}\right)^n \sum_{k=0}^{n}\binom{n}{k}^2\left(\frac{z+1}{z-1}\right)^k$$

$$+\sum_{k=0}^{\lfloor\frac{1}{2}(n-1)\rfloor}\frac{1+4k-2n}{(k-n)(1+2k)}\sum_{m=0}^{n-1-2k}\left(\frac{z-1}{2}\right)^{n-1-2k}\binom{n-1-2k}{m}^2\left(\frac{z+1}{z-1}\right)^m, \quad n\in\mathbb{N}$$

Associated Legendre Polynomials of the 2nd Kind:

$$Q_n^\mu(z) = \frac{\pi}{2}\left(\cot(\pi\mu)+(-1)^{\mu+1}\csc(\pi\mu)\right)\left(\frac{z+1}{1-z}\right)^{\mu/2}\sum_{k=0}^{n}\frac{\Gamma(k-n)\,\Gamma(k+n+1)}{k!\,\Gamma(k-\mu+1)\,\Gamma(-n)\,\Gamma(n+1)}\left(\frac{1-z}{2}\right)^k, \quad \begin{matrix} n\in\mathbb{N}, \\ \mu\notin\mathbb{Z}\end{matrix}$$

Spherical Harmonics:

$$Y_n^m(\theta,\phi) = \sqrt{\frac{(2n+1)\,(n-m)!}{4\pi(n+m)!}}\,e^{im\phi}\cot^m\left(\frac{\theta}{2}\right)\sum_{k=0}^{n}\frac{\Gamma(k-n)\,\Gamma(k+n+1)}{k!\,\Gamma(k-m+1)\,\Gamma(-n)\,\Gamma(n+1)}\sin^{2k}\left(\frac{\theta}{2}\right), \quad \begin{matrix} n\in\mathbb{N}, \\ m\in\mathbb{Z}, \\ |m|\le n\end{matrix}$$

Chebyshev Polynomials of the 1st Kind:

$$T_n(z) = \sum_{k=0}^{\lfloor\frac{n}{2}\rfloor}\binom{n}{2k}z^{n-2k}(z^2-1)^k, \quad n\in\mathbb{N}$$

Chebyshev Polynomials of the 2nd Kind:

$$U_n(z) = \sum_{k=0}^{\lfloor\frac{n}{2}\rfloor}\frac{(-1)^k\,(n-k)!\,(2z)^{n-2k}}{k!\,(n-2k)!}, \quad n\in\mathbb{N}$$

Legendre Functions of the 1st Kind:

$$P_\nu(z) = \sum_{k=0}^{\infty}\frac{\Gamma(k-\nu)\,\Gamma(k+\nu+1)\,(1-z)^k}{\Gamma(k+1)\,\Gamma(-\nu)\,\Gamma(\nu+1)\,k!\,2^k}$$

Associated Legendre Functions of the 1st Kind:

$$P_\lambda^\mu(z) = \left[\frac{1+z}{1-z}\right]^{\mu/2}\sum_{k=0}^{\infty}\frac{\Gamma(k-\lambda)\,\Gamma(k+\lambda+1)\,(1-z)^k}{\Gamma(\lambda+1)\,\Gamma(-\lambda)\,\Gamma(k+1-\mu)\,k!\,2^k}$$

Legendre Functions of the 2nd Kind:

$$Q_\nu(z) = \frac{\pi}{2}\sum_{k=0}^{\infty}\frac{z^{2k}}{k!}\left(\frac{z\cos(\frac{\pi\nu}{2})\,\Gamma(k+\frac{1-\nu}{2})\,\Gamma(k+\frac{\nu}{2}+1)}{\Gamma(k+\frac{1}{2})\,\Gamma(\frac{1-\nu}{2})\,\Gamma(\frac{\nu+1}{2})\,(k+\frac{1}{2})} - \frac{\sin(\frac{\pi\nu}{2})\,\Gamma(k+\frac{1+\nu}{2})\,\Gamma(k-\frac{\nu}{2})}{\Gamma(k+\frac{1}{2})\,\Gamma(\frac{-\nu}{2})\,\Gamma(\frac{\nu}{2}+1)}\right)$$

Associated Legendre Functions of the 2nd Kind:

$$Q_\nu^\mu(z) = \frac{\pi}{2}\left(\cot(\pi\mu)+(-1)^{\mu+1}\csc(\pi\mu)\right)\left(\frac{z+1}{1-z}\right)^{\mu/2}\sum_{k=0}^{\nu}\frac{\Gamma(k-\nu)\,\Gamma(k+\nu+1)}{k!\,\Gamma(k-\mu+1)\,\Gamma(-\nu)\,\Gamma(\nu+1)}\left(\frac{1-z}{2}\right)^k, \quad \mu\notin\mathbb{Z}$$

Spherical Harmonic Functions:

$$Y_\lambda^\mu(\theta,\phi) = \sqrt{\frac{(2\lambda+1)\,\Gamma(\lambda-\mu+1)}{4\pi\Gamma(\lambda+\mu+1)}}\,e^{i\mu\phi}\cot^\mu\left(\frac{\theta}{2}\right)\sum_{k=0}^{\infty}\frac{\Gamma(k-\lambda)\,\Gamma(k+\lambda+1)}{\Gamma(\lambda+1)\,\Gamma(-\lambda)\,\Gamma(k+1-\mu)\,k!}\sin^{2k}\left(\frac{\theta}{2}\right), \quad \begin{matrix}\lambda\notin\mathbb{N}, \\ \mu\notin\mathbb{Z}\end{matrix}$$

Chebyshev Functions of the 1st Kind:

$$T_\nu(z) = \sum_{k=0}^{\infty} \frac{\sqrt{\pi}\,\Gamma(k-\nu)\,\Gamma(k+\nu)(1-z)^k}{\Gamma(-\nu)\,\Gamma(\nu)\,\Gamma(k+1/2)\,k!\,2^k}$$

Chebyshev Functions of the 2nd Kind:

$$U_\nu(z) = (\nu+1)\sum_{k=0}^{\infty} \frac{\sqrt{\pi}\,\Gamma(k-\nu)\,\Gamma(k+\nu+2)(1-z)^k}{\Gamma(-\nu)\,\Gamma(\nu+2)\,\Gamma(k+3/2)\,k!\,2^{k+1}}$$

Hermite Functions:

$$H_\nu(z) = \frac{2^\nu \pi}{\Gamma(\frac{1-\nu}{2})\,\Gamma(-\frac{\nu}{2})} \sum_{k=0}^{\infty} \frac{z^{2k}}{\Gamma(k+1/2)\,k!} \left(\Gamma(k-\nu/2) - \frac{z\,\Gamma(k+\frac{1-\nu}{2})}{k+\frac{1}{2}} \right)$$

Laguerre Functions:

$$L_\nu(z) = \sum_{k=0}^{\infty} \frac{\Gamma(k-\nu)\,z^k}{\Gamma(k+1)\,\Gamma(-\nu)\,k!}$$

Associated Laguerre Functions:

$$L_\nu^\lambda(z) = \frac{\Gamma(\nu+\lambda+1)}{\Gamma(\nu+1)} \sum_{k=0}^{\infty} \frac{\Gamma(k-\nu)\,\Gamma(\lambda+1)\,z^k}{\Gamma(\lambda+k+1)\,\Gamma(-\nu)\,k!}$$

Series with Special Functions:

1.10 Properties of Products

The product for a sequence of complex numbers $a_1, a_2, \ldots$ is given by:

$$\prod_{k=1}^{n} a_k = a_1\, a_2\, a_3 \ldots \tag{26}$$

and in the limit $n \to \infty$, the product converges to a non-zero complex number if the series $\sum_{k=1}^{\infty} \log(a_n)$ converges also (and assuming we use the branch cut corresponding to $\ln(1) = 0$).

We can replace the starting index k with $-\infty$ and define:

$$\prod_{k=-\infty}^{\infty} x_k = \left(\lim_{m\to-\infty} \prod_{k=m}^{0} x_k \right) \cdot \left(\lim_{n\to\infty} \prod_{k=1}^{n} x_k \right), \tag{27}$$

provided both limits exist.

An important result from complex analysis states that every entire function can be factored into an infinite product of entire functions, each with at most a single root. For a function f with a root of order m at the origin and other complex roots $u_1, u_2, u_3, \ldots$:

$$f(z) = z^m e^{\phi(z)} \prod_{n=1}^{\infty} \left(1 - \frac{z}{u_n}\right) \exp\left(\frac{z}{u_n} + \frac{1}{2}\left(\frac{z}{u_n}\right)^2 + \cdots + \frac{1}{\lambda_n}\left(\frac{z}{u_n}\right)^{\lambda_n} \right), \tag{28}$$

where λ_n are non-negative integers that can be chosen to make the product converge and $\phi(z)$ is some entire function (term in front of product has no roots in complex plane).

Some useful identities include:

$$\ln\left(\prod_{i=1}^{\infty} f_i\right) = \sum_{i=1}^{\infty} \ln(f_i) \quad \text{and} \quad \prod_{i=1}^{\infty} f_i = \exp\left(\sum_{i=1}^{\infty} \ln(f_i)\right).$$

Some common products are:

$$\frac{(k!)^2}{(2k)!} = \prod_{n=k+1}^{\infty}\left(1-\frac{k^2}{n^2}\right) \qquad \frac{\Gamma(c)\,\Gamma(d)}{\Gamma(a)\,\Gamma(b)} = \prod_{n=0}^{\infty}\frac{(n+a)\,(n+b)}{(n+c)\,(n+d)}$$

$$\prod_{\substack{n=1\\ n\neq m}}^{\infty}\frac{n^r-m^r}{n^r+m^r} = (-1)^{m+1}\,\frac{2\,m\,m!}{r}\prod_{j=1}^{2r-1}\left[\Gamma(-me^{i\pi j/k})\right]^{(-1)^{j+1}}, \quad r>1$$

$$\prod_{k=1}^{\infty}\frac{\sum_{i=1}^{p}\frac{a_i}{k^i}}{\sum_{i=0}^{q}\frac{b_i}{k^i}} = \frac{b_q}{a_p}\,\frac{\prod_{i=0}^{q}\Gamma(-s_i)}{\prod_{i=0}^{p}\Gamma(-r_i)}$$

(with requirement that $a_0 = b_0 = 1$ and $a_1 = b_1$, where r_i is the ith root of $\sum_{j=0}^{p} a_j/k^j$ and s_i is the ith root of $\sum_{j=0}^{q} b_j/k^j$)

1.10.1 Rational and Irrational Functions

$$\prod_{n=0}^{\infty}\left(1+z^{2^n}\right) = \frac{1}{1-z}$$

$$\prod_{n=1}^{\infty} e^{\frac{1}{n}\left(\frac{z}{c}\right)^n} = \frac{c}{c-z}$$

$$\prod_{n=1}^{\infty}(1-z^n) = \sum_{k=1}^{\infty}(-1)^k\,z^{k(3k-1)/2}$$

$$\prod_{n=1}^{\infty} z^{1/2^n} = z$$

$$\prod_{n=1}^{\infty}\left[1-\left(\frac{z}{n}\right)^{2N}\right] = \frac{\sin(\pi z)}{\pi z^{2N-1}}\prod_{k=1}^{N-1}\left|\Gamma\left(z\,e^{2\pi i(k-N)/(2N)}\right)\right|^{-2}$$

$$\prod_{n=1}^{\infty}\left[1+\left(\frac{z}{n}\right)^{2N}\right] = \frac{1}{z^{2N}}\prod_{k=1}^{N}\left|\Gamma\left(z\,e^{\pi i(2(k-N)-1)/(2N)}\right)\right|^{-2}$$

$$\prod_{n=1}^{\infty}\left[1-\left(\frac{z}{n}\right)^{2N+1}\right] = \frac{1}{\Gamma(1-z)\,z^{2N}}\prod_{k=1}^{N}\left|\Gamma\left(z\,e^{\pi i(2(k-N)-1)/(2N+1)}\right)\right|^{-2}$$

$$\prod_{n=1}^{\infty}\left[1+\left(\frac{z}{n}\right)^{2N+1}\right] = \frac{1}{\Gamma(1+z)\,z^{2N}}\prod_{k=1}^{N}\left|\Gamma\left(z\,e^{2\pi i(k-N-1)/(2N+1)}\right)\right|^{-2}$$

1.10.2 Trigonometric Functions

$$\sin(z) = z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2\pi^2}\right) = z\prod_{n=1}^{\infty}\cos\left(2^{-n}z\right) \qquad \cos(z) = \prod_{n=1}^{\infty}\left(1-\frac{4z^2}{(2n-1)^2\pi^2}\right)$$

$$\csc(z) = \frac{1}{z}\prod_{n=1}^{\infty}\frac{n^2\pi^2}{n^2\pi^2-z^2} \qquad \sec(z) = \prod_{n=1}^{\infty}\frac{(2n-1)^2\pi^2}{\pi^2(2n-1)^2-4z^2}$$

$$\sin(nz) = 2^{n-1}\prod_{k=0}^{n-1}\sin\left(z+\frac{k\pi}{n}\right) \qquad \frac{\sin(2^k\,z)}{2^k\,\sin(z)} = \prod_{i=0}^{k-1}\cos(2^i\,z)$$

$$\tan(z) = z\prod_{n=1}^{\infty}\frac{(1-2n)^2(n^2\pi^2-z^2)}{n^2(\pi^2(1-2n)^2-4z^2)} \qquad \cot(z) = \frac{1}{z}\prod_{n=1}^{\infty}\frac{n^2((2n-1)^2\pi^2-4z^2)}{(2n-1)^2(\pi^2n^2-z^2)}$$

$$\frac{\sin(\pi z)}{\pi z} = \operatorname{sinc}(\pi z) = \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right) = \prod_{n=1}^{\infty}\cos\left(\frac{\pi z}{2^n}\right)$$

1.10.3 Hyperbolic Functions

$$\sinh(z) = z\prod_{n=1}^{\infty}\left(1 + \frac{z^2}{n^2\pi^2}\right) \qquad \cosh(z) = \prod_{n=1}^{\infty}\left(1 + \frac{4z^2}{(2n-1)^2\pi^2}\right) \qquad \tanh(z) = z\prod_{n=1}^{\infty}\frac{(1-2n)^2(n^2\pi^2 + z^2)}{n^2(\pi^2(1-2n)^2 + 4z^2)}$$

$$\operatorname{csch}(z) = \frac{1}{z}\prod_{n=1}^{\infty}\frac{n^2\pi^2}{n^2\pi^2 + z^2} \qquad \operatorname{sech}(z) = \prod_{n=1}^{\infty}\frac{(2n-1)^2\pi^2}{\pi^2(2n-1)^2 + 4z^2}$$

1.10.4 Exponential and Logarithmic Functions

$$\ln(z) = \frac{z-1}{e-1}\prod_{n=1}^{\infty}\frac{1 + e^{2^{-n}}}{1 + z^{2^{-n}}}$$

1.10.5 Special Functions

Gamma Function:

$$\frac{1}{\Gamma(z)} = ze^{\gamma z}\prod_{n=1}^{\infty}\left(1 + \frac{z}{n}\right)e^{-\frac{z}{n}} = z\prod_{n=1}^{\infty}\frac{1 + \frac{z}{n}}{\left(1 + \frac{1}{n}\right)^z}$$

$$\Gamma(1+z) = 2^{2z}\prod_{n=1}^{\infty}\left[\pi^{-1/2}\,\Gamma\left(\frac{1}{2} + 2^{-n}z\right)\right]$$

Beta Function:

$$B(a,b) = \frac{1}{a+b-1}\prod_{n=1}^{\infty}\frac{n(a+b+n-2)}{(a+n-1)\,(b+n-1)}$$

Q-Pochhammer Symbol:

$$(z;q)_\infty = \prod_{n=0}^{\infty}(1 - zq^n)$$

Riemann Zeta Function:

$$\zeta(z) = \prod_{n=1}^{\infty}\frac{1}{1 - p_n^{-z}}$$

Mellin's Formula:

$$\frac{e^{y\,\psi_0(x)}\,\Gamma(x)}{\Gamma(x+y)} = \prod_{n=0}^{\infty}\left(1 + \frac{y}{n+x}\right)e^{-y(n+x)}$$

Bessel Function:

$$J_\alpha(z) = \frac{z^\alpha}{2^\alpha\Gamma(\alpha+1)}\prod_{n=1}^{\infty}\left(1 - \frac{z^2}{j_{\alpha,n}^2}\right)$$

1.11 Trigonometric Identities

The mnemonic "SOHCAHTOA" is used frequently to remember the definitions of the trigonometric functions in terms of right-angled triangles:

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} = \frac{\text{opposite}}{\text{adjacent}}$$

where $\sin(\theta) > 0$ for $2n\pi \leq \theta \leq (2n+1)\pi$ and $\cos(\theta) > 0$ for $-(2n+1)\pi/2 \leq \theta \leq (2n+1)\pi/2$ and $\tan(\theta) > 0$ for $(2n\pi \leq \theta \leq (2n+1/2)\pi) \cup ((2n+1)\pi \leq \theta \leq (2n+1/2)\pi)$.

The 6 defining trigonometric functions, and their relationship to the hyperbolic functions, are:

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i} = -i\sinh(iz) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!} = z\prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2\pi^2}\right)$$

$$\csc(z) = \frac{1}{\sin(z)} = \frac{2i}{e^{iz} - e^{-iz}} = i\,\mathrm{csch}(iz) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}2(2^{2n-1}-1)B_{2n}z^{2n-1}}{(2n)!}, \quad 0 < |z| < \pi$$

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} = \cosh(iz) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!} = \prod_{n=1}^{\infty}\left(1 - \frac{4z^2}{(2n-1)^2\pi^2}\right)$$

$$\sec(z) = \frac{1}{\cos(z)} = \frac{2}{e^{iz} + e^{-iz}} = \mathrm{sech}(iz) = \sum_{n=0}^{\infty} \frac{(-1)^n E_{2n} z^{2n}}{(2n)!}, \quad |z| < \frac{\pi}{2}$$

$$\tan(z) = \frac{\sin(z)}{\cos(z)} = -i\frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}} = -i\tanh(iz) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}2^{2n}(2^{2n}-1)B_{2n}z^{2n-1}}{(2n)!}, \quad |z| < \frac{\pi}{2}$$

$$\cot(z) = \frac{1}{\tan(z)} = \frac{\cos(z)}{\sin(z)} = i\frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}}, \quad (z \neq 0) = i\coth(iz) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} B_{2n} z^{2n-1}}{(2n)!}, \quad 0 < |z| < \pi$$

where E_n and B_n are the nth Euler and Bernoulli numbers respectively.

Some useful relations are:

Parity:

$$\sin(-z) = -\sin(z)$$
$$\cos(-z) = \cos(z)$$
$$\tan(-z) = -\tan(z)$$
$$\cot(-z) = -\cot(z)$$
$$\sec(-z) = \sec(z)$$
$$\csc(-z) = -\csc(z)$$

Periodicity:

$$\sin(z \pm 2n\pi) = \sin(z)$$
$$\cos(z \pm 2n\pi) = \cos(z)$$
$$\tan(z \pm n\pi) = \tan(z)$$
$$\cot(z \pm n\pi) = \cot(z)$$
$$\sec(z \pm 2n\pi) = \sec(z)$$
$$\csc(z \pm 2n\pi) = \csc(z)$$

Pythagorean Identities:

$$\cos^2(z) + \sin^2(z) = 1$$
$$\cos(iz) - i\sin(iz) = e^z$$
$$\cos(z) + i\sin(z) = e^{iz}$$
$$\sec^2(z) - \tan^2(z) = 1$$
$$\csc^2(z) - \cot^2(z) = 1$$

Quarter Period Shift:

$$\sin\left(z \pm \frac{\pi}{2}\right) = \pm\cos(z)$$

$$\cos\left(z \pm \frac{\pi}{2}\right) = \mp\sin(z)$$

$$\tan\left(z \pm \frac{\pi}{4}\right) = \frac{\tan(z) \pm 1}{1 \mp \tan(z)}$$

$$\cot\left(z \pm \frac{\pi}{4}\right) = \frac{\cot(z) \pm 1}{1 \mp \cot(z)}$$

$$\sec\left(z \pm \frac{\pi}{2}\right) = \mp\csc(z)$$

$$\csc\left(z \pm \frac{\pi}{2}\right) = \pm\sec(z)$$

Half Period Shift:

$$\sin(z \pm \pi) = -\sin(z)$$

$$\cos(z \pm \pi) = -\cos(z)$$

$$\tan\left(z \pm \frac{\pi}{2}\right) = -\cot(z)$$

$$\cot\left(z \pm \frac{\pi}{2}\right) = -\tan(z)$$

$$\sec(z \pm \pi) = -\sec(z)$$

$$\csc(z \pm \pi) = -\csc(z)$$

Law of Sines:

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

Law of Cosines:

$$c^2 = a^2 + b^2 - 2\,a\,b\cos(C)$$

Law of Tangents:

$$\frac{\tan(\frac{A-B}{2})}{\tan(\frac{A+B}{2})} = \frac{a-b}{a+b}$$

$$\frac{\tan(\frac{A-C}{2})}{\tan(\frac{A+C}{2})} = \frac{a-c}{a+c}$$

$$\frac{\tan(\frac{B-C}{2})}{\tan(\frac{B+C}{2})} = \frac{b-c}{b+c}$$

Sums/Differences of Arguments:

$$\sin(z \pm w) = \sin(z)\cos(w) \pm \cos(z)\sin(w)$$

$$\cos(z \pm w) = \cos(z)\cos(w) \mp \sin(z)\sin(w)$$

$$\tan(z \pm w) = \frac{\tan(z) \pm \tan(w)}{1 \mp \tan(z)\tan(w)}$$

$$\csc(z \pm w) = \frac{\sec(z)\sec(w)\csc(z)\csc(w)}{\csc(z)\csc(w) \pm \sec(z)\sec(w)}$$

$$\sec(z \pm w) = \frac{\sec(z)\sec(w)\csc(z)\csc(w)}{\csc(z)\csc(w) \mp \sec(z)\sec(w)}$$

$$\cot(z \pm w) = \frac{\cot(z)\cot(w) \mp 1}{\cot(w) \pm \cot(z)}$$

$$\sin(z + iw) = \sin(z)\cosh(w) + i\cos(z)\sinh(w)$$

$$\cos(z + iw) = \cosh(w)\cos(z) - i\sinh(w)\sin(z)$$

$$\tan(z + iw) = \frac{\sin(2z) + i\sinh(2w)}{\cos(2z) + \cosh(2w)}$$

Product to Sum/Difference:

$$2\sin(z)\cos(w) = \sin(z+w) + \sin(z-w)$$

$$2\sin(w)\cos(z) = \sin(z+w) - \sin(z-w)$$

$$2\sin(z)\sin(w) = \cos(z-w) - \cos(z+w)$$

$$2\cos(z)\cos(w) = \cos(z-w) + \cos(z+w)$$

$$\tan(z)\tan(w) = \frac{\cos(z-w) - \cos(z+w)}{\cos(z-w) + \cos(z+w)}$$

Sum/Difference to Product:

$$\sin(z) \pm \sin(w) = 2\sin\left(\frac{z \pm w}{2}\right)\cos\left(\frac{z \mp w}{2}\right)$$

$$\cos(z) + \cos(w) = 2\cos\left(\frac{z+w}{2}\right)\cos\left(\frac{z-w}{2}\right)$$

$$\cos(z) - \cos(w) = -2\sin\left(\frac{z+w}{2}\right)\sin\left(\frac{z-w}{2}\right)$$

$$\tan(z) \pm \tan(w) = \frac{\sin(z \pm w)}{\cos(z)\cos(w)}$$

$$\cot(z) \pm \cot(w) = \frac{\sin(z \pm w)}{\sin(z)\sin(w)}$$

Sums of Many Arguments:

$$\sin\left(\sum_{i=1}^{N}\theta_i\right) = \sum_{\text{odd } k\geq 1}(-1)^{\frac{k-1}{2}}\sum_{\substack{A\subseteq\{1,2,3,\ldots\}\\|A|=k}}\left(\prod_{i\in A}\sin(\theta_i)\prod_{i\notin A}\cos(\theta_i)\right)$$

$$\cos\left(\sum_{i=1}^{N}\theta_i\right) = \sum_{\text{even } k\geq 0}(-1)^{\frac{k}{2}}\sum_{\substack{A\subseteq\{1,2,3,\ldots\}\\|A|=k}}\left(\prod_{i\in A}\sin(\theta_i)\prod_{i\notin A}\cos(\theta_i)\right)$$

$$\tan\left(\sum_{i=1}^{N}\theta_i\right) = \frac{\sum_{\text{odd } k\geq 1}(-1)^{\frac{k-1}{2}}\sum_{\substack{A\subseteq\{1,2,3,\ldots\}\\|A|=k}}\prod_{i\in A}\tan(\theta_i)}{\sum_{\text{even } k\geq 0}(-1)^{\frac{k}{2}}\sum_{\substack{A\subseteq\{1,2,3,\ldots\}\\|A|=k}}\prod_{i\in A}\tan(\theta_i)}$$

(Note that $\sum_{i=1}^{N}\theta_i$ must converge absolutely)

Double Argument Identities:

$$\sin(2z) = 2\sin(z)\cos(z) = \frac{2\tan(z)}{1+\tan^2(z)}$$

$$\cos(2z) = \cos^2(z) - \sin^2(z) = 2\cos^2(z) - 1$$

$$= 1 - 2\sin^2(z) = \frac{1-\tan^2(z)}{1+\tan^2(z)}$$

$$\tan(2z) = \frac{2\tan(z)}{1-\tan^2(z)}$$

$$\cot(2z) = \frac{\cot^2(z)-1}{2\cot(z)}$$

$$\sec(2z) = \frac{\sec^2(z)}{2-\sec^2(z)}$$

$$\csc(2z) = \frac{\sec(z)\csc(z)}{2}$$

Triple Argument Identities:

$$\sin(3z) = 3\sin(z) - 4\sin^3(z)$$

$$\cos(3z) = 4\cos^3(z) - 3\cos(z)$$

$$\tan(3z) = \frac{3\tan(z)-\tan^3(z)}{1-3\tan^2(z)}$$

$$\cot(3z) = \frac{3\cot(z)-\cot^3(z)}{1-3\cot^2(z)}$$

$$\sec(3z) = \frac{\sec^3(z)}{4-3\sec^2(z)}$$

$$\csc(3z) = \frac{\csc^3(z)}{3\csc^2(z)-4}$$

Half Argument Identities:

$$\sin\left(\frac{z}{2}\right) = \operatorname{sgn}\left(2\pi - z + 4\pi\left\lfloor\frac{z}{4\pi}\right\rfloor\right)\sqrt{\frac{1-\cos(z)}{2}}$$

$$\cos\left(\frac{z}{2}\right) = \operatorname{sgn}\left(\pi + z + 4\pi\left\lfloor\frac{\pi-z}{4\pi}\right\rfloor\right)\sqrt{\frac{\cos(z)+1}{2}}$$

$$\tan\left(\frac{z}{2}\right) = \csc(z) - \cot(z) = \frac{\sin(z)}{\cos(z)+1} = \pm\sqrt{\frac{1-\cos(z)}{\cos(z)+1}}$$

$$= \frac{1-\cos(z)}{\sin(z)} = \frac{-1\pm\sqrt{1+\tan^2(z)}}{\tan(z)} = \frac{\tan(z)}{1+\sec(z)}$$

$$\cot\left(\frac{z}{2}\right) = \csc(z) + \cot(z) = \frac{\sin(z)}{1-\cos(z)} = \pm\sqrt{\frac{1+\cos(z)}{1-\cos(z)}}$$

$$= \frac{1+\cos(z)}{\sin(z)}$$

$$\tan\left(\frac{z}{2}+\frac{\pi}{4}\right) = \sec(z) + \tan(z)$$

Multiple Argument Identities:

$$\sin(nz) = \sum_{\text{k odd}}(-1)^{\frac{k-1}{2}}\frac{n!}{k!(n-k)!}\cos^{n-k}(z)\sin^k(z)$$

$$\cos(nz) = \sum_{\text{k even}}(-1)^{\frac{k}{2}}\frac{n!}{k!(n-k)!}\cos^{n-k}(z)\sin^k(z)$$

$$\tan(nz) = \frac{\sum_{\text{k odd}}(-1)^{\frac{k-1}{2}}\frac{n!}{k!(n-k)!}\tan^k(z)}{\sum_{\text{k even}}(-1)^{\frac{k}{2}}\frac{n!}{k!(n-k)!}\tan^k(z)}$$

Lagrange's Trigonometric Identities:

$$\sum_{n=1}^{N} \sin(n\theta) = \frac{1}{2}\cot\left(\frac{\theta}{2}\right) - \frac{\cos\left(\left(N+\frac{1}{2}\right)\theta\right)}{2\,\sin\left(\frac{\theta}{2}\right)}$$

$$\sum_{n=1}^{N} \cos(n\theta) = -\frac{1}{2} + \frac{\sin\left(\left(N+\frac{1}{2}\right)\theta\right)}{2\,\sin\left(\frac{\theta}{2}\right)}$$

Chebyshev Multiple Argument Identity:

$$\sin(nz) = 2\cos(z)\sin((n-1)z) - \sin((n-2)z)$$

Recursive Algorithm:

$$\cos(nz) = 2\cos(z)\cos((n-1)z) - \cos((n-2)z)$$

$$\tan(nz) = \frac{\tan((n-1)z) + \tan(z)}{1 - \tan((n-1)z)\tan(z)}$$

Power Reduction Formulae:

$$\sin^2(z) = \frac{1-\cos(2z)}{2} \qquad \cos^2(z) = \frac{1+\cos(2z)}{2}$$

$$\sin^3(z) = \frac{3\sin(z) - \sin(3z)}{4} \qquad \cos^3(z) = \frac{3\cos(z) + \cos(3z)}{2}$$

$$\sin^4(z) = \frac{3 - 4\cos(2z) + \cos(4z)}{8} \qquad \cos^4(z) = \frac{3 + 4\cos(2z) + \cos(4z)}{8}$$

$$\sin^5(z) = \frac{10\sin(z) - 5\sin(3z) + \sin(5z)}{16} \qquad \cos^5(z) = \frac{10\cos(z) + 5\cos(3z) + \cos(5z)}{16}$$

$$\sin^2(z)\cos^2(z) = \frac{1-\cos(4z)}{8} \qquad \sin^3(z)\cos^3(z) = \frac{3\sin(2z) - \sin(6z)}{32}$$

$$\sin^4(z)\cos^4(z) = \frac{3 - 4\cos(4z) + \cos(8z)}{128} \qquad \sin^5(z)\cos^5(z) = \frac{10\sin(2z) - 5\sin(6z) + \sin(10z)}{512}$$

Generalized Power Reduction Formulae:

$$\sin^n(z) = \begin{cases} \dfrac{2}{2^n}\displaystyle\sum_{k=0}^{\frac{n-1}{2}} (-1)^{\frac{n-1}{2}-k} \frac{n!}{k!(n-k)!}\sin((n-2k)z) & n \text{ odd} \\[2ex] \dfrac{1}{2^n}\dfrac{n!}{((\frac{n}{2})!)^2} + \dfrac{2}{2^n}\displaystyle\sum_{k=0}^{\frac{n}{2}-1} (-1)^{\frac{n}{2}-k} \frac{n!}{k!(n-k)!}\cos((n-2k)z) & n \text{ even} \end{cases}$$

$$\cos^n(z) = \begin{cases} \dfrac{2}{2^n}\displaystyle\sum_{k=0}^{\frac{n-1}{2}} \frac{n!}{k!(n-k)!}\cos((n-2k)z) & n \text{ odd} \\[2ex] \dfrac{1}{2^n}\dfrac{n!}{((\frac{n}{2})!)^2} + \dfrac{2}{2^n}\displaystyle\sum_{k=0}^{\frac{n}{2}-1} \frac{n!}{k!(n-k)!}\cos((n-2k)z) & n \text{ even} \end{cases}$$

De Moivre's Theorem:

$$(\cos(z) + i\sin(z))^n = \cos(nz) + i\sin(nz)$$

Some special values include:

Degree (°)	Radian	Sin(z)	Cos(z)	Tan(z)
0	0	0	1	0
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90	$\frac{\pi}{2}$	1	0	Undefined
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180	π	0	-1	0
210	$\frac{7\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
225	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1
240	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
270	$\frac{3\pi}{2}$	-1	0	Undefined
300	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
315	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
330	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
360	2π	0	1	0

For a value a, the relationships between different trigonometric functions are:

	$\sin(z) = a$	$\cos(z) = a$	$\tan(z) = a$	$\csc(z) = a$	$\sec(z) = a$	$\cot(z) = a$
$\sin(z)$	a	$\sqrt{1-a^2}$	$a(1+a^2)^{-1/2}$	a^{-1}	$a^{-1}\sqrt{a^2-1}$	$(a^2+1)^{-1/2}$
$\cos(z)$	$\sqrt{1-a^2}$	a	$(1+a^2)^{-1/2}$	$a^{-1}\sqrt{a^2-1}$	a^{-1}	$a(a^2+1)^{-1/2}$
$\tan(z)$	$a(1-a^2)^{-1/2}$	$a^{-1}\sqrt{1-a^2}$	a	$(a^2-1)^{-1/2}$	$\sqrt{a^2-1}$	a^{-1}
$\csc(z)$	a^{-1}	$(1-a^2)^{-1/2}$	$a^{-1}\sqrt{1+a^2}$	a	$a(a^2-1)^{-1/2}$	$\sqrt{a^2+1}$
$\sec(z)$	$(1-a^2)^{-1/2}$	a^{-1}	$\sqrt{1+a^2}$	$a(a^2-1)^{-1/2}$	a	$a^{-1}\sqrt{a^2+1}$
$\cot(z)$	$a^{-1}\sqrt{1-a^2}$	$a(1-a^2)^{-1/2}$	a^{-1}	$\sqrt{a^2-1}$	$(a^2-1)^{-1/2}$	a

1.11.1 Inverse Trigonometric Identities

The 6 defining inverse trigonometric functions, and their relationship to the inverse hyperbolic functions, are:

$$\arcsin(z) = -i\ln\left(iz + \sqrt{1-z^2}\right) = -i\,\mathrm{arsinh}(iz) = \mathrm{arccsc}\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty}\frac{(2n-1)!!}{(2n)!!}\frac{z^{2n+1}}{(2n+1)}\,, \quad |z| \le 1$$

$$\mathrm{arccsc}(z) = -i\ln\left(\frac{i}{z} + \sqrt{1-\frac{1}{z^2}}\right) = i\,\mathrm{arcsch}(iz) = \arcsin\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty}\left(\frac{\Gamma(n+1/2)}{\sqrt{\pi}}\right)\frac{z^{-(2n+1)}}{(2n+1)\,n!}\,, \quad |z| > 1$$

$$\arccos(z) = -i\ln\left(z + i\sqrt{z^2-1}\right) = \pm i\,\mathrm{arcosh}(z) = \mathrm{arcsec}\left(\frac{1}{z}\right) = \frac{\pi}{2} - \sum_{n=0}^{\infty}\left(\frac{\Gamma(n+1/2)}{\sqrt{\pi}}\right)\frac{z^{2n+1}}{(2n+1)\,n!}\,, \quad |z| < 1$$

$$\mathrm{arcsec}(z) = -i\ln\left(\frac{1}{z} + \sqrt{\frac{1}{z^2}-1}\right) = \pm i\,\mathrm{arsech}(z) = \arccos\left(\frac{1}{z}\right) = \ln\left(\frac{2}{z}\right) = \frac{\pi}{2} - \sum_{n=0}^{\infty}\left(\frac{\Gamma(n+1/2)}{2n+1}\right)\frac{z^{-(2n+1)}}{\sqrt{\pi}\,n!}\,, \quad |z| < 1$$

$$\arctan(z) = \frac{i}{2}\ln\left(\frac{i+z}{i-z}\right) = -i\,\mathrm{artanh}(iz) = \mathrm{arccot}\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty}\frac{(-1)^n z^{(2n+1)}}{(2n+1)}\,, \quad |z| \le 1$$

$$\mathrm{arccot}(z) = \frac{i}{2}\ln\left(\frac{z-i}{z+i}\right) = i\,\mathrm{arcoth}(iz) = \arctan\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty}\frac{(-1)^n z^{-(2n+1)}}{(2n+1)}\,, \quad |z| > 1$$

Some useful relations are:

Parity:

$$\arcsin(-z) = -\arcsin(z)$$
$$\arccos(-z) = \pi - \arccos(z)$$
$$\arctan(-z) = -\arctan(z)$$
$$\operatorname{arccot}(-z) = \pi - \operatorname{arccot}(z)$$
$$\operatorname{arcsec}(-z) = \pi - \operatorname{arcsec}(z)$$
$$\operatorname{arccsc}(-z) = -\operatorname{arccsc}(z)$$

Sum/Difference Formulae:

$$\arcsin(z) \pm \arcsin(w) = \arcsin\left(z\sqrt{1-w^2} \pm w\sqrt{1-z^2}\right)$$
$$\arccos(z) \pm \arccos(w) = \arccos\left(z\,w \mp \sqrt{(1-w^2)(1-z^2)}\right)$$
$$\arctan(z) \pm \arctan(w) = \arctan\left(\frac{z \pm w}{1 \mp z\,w}\right)$$
$$\begin{aligned}\arcsin(z) \pm \arccos(w) &= \arcsin\left(z\,w \pm \sqrt{(1-w^2)(1-z^2)}\right)\\ &= \arccos\left(w\sqrt{1-z^2} \mp z\sqrt{1-w^2}\right)\end{aligned}$$

Complimentary Angles:

$$\arccos(z) = \frac{\pi}{2} - \arcsin(z)$$
$$\operatorname{arccot}(z) = \frac{\pi}{2} - \arctan(z)$$
$$\operatorname{arccsc}(z) = \frac{\pi}{2} - \operatorname{arcsec}(z)$$

Conversions:

$$\arctan(z) = 2\arctan\left(\frac{z}{1+\sqrt{1+z^2}}\right) = \arcsin\left(\frac{z}{\sqrt{1+z^2}}\right)$$
$$\arcsin(z) = 2\arctan\left(\frac{z}{1+\sqrt{1-z^2}}\right) = \arctan\left(\frac{z}{\sqrt{1-z^2}}\right)$$
$$\arccos(z) = 2\arctan\left(\frac{\sqrt{1-z^2}}{1+z}\right), \quad -1 < z \leq 1$$

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$	$\csc(\theta)$	$\sec(\theta)$	$\cot(\theta)$
$\arcsin(z)$	z	$\sqrt{1-z^2}$	$(1+z^{-2})^{-\frac{1}{2}}$	z^{-1}	$\sqrt{1-z^{-2}}$	$(z^2+1)^{-\frac{1}{2}}$
domain:	$[-1,1]$	$[-1,1]$	$(-1,1)$	$[-1,1]\backslash\{0\}$	$(-1,1)$	$[-1,1]$
$\arccos(z)$	$\sqrt{1-z^2}$	z	$\sqrt{(z^{-2}-1)}$	$(1-z^2)^{-\frac{1}{2}}$	z^{-1}	$(z^{-2}-1)^{-\frac{1}{2}}$
domain:	$[-1,1]$	$[-1,1]$	$[-1,1]\backslash\{0\}$	$(-1,1)$	$[-1,1]\backslash\{0\}$	$(-1,1)$
$\arctan(z)$	$(1+z^{-2})^{-\frac{1}{2}}$	$(1+z^2)^{-\frac{1}{2}}$	z	$\sqrt{z^{-2}+1}$	$\sqrt{z^2+1}$	z^{-1}
domain:	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R}\backslash\{0\}$	$\mathbb{R}$	$\mathbb{R}\backslash\{0\}$
$\operatorname{arccsc}(z)$	z^{-1}	$\sqrt{1-z^{-2}}$	$(z^2-1)^{-\frac{1}{2}}$	z	$(1-z^{-2})^{-\frac{1}{2}}$	$\pm\sqrt{z^2-1}$
domain:	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$
$\operatorname{arcsec}(z)$	$\sqrt{1-z^{-2}}$	z^{-1}	$\sqrt{z^2-1}$	$(1-z^{-2})^{-\frac{1}{2}}$	z	$\pm(z^2-1)^{-\frac{1}{2}}$
domain:	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$	$\mathbb{R}\backslash(-1,1)$
$\operatorname{arccot}(z)$	$(1+z^2)^{-\frac{1}{2}}$	$(1+z^{-2})^{-\frac{1}{2}}$	z^{-1}	$\pm\sqrt{z^2+1}$	$\sqrt{z^{-2}+1}$	z
domain:	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R}\backslash\{0\}$	$\mathbb{R}$	$\mathbb{R}\backslash\{0\}$	$\mathbb{R}$

Useful approximations for $sin(z)$ and $cos(z)$ when z is small are:

$$\sin(z) \approx z \quad \text{and} \quad \cos(z) \approx 1 - \frac{z^2}{2}$$

1.12 Hyperbolic Identities

The 6 defining hyperbolic functions, and their relationship to the trigonometric functions, are:

$$\sinh(z) = \frac{e^z - e^{-z}}{2} = -i\sin(iz) = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = z\prod_{n=1}^{\infty}\left(1 + \frac{z^2}{n^2\pi^2}\right)$$

$$\operatorname{csch}(z) = \frac{1}{\sinh(z)} = \frac{2}{e^z - e^{-z}} = i\csc(iz) = z^{-1} + \sum_{n=1}^{\infty} \frac{2(1-2^{2n-1})B_{2n}z^{2n-1}}{(2n)!}\ , \quad 0 < |z| < \pi$$

$$\cosh(z) = \frac{e^z + e^{-z}}{2} = \cos(iz) = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = \prod_{n=1}^{\infty}\left(1 + \frac{4z^2}{(2n-1)^2\pi^2}\right)$$

$$\operatorname{sech}(z) = \frac{1}{\cosh(z)} = \frac{2}{e^z + e^{-z}} = \sec(iz) = \sum_{n=0}^{\infty} \frac{E_{2n}z^{2n}}{(2n)!}\ , \quad |z| < \frac{\pi}{2}$$

$$\tanh(z) = \frac{\sinh(z)}{\cosh(z)} = \frac{e^z - e^{-z}}{e^z + e^{-z}} = -i\tan(iz) = \sum_{n=1}^{\infty} \frac{2^{2n}(2^{2n}-1)B_{2n}z^{2n-1}}{(2n)!}\ , \quad |z| < \frac{\pi}{2}$$

$$\coth(z) = \frac{1}{\tanh(z)} = \frac{\cosh(z)}{\sinh(z)} = \frac{e^z + e^{-z}}{e^z - e^{-z}}\ , \ (z \neq 0) = \ i\ \cot(iz) = z^{-1} + \sum_{n=1}^{\infty} \frac{2^{2n}B_{2n}z^{2n-1}}{(2n)!}\ , \quad 0 < |z| < \pi$$

where E_n and B_n are the n-th Euler and Bernoulli numbers respectively.

Some useful relations are:

Parity:

$$\sinh(-z) = -\sinh(z)$$
$$\cosh(-z) = \cosh(z)$$
$$\tanh(-z) = -\tanh(z)$$
$$\coth(-z) = -\coth(z)$$
$$\operatorname{sech}(-z) = \operatorname{sech}(z)$$
$$\operatorname{csch}(-z) = -\operatorname{csch}(z)$$

Periodicity:

$$\sinh(z + 2n\pi i) = \sinh(z)$$
$$\cosh(z + 2n\pi i) = \cosh(z)$$
$$\tanh(z + n\pi i) = \tanh(z)$$

Pseudo-Pythagorean Identities:

$$1 = \cosh^2(z) - \sinh^2(z)$$
$$e^z = \cosh(z) + \sinh(z)$$
$$e^{-z} = \cosh(z) - \sinh(z)$$
$$1 = \operatorname{sech}^2(z) + \tanh^2(z)$$
$$1 = \coth^2(z) - \operatorname{csch}^2(z)$$

Sums/Differences of Arguments:

$$\sinh(z \pm w) = \sinh(z)\cosh(w) \pm \cosh(z)\sinh(w)$$

$$\cosh(z \pm w) = \cosh(z)\cosh(w) \pm \sinh(z)\sinh(w)$$

$$\tanh(z \pm w) = \frac{\tanh(z) \pm \tanh(w)}{1 \pm \tanh(z)\tanh(w)}$$

$$\sinh(z + iw) = \sinh(z)\cos(w) + i\cosh(z)\sin(w)$$

$$\cosh(z + iw) = \cosh(z)\cos(w) + i\sinh(z)\sin(w)$$

$$\tanh(z + iw) = \frac{\sinh(2z) + i\sin(2w)}{\cosh(2z) - \cos(2w)}$$

Product to Sum/Difference:

$$\sinh(z) \pm \sinh(w) = 2\sinh\left(\frac{z \pm w}{2}\right)\cosh\left(\frac{z \mp w}{2}\right)$$

$$\cosh(z) + \cosh(w) = 2\cosh\left(\frac{z + w}{2}\right)\cosh\left(\frac{z - w}{2}\right)$$

$$\cosh(z) - \cosh(w) = 2\sinh\left(\frac{z + w}{2}\right)\sinh\left(\frac{z - w}{2}\right)$$

$$\tanh(z) + \tanh(w) = \frac{\sinh(z + w)}{\cosh(z)\cosh(w)}$$

Double Argument Identities:

$$\sinh(2z) = 2\sinh(z)\cosh(z)$$

$$\cosh(2z) = \sinh^2(z) + \cosh^2(z) = 2\sinh^2(z) + 1$$

$$= 2\cosh^2(z) - 1$$

$$\tanh(2z) = \frac{2\tanh(z)}{1 + \tanh^2(z)}$$

Half Argument Identities:

$$\sinh\left(\frac{z}{2}\right) = \frac{\sinh(z)}{\sqrt{2(\cosh(z) + 1)}} = \operatorname{sgn}(z)\sqrt{\frac{\cosh(z) - 1}{2}}$$

$$\cosh\left(\frac{z}{2}\right) = \sqrt{\frac{\cosh(z) + 1}{2}}$$

$$\tanh\left(\frac{z}{2}\right) = \frac{\sinh(z)}{\cosh(z) + 1} = \operatorname{sgn}(z)\sqrt{\frac{\cosh(z) - 1}{\cosh(z) + 1}} = \frac{e^z - 1}{e^z + 1}$$

De Moivre's Theorem:

$$(\cosh(z) + \sinh(z))^n = \cosh(nz) + \sinh(nz)$$

Some special values include:

z	0	$\frac{i\pi}{2}$	$i\pi$	$\frac{3\pi i}{2}$	∞
$\sinh(z)$	0	i	0	$-i$	∞
$\cosh(z)$	1	0	-1	0	∞
$\tanh(z)$	0	$i\infty$	0	$-i\infty$	1
$\operatorname{csch}(z)$	∞	$-i$	∞	i	0
$\operatorname{sech}(z)$	1	∞	-1	∞	0
$\coth(z)$	∞	0	∞	0	1

For a value a, the relationships between different hyperbolic functions are:

	$\sinh(z) = a$	$\cosh(z) = a$	$\tanh(z) = a$	$\operatorname{csch}(z) = a$	$\operatorname{sech}(z) = a$	$\coth(z) = a$
$\sinh(z)$	a	$\sqrt{a^2-1}$	$a(1-a^2)^{-1/2}$	a^{-1}	$a^{-1}\sqrt{1-a^2}$	$(a^2-1)^{-1/2}$
$\cosh(z)$	$\sqrt{1+a^2}$	a	$(1-a^2)^{-1/2}$	$a^{-1}\sqrt{1+a^2}$	a^{-1}	$a(a^2-1)^{-1/2}$
$\tanh(z)$	$a(1+a^2)^{-1/2}$	$a^{-1}\sqrt{a^2-1}$	a	$(1+a^2)^{-1/2}$	$\sqrt{1-a^2}$	a^{-1}
$\operatorname{csch}(z)$	a^{-1}	$(a^2-1)^{-1/2}$	$a^{-1}\sqrt{1-a^2}$	a	$a(1-a^2)^{-1/2}$	$\sqrt{a^2-1}$
$\operatorname{sech}(z)$	$(1+a^2)^{-1/2}$	a^{-1}	$\sqrt{1-a^2}$	$a(1+a^2)^{-1/2}$	a	$a^{-1}\sqrt{a^2-1}$
$\coth(z)$	$a^{-1}\sqrt{1+a^2}$	$a(a^2-1)^{-1/2}$	a^{-1}	$\sqrt{1+a^2}$	$(1-a^2)^{-1/2}$	a

1.12.1 Inverse Hyperbolic Identities

The 6 defining inverse hyperbolic functions, and their relationship to the inverse trigonometric functions, are:

$$\operatorname{arsinh}(z) = \ln\left(z + \sqrt{z^2+1}\right)\ ,\ z \in \mathbb{R} = -i\ \arcsin(iz) = \sum_{n=0}^{\infty}\left(\frac{(-1)^n(2n)!}{2^{2n}(n!)^2}\right)\frac{z^{2n+1}}{(2n+1)}\ ,\quad |z| < 1$$

$$\operatorname{arcsch}(z) = \ln\left(\frac{1}{z} + \sqrt{\frac{1}{z^2}+1}\right)\ ,\ z \in \mathbb{R}\backslash\{0\} = i\operatorname{arccsc}(iz) = \operatorname{arsinh}\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty}\left(\frac{(-1)^n(2n)!}{2^{2n}(n!)^2}\right)\frac{z^{-(2n+1)}}{(2n+1)}\ ,\quad |z| > 1$$

$$\operatorname{arcosh}(z) = \ln\left(z + \sqrt{z^2-1}\right)\ ,\ (1 \le z < \infty) = \pm i\ \arccos(z) = \ln(2z) - \sum_{n=1}^{\infty}\left(\frac{(2n)!}{2^{2n}(n!)^2}\right)\frac{z^{-2n}}{(2n)}\ ,\quad |z| > 1$$

$$\operatorname{arsech}(z) = \ln\left(\frac{1}{z} + \sqrt{\frac{1}{z^2}-1}\right)\ ,\ (0 < z \le 1) = \pm i\operatorname{arcsec}(z) = \operatorname{arcosh}\left(\frac{1}{z}\right) = \ln\left(\frac{2}{z}\right) - \sum_{n=1}^{\infty}\frac{z^{2n}(2n-1)!}{2^{2n}(n!)^2}\ ,\quad 0 < z \le 1$$

$$\operatorname{artanh}(z) = \frac{1}{2}\ln\left(\frac{1+z}{1-z}\right)\ ,\ (-1 < z < 1) = -i\ \arctan(iz) = \sum_{n=0}^{\infty}\frac{z^{(2n+1)}}{(2n+1)}\ ,\quad |z| < 1$$

$$\operatorname{arcoth}(z) = \frac{1}{2}\ln\left(\frac{z+1}{z-1}\right)\ ,\ z \in \mathbb{R}\backslash\{[-1,1]\} = i\operatorname{arccot}(iz) = \operatorname{artanh}\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty}\frac{z^{-(2n+1)}}{(2n+1)}\ ,\quad |z| > 1$$

Some useful relations are:

Parity:

$$\operatorname{arsinh}(-z) = -\operatorname{arsinh}(z)$$

$$\operatorname{arcosh}(-z) = \pi i - \operatorname{arcosh}(z)$$

$$\operatorname{artanh}(-z) = -\operatorname{artanh}(z)$$

$$\operatorname{arcoth}(-z) = -\operatorname{arcoth}(z)$$

$$\operatorname{arsech}(-z) = \pi i - \operatorname{arsech}(z)$$

$$\operatorname{arcsch}(-z) = -\operatorname{arcsch}(z)$$

Sum/Difference Formulae:

$$\operatorname{arsinh}(z) \pm \operatorname{arsinh}(w) = \operatorname{arsinh}\left(z\sqrt{1+w^2} \pm w\sqrt{1+z^2}\right)$$

$$\operatorname{arcosh}(z) \pm \operatorname{arcosh}(w) = \operatorname{arcosh}\left(z\,w \pm \sqrt{(w^2-1)\,(z^2-1)}\right)$$

$$\operatorname{artanh}(z) \pm \operatorname{artanh}(w) = \operatorname{artanh}\left(\frac{z \pm w}{1 \pm z\,w}\right)$$

$$\begin{aligned}\operatorname{arsinh}(z) + \operatorname{arcosh}(w) &= \operatorname{arsinh}\left(z\,w + \sqrt{(w^2-1)\,(z^2+1)}\right)\\ &= \operatorname{arcosh}\left(w\sqrt{1+z^2} + z\sqrt{w^2-1}\right)\end{aligned}$$

Hyperbolic Composition with Inverse Hyperbolic Functions:

$$\sinh(\operatorname{arcosh}(z)) = \sqrt{z^2-1}\,, \quad |z| > 1$$

$$\sinh(\operatorname{artanh}(z)) = \frac{z}{\sqrt{1-z^2}}\,, \quad -1 < z < 1$$

$$\cosh(\operatorname{arsinh}(z)) = \sqrt{z^2+1}$$

$$\cosh(\operatorname{artanh}(z)) = \frac{1}{\sqrt{1-z^2}}\,, \quad -1 < z < 1$$

$$\tanh(\operatorname{arsinh}(z)) = \frac{z}{\sqrt{1+z^2}}$$

$$\tanh(\operatorname{arcosh}(z)) = \frac{\sqrt{z^2-1}}{z}\,, \quad |z| > 1$$

Inverse Hyperbolic Composition with Trigonometric Functions:

$$\begin{aligned}\operatorname{arsinh}(\tan(z)) = \operatorname{artanh}(\sin(z)) &= \ln\left(\frac{1+\sin(z)}{\cos(z)}\right)\\ &= \pm\operatorname{arcosh}\left(\frac{1}{\cos(z)}\right)\end{aligned}$$

$$\ln|\tan(z)| = -\operatorname{artanh}(\cos(2z))$$

Alternate Ln Definition:

$$\begin{aligned}\ln(z) &= \operatorname{arcosh}\left(\frac{z^2+1}{2z}\right) = \operatorname{arsinh}\left(\frac{z^2-1}{2z}\right)\\ &= \operatorname{artanh}\left(\frac{z^2-1}{z^2+1}\right)\end{aligned}$$

Conversions:

$$\operatorname{artanh}(z) = \operatorname{arsinh}\left(\frac{z}{\sqrt{1-z^2}}\right) = \pm\operatorname{arcosh}\left(\frac{1}{\sqrt{1-z^2}}\right) \qquad \operatorname{arsinh}(z) = \operatorname{artanh}\left(\frac{z}{\sqrt{1+z^2}}\right) = \pm\operatorname{arcosh}\left(\sqrt{1+z^2}\right)$$

$$\operatorname{arcosh}(z) = \left|\operatorname{artanh}\left(\frac{\sqrt{z^2-1}}{z}\right)\right| = \left|\operatorname{arsinh}\left(\sqrt{z^2-1}\right)\right|$$

1.13 Common Derivatives

In the following lists, the functions are assumed to be differentiable everywhere and for the requisite order, unless otherwise stated. For nth-order differentiation, assume that $n \in \mathbb{N}$ unless stated otherwise. Some general differentiation rules include:

Linearity:
$$\frac{\mathrm{d}}{\mathrm{d}z}\left[a\,f(z) + b\,g(z)\right] = a\frac{\mathrm{d}}{\mathrm{d}z}\,f(z) + b\frac{\mathrm{d}}{\mathrm{d}z}\,g(z)$$

Product Rule:
$$\frac{\mathrm{d}}{\mathrm{d}z}\left[f(z)\,g(z)\right] = \frac{\mathrm{d}f}{\mathrm{d}z}\,g(z) + f(z)\frac{\mathrm{d}g}{\mathrm{d}z}$$

Generalized Product Rule: $$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\left[f(z)\,g(z)\right] = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!}\frac{\mathrm{d}^{n-k}}{\mathrm{d}z^{n-k}} f(z) \frac{\mathrm{d}^k}{\mathrm{d}z^k} g(z)$$

Quotient Rule: $$\frac{\mathrm{d}}{\mathrm{d}z}\left[\frac{f(z)}{g(z)}\right] = \frac{1}{(g(z))^2}\left(\frac{\mathrm{d}f}{\mathrm{d}z}\,g(z) - \frac{\mathrm{d}g}{\mathrm{d}z} f(z)\right)$$

Chain Rule: $$\frac{\mathrm{d}}{\mathrm{d}z} f(g(z)) = \frac{\mathrm{d}f(g(z))}{\mathrm{d}g(z)} \cdot \frac{\mathrm{d}g(z)}{\mathrm{d}z}$$

Generalized Chain Rule: $$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\left[f(g(z))\right] = n! \sum_{\{k_m\}} f^{(r)}(g(z)) \prod_{m=1}^{n} \frac{1}{k_m!}\left(g^{(m)}(z)\right)^{k_m}$$

(where $r = \sum_{m=1}^{n-1} k_m$ and the set $\{k_m\}$ consists of all non-negative integer solutions of the Diophantine equation $\sum_{m=1}^{n} m k_m = n$)

Reciprocal Rule: $$\frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{f(z)} = -\frac{1}{(f(z))^2}\frac{\mathrm{d}f}{\mathrm{d}z}$$

Power Rule: $$\frac{\mathrm{d}}{\mathrm{d}z} z^r = r z^{r-1}$$

Generalized Power Rule: $$\frac{\mathrm{d}}{\mathrm{d}z} f(z)^{g(z)} = \frac{\mathrm{d}}{\mathrm{d}z} e^{g(z)\,\ln(f(z))} = f(z)^{g(z)}\left(\frac{\mathrm{d}f}{\mathrm{d}z}\frac{g(z)}{f(z)} + \frac{\mathrm{d}g}{\mathrm{d}z}\,\ln(f(z))\right)$$

1.13.1 Trigonometric and Inverse Trigonometric Functions

$$\frac{\mathrm{d}}{\mathrm{d}z}\sin(az) = a\cos(z)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\cos(az) = -a\sin(az)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\sin(az) = a^n \sin\left(az + \frac{n\pi}{2}\right)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\cos(az) = a^n \cos\left(az + \frac{n\pi}{2}\right)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\csc(az) = \frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{\sin(az)} = -a\csc(az)\cot(az)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\sec(az) = \frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{\cos(az)} = a\sec(az)\tan(az)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\tan(az) = a\sec^2(az) = a(1+\tan^2(az))$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\cot(az) = -a\csc^2(az) = -a(1+\cot^2(az))$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\arcsin(az) = \frac{a}{\sqrt{1-(az)^2}}\,,\quad z \neq \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\arccos(az) = -\frac{a}{\sqrt{1-(za)^2}}\,,\quad z \neq \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arccsc}(az) = -\frac{a}{z\sqrt{(az)^2-1}}\,,\quad z \neq -1,\ 0,\ 1$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arcsec}(az) = \frac{a}{z\sqrt{(az)^2-1}}\,,\quad z \neq 0,\ \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\arctan(az) = \frac{a}{1+(az)^2}\,,\quad z \neq \pm i a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arccot}(az) = -\frac{a}{1+(az)^2}\,,\quad z \neq \pm i a^{-1}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\arctan(z) = \frac{i(-1)^n(n-1)!}{2}\left(\frac{1}{(z-i)^n} - \frac{1}{(z+i)^n}\right)\,,\quad z \neq -i,\ i\,;\ n \in \mathbb{N}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\operatorname{arccot}(z) = \frac{i(-1)^n(n-1)!}{2}\left(\frac{1}{(z+i)^n} - \frac{1}{(z-i)^n}\right), \quad z \neq -i,\, i\,;\, n \in \mathbb{N}$$

1.13.2 Hyperbolic and Inverse Hyperbolic Functions

$$\frac{\mathrm{d}}{\mathrm{d}z}\sinh(az) = a\cosh(z)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\cosh(az) = -a\sinh(az)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\sinh(az) = (-ia)^n \sinh\left(az + \frac{n\pi}{2}\right)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\cosh(az) = a^n \cos\left(az + \frac{n\pi}{2}\right)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{csch}(az) = \frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{\sinh(az)} = -a\operatorname{csch}(az)\coth(az)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{sech}(az) = \frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{\cosh(az)} = a\operatorname{sech}(az)\,\tanh(az)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\tanh(az) = a\operatorname{sech}^2(az) = a(1-\tanh^2(az))$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\coth(az) = -a\operatorname{csch}^2(az) = a(1-\coth^2(az))$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arsinh}(az) = \frac{a}{\sqrt{1+(az)^2}}, \quad z \neq \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arcosh}(az) = \frac{a}{\sqrt{(az)^2-1}}, \quad z \neq \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arcsch}(az) = -\frac{a}{z\sqrt{(az)^2+1}}, \quad z \neq 0,\, \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arsech}(az) = -\frac{a}{z\sqrt{1-(az)^2}}, \quad z \neq 0,\, \pm a^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{artanh}(az) = \frac{a}{1-(az)^2}, \quad z \neq \pm ia^{-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{arcoth}(az) = \frac{a}{1-(az)^2}, \quad z \neq \pm ia^{-1}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\operatorname{artanh}(z) = \frac{(n-1)!}{2}\left(\frac{1}{(1-z)^n} + (-1)^{n-1}\frac{1}{(z+1)^n}\right), \quad z \neq -i,\, i\,;\, n \in \mathbb{N}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\operatorname{arcoth}(z) = \frac{(n-1)!}{2}\left(\frac{1}{(1-z)^n} + (-1)^{n-1}\frac{1}{(z+1)^n}\right), \quad z \neq -i,\, i\,;\, n \in \mathbb{N}$$

1.13.3 Exponential and Logarithmic Functions

$$\frac{\mathrm{d}}{\mathrm{d}z}e^{az} = a\,e^{az}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\ln(f(z)) = \frac{1}{f(z)}\frac{\mathrm{d}f}{\mathrm{d}z}, \quad f(z) > 0$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}e^{az} = a^n\,e^{az}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\ln(f(z)) = \sum_{k=1}^{n}\frac{(-1)^k}{k\,(f(z))^k}\frac{n!}{k!(n-k)!}\quad\frac{\mathrm{d}^n}{\mathrm{d}z^n}(f(z))^k, \quad f(z) > 0\,;\, n \in \mathbb{N}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}a^z = a^z\,\ln(a)$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\ln(z) = \frac{1}{z}, \quad z > 0$$

$$\frac{\mathrm{d}}{\mathrm{d}z}z^z = z^z(1+\ln(z))$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\log_a(z) = \frac{1}{z\,\ln(a)}, \quad z > 0\,;\, a \neq 1$$

1.13.4 Special Functions

Heaviside Step-Function:

$$\frac{\mathrm{d}}{\mathrm{d}z}\theta(az) = a\,\delta(az)$$

Dirac Delta Function:

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\delta(az) = a^{-(n+1)}\frac{\mathrm{d}^n}{\mathrm{d}z^n}\delta(z)$$

Gamma Function:

$$\frac{\mathrm{d}}{\mathrm{d}z}\Gamma(z) = \Gamma(z)\,\psi(z)$$

Zeta Function:

$$\frac{\mathrm{d}}{\mathrm{d}z}\zeta(z) = -\sum_{k=2}^{\infty}\frac{\ln(k)}{k^z}, \quad \mathrm{Re}(z) > 1$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\zeta(z) = (-1)^n\sum_{k=2}^{\infty}\frac{\ln^n(k)}{k^z}, \quad \mathrm{Re}(z) > 1$$

Hermite Polynomials:

$$\frac{\mathrm{d}}{\mathrm{d}z}H_n(z) = 2\,n\,H_{n-1}(z)$$

$$\frac{\mathrm{d}^m}{\mathrm{d}z^m}H_n(z) = \frac{2^m\,n!}{(n-m)!}H_{n-m}(z)$$

Error Function:

$$\frac{\mathrm{d}}{\mathrm{d}z}\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}}e^{-z^2}$$

Bessel Function of the 1st Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}J_\alpha(z) = \frac{\alpha}{2}J_\alpha(z) - J_{\alpha+1}(z)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}J_\alpha(z) = 2^{-n}\sum_{k=0}^{n}(-1)^k\frac{n!}{k!(n-k)!}J_{2k-n+\alpha}(z)$$

Bessel Function of the 2nd Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}Y_\alpha(z) = \frac{\alpha}{2}Y_\alpha(z) - Y_{\alpha+1}(z)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}Y_\alpha(z) = 2^{-n}\sum_{k=0}^{n}(-1)^k\frac{n!}{k!(n-k)!}Y_{2k-n+\alpha}(z)$$

Modified Bessel Function of the 1st Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}I_\alpha(z) = \frac{\alpha}{2}I_\alpha(z) + I_{\alpha+1}(z)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}I_\alpha(z) = 2^{-n}\sum_{k=0}^{n}\frac{n!}{k!(n-k)!}I_{2k-n+\alpha}(z)$$

Modified Bessel Function of the 2nd Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}K_\alpha(z) = \frac{\alpha}{2}K_\alpha(z) - K_{\alpha+1}(z)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}K_\alpha(z) = (-1)^n 2^{-n}\sum_{k=0}^{n}\frac{n!}{k!(n-k)!}K_{2k-n+\alpha}(z)$$

Spherical Bessel Function of the 1st Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}j_\alpha(z) = \frac{\alpha}{2}j_\alpha(z) - j_{\alpha+1}(z)$$

Associated Legendre Polynomials:

$$\frac{\mathrm{d}}{\mathrm{d}z}P_n^\mu(z) = \frac{1}{z^2-1}(z\,n\,P_n^\mu(z) - (\mu+n)\,P_{n-1}^\mu(z))$$

Associated Laguerre Polynomials:

$$\frac{\mathrm{d}}{\mathrm{d}z}L_n^\lambda(z) = -L_{n-1}^{\lambda+1}(z)$$

$$\frac{\mathrm{d}^m}{\mathrm{d}z^m}L_n^\lambda(z) = (-1)^m L_{n-m}^{\lambda+m}(z)$$

Hermite Function:

$$\frac{\mathrm{d}}{\mathrm{d}z}H_\alpha(z) = 2\,\alpha\,H_{\alpha-1}(z)$$

Spherical Harmonics:

$$\frac{\partial}{\partial\theta}Y_n^m(\theta,\phi) = m\,\cot(\theta)Y_n^m(\theta,\phi) + \sqrt{(n-m)\,(n+m+1)}\frac{Y_n^{m+1}(\theta,\phi)}{e^{i\phi}}$$

$$\frac{\partial}{\partial\phi}Y_n^m(\theta,\phi) = i\,m\,Y_n^m(\theta,\phi)$$

$$\frac{\partial^p}{\partial\phi^p}Y_n^m(\theta,\phi) = (i\,m)^p\,Y_n^m(\theta,\phi)$$

Legendre Function of the 1st Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}P_\alpha(z) = \frac{\alpha}{z^2-1}(z\,P_\alpha(z) - P_{\alpha-1}(z))$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}P_\alpha(z) = (-1)^n(1-z^2)^{-n/2}P_\alpha^n(z)$$

Legendre Function of the 2nd Kind:

$$\frac{\mathrm{d}}{\mathrm{d}z}Q_\alpha(z) = \frac{\alpha}{z^2-1}(z\,Q_\alpha(z) - Q_{\alpha-1}(z))$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}Q_\alpha(z) = (-1)^n(1-z^2)^{-n/2}Q_\alpha^n(z)$$

Spherical Harmonic Function:

$$\frac{\partial}{\partial\theta}Y_\alpha^\mu(\theta,\phi) = \mu\,\cot(\theta)Y_\alpha^\mu(\theta,\phi) + \frac{\sqrt{\Gamma(\alpha-\mu+1)\,\Gamma(\alpha+\mu+2)}}{\sqrt{\Gamma(\alpha+\mu+1)\,\Gamma(\alpha-\mu)}}e^{-i\phi}Y_\alpha^{\mu+1}(\theta,\phi)$$

$$\frac{\partial}{\partial\phi}Y_\alpha^\mu(\theta,\phi) = i\,\mu\,Y_\alpha^\mu(\theta,\phi)$$

$$\frac{\partial^n}{\partial\phi^n}Y_\alpha^\mu(\theta,\phi) = (i\,\mu)^n\,Y_\alpha^\mu(\theta,\phi)$$

Associated Laguerre Function:

$$\frac{\mathrm{d}}{\mathrm{d}z}L_\alpha^\lambda(z) = -L_{\alpha-1}^{\lambda+1}(z)$$

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}L_\alpha^\lambda(z) = (-1)^n L_{\alpha-n}^{\lambda+n}(z)$$

Generalized Hypergeometric Function:

$$\frac{\partial}{\partial z}\,{}_pF_q(a_1,\ldots,a_p;b_1,\ldots,b_q;z) = \frac{\prod_{j=1}^p a_j}{\prod_{j=1}^q b_j}\,{}_pF_q(a_1+1,\ldots,a_p+1;b_1+1,\ldots,b_q+1;z)$$

$$\frac{\partial^n}{\partial z^n}\,{}_pF_q(a_1,\ldots,a_p;b_1,\ldots,b_q;z) = \frac{\prod_{j=1}^p \frac{\Gamma(a_j+n)}{\Gamma(a_j)}}{\prod_{j=1}^q \frac{\Gamma(b_j+n)}{\Gamma(b_j)}}\,{}_pF_q(a_1+n,\ldots,a_p+n;b_1+n,\ldots,b_q+n;z)\,,\ n\in\mathbb{N}$$

1.14 Common Integrals

In the following lists it should be understood that any formula for an indefinite integral should have a constant of integration added to the right-hand side. Some miscellaneous relations are:

$$\int \frac{f'(z)}{f(z)}\,\mathrm{d}z = \ln|f(z)|$$

$$\int f'(z)f(z)\,\mathrm{d}z = \frac{1}{2}[f(z)]^2$$

$$\int f'(z)e^{f(z)}\,\mathrm{d}z = e^{f(z)}$$

$$\int \frac{f'(z)}{\sqrt{f(z)}}\,\mathrm{d}z = 2\sqrt{f(z)}$$

1.14.1 Rational Functions

$$\int (az+b)^n \, \mathrm{d}z = \begin{cases} \dfrac{(az+b)^{n+1}}{a(n+1)} & n \neq -1 \\ \dfrac{1}{a} ln\,|az+b| & n = -1 \end{cases}$$

$$\int \frac{1}{z^2-a^2} \, \mathrm{d}z = \begin{cases} -\dfrac{1}{a}\mathrm{arctanh}\left(\dfrac{z}{a}\right) = \dfrac{1}{2a}\ln\left(\dfrac{a-z}{a+z}\right) & |z| < |a| \\ -\dfrac{1}{a}\mathrm{arccoth}\left(\dfrac{z}{a}\right) = \dfrac{1}{2a}\ln\left(\dfrac{z-a}{a+z}\right) & |z| > |a| \end{cases}$$

$$\int \frac{1}{z^2+z} \, \mathrm{d}z = \ln\left(\frac{z}{1+z}\right)$$

$$\int \frac{1}{z^2+a^2} \, \mathrm{d}z = \frac{1}{a}\arctan\left(\frac{z}{a}\right)$$

$$\int \frac{z}{z^2+a^2} \, \mathrm{d}z = \frac{1}{2}\ln(a^2+z^2)$$

$$\int \frac{z}{(az+b)^n} \, \mathrm{d}z = \frac{a(1-n)z-b}{a^2(n-1)\,(n-2)\,(az+b)^{n-1}}\,, \quad n \neq 1,2$$

$$\int \frac{z^2}{(az+b)^n} \, \mathrm{d}z = \frac{1}{a^3}\left(-\frac{(az+b)^{3-n}}{n-3} + \frac{2b(az+b)^{2-n}}{n-2} - \frac{b^2(az+b)^{1-n}}{n-1}\right), \quad n \neq 1,2,3$$

1.14.2 Irrational Functions

$$\int \sqrt{a^2+z^2} \, \mathrm{d}z = \frac{1}{2}\left(z\sqrt{a^2+z^2} + a^2\ln(z+\sqrt{a^2+z^2})\right)$$

$$\int \frac{z}{\sqrt{a^2+z^2}} \, \mathrm{d}z = \sqrt{a^2+z^2}$$

$$\int \frac{1}{\sqrt{a^2+z^2}} \, \mathrm{d}z = \mathrm{arcsinh}\left(\frac{z}{a}\right) = \ln\left(\frac{z+\sqrt{a^2+z^2}}{a}\right)$$

$$\int \frac{z}{\left(\sqrt{a^2+z^2}\right)^3} \, \mathrm{d}z = -\frac{1}{\sqrt{a^2+z^2}}$$

$$\int \frac{1}{\left(\sqrt{a^2+z^2}\right)^3} \, \mathrm{d}z = \frac{z}{a^2\sqrt{a^2+z^2}}$$

$$\int \frac{1}{z\sqrt{a^2+z^2}} \, \mathrm{d}z = -\frac{1}{a}\mathrm{arcsch}\left|\frac{z}{a}\right|, \quad |z| \leq |a|$$

$$\int \sqrt{a^2-z^2} \, \mathrm{d}z = \begin{cases} \dfrac{1}{2}\left(z\sqrt{a^2-z^2} + a^2\arcsin\left(\dfrac{z}{a}\right)\right) & |z| \leq |a| \\ \dfrac{1}{2}\left(z\sqrt{a^2-z^2} - \mathrm{sgn}(z)\mathrm{arcosh}\left|\dfrac{z}{a}\right|\right) & |z| \geq |a| \end{cases}$$

$$\int \frac{1}{\sqrt{a^2-z^2}} \, \mathrm{d}z = \arcsin\left(\frac{z}{a}\right), \quad |z| \leq |a|$$

$$\int \frac{z}{\sqrt{a^2-z^2}} \, \mathrm{d}z = -\sqrt{a^2-z^2}\,, \quad |z| \leq |a|$$

$$\int \frac{1}{\left(\sqrt{a^2-z^2}\right)^3} \, \mathrm{d}z = \frac{z}{a^2\sqrt{a^2-z^2}}\,, \quad |z| \leq |a|$$

$$\int \frac{z}{\left(\sqrt{a^2-z^2}\right)^3} \, \mathrm{d}z = \frac{1}{\sqrt{a^2-z^2}}\,, \quad |z| \leq |a|$$

$$\int \sqrt{\frac{z}{a-z}} \, \mathrm{d}z = \arctan\left(\sqrt{\frac{z}{a-z}}\right) a - \sqrt{z(a-z)}\,, \quad |z| \leq |a|$$

$$\int \frac{1}{z\sqrt{a^2-z^2}} \, \mathrm{d}z = -\frac{1}{a}\mathrm{arsech}\left(\frac{z}{a}\right), \quad |z| \leq |a|$$

$$\int \sqrt{z^2 - a^2}\,\mathrm{d}z = \frac{1}{2}\left(z\sqrt{z^2 - a^2} + a^2 \ln\left|z + \sqrt{z^2 - a^2}\right|\right)$$

$$\int \frac{1}{\left(\sqrt{z^2 - a^2}\right)^3}\,\mathrm{d}z = -\frac{z}{a^2\sqrt{z^2 - a^2}}$$

$$\int \frac{1}{\sqrt{z^2 - a^2}}\,\mathrm{d}z = \ln\left|\frac{z + \sqrt{z^2 - a^2}}{a}\right|$$

$$\int \frac{z}{\sqrt{z^2 - a^2}}\,\mathrm{d}z = \sqrt{z^2 - a^2}$$

$$\int \frac{z}{\sqrt{z^2 - a^2}}\,\mathrm{d}z = -\frac{1}{\sqrt{z^2 - a^2}}$$

$$\int \frac{1}{\left(\sqrt{az^2 + bz + c}\right)^3}\,\mathrm{d}z = \frac{4az + 2b}{(4ac - b^2)\sqrt{az^2 + bz + c}}$$

$$\int \frac{1}{z\sqrt{z^2 - 1}}\,\mathrm{d}z = \arccos\left(\frac{1}{z}\right)$$

$$\int \frac{z}{\left(\sqrt{az^2 + bz + c}\right)^3}\,\mathrm{d}z = -\frac{2bz + 4c}{(4ac - b^2)\sqrt{az^2 + bz + c}}$$

$$\int \sqrt{az^2 + bz + c}\,\mathrm{d}z = \frac{2az - 3b}{4a^2}\sqrt{az^2 + bz + c} + \frac{3b^2 - 4ac}{8a^2}\frac{1}{\sqrt{a}}\ln\left|2\sqrt{a}\sqrt{az^2 + bz + c} + 2az + b\right|\,,\quad a > 0$$

$$\int \frac{1}{\sqrt{az^2 + bz + c}}\,\mathrm{d}z = \frac{1}{\sqrt{a}}\ln\left|2\sqrt{a}\sqrt{az^2 + bz + c} + 2az + b\right|\,,\quad a > 0$$

$$\int \frac{z}{\sqrt{az^2 + bz + c}}\,\mathrm{d}z = \frac{1}{a}\sqrt{az^2 + bz + c} - \frac{b}{2a}\frac{1}{\sqrt{a}}\ln\left|2\sqrt{a}\sqrt{az^2 + bz + c} + 2az + b\right|\,,\quad a > 0$$

1.14.3 Trigonometric Functions

$$\int \sin(az)\,\mathrm{d}z = -\frac{1}{a}\cos(az) \qquad \int \cos(az)\,\mathrm{d}z = \frac{1}{a}\sin(az) \qquad \int \tan(az)\,\mathrm{d}z = -\frac{1}{a}\ln|\cos(az)|$$

$$\int \csc(az)\,\mathrm{d}z = -\frac{1}{a}\ln|\csc(az) + \cot(az)| \qquad \int \sec(az)\,\mathrm{d}z = \frac{1}{a}\ln\left|\tan\left(\frac{az}{2} + \frac{\pi}{4}\right)\right| \qquad \int \cot(az)\,\mathrm{d}z = \frac{1}{a}\ln|\sin(az)|$$

$$\int \sin^2(az)\,\mathrm{d}z = \frac{z}{2} - \frac{1}{4a}\sin(2az) \qquad \int \cos^2(az)\,\mathrm{d}z = \frac{z}{2} + \frac{1}{4a}\sin(2az) \qquad \int \tan^2(az)\,\mathrm{d}z = \frac{1}{a}\tan(az) - z$$

$$\int \csc^2(az)\,\mathrm{d}z = -\frac{1}{a}\cot(az) \qquad \int \sec^2(az)\,\mathrm{d}z = \frac{1}{a}\tan(az) \qquad \int \cot^2(az)\,\mathrm{d}z = -\frac{1}{a}\cot(az) - z$$

$$\int \sin^n(az)\,\mathrm{d}z = -\frac{1}{an}(\sin^{n-1}(az))\,(\cos(az)) + \frac{n-1}{n}\int \sin^{n-2}(az)\,\mathrm{d}z\,,\quad n > 0$$

$$\int \csc^n(az)\,\mathrm{d}z = \frac{\cos(az)\csc^{n-1}(az)}{a(1-n)} + \frac{n-2}{n-1}\int \csc^{n-2}(az)\,\mathrm{d}z\,,\quad n \neq 1$$

$$\int \cos^n(az)\,\mathrm{d}z = \frac{1}{an}(\cos^{n-1}(az))\,(\sin(az)) + \frac{n-1}{n}\int \cos^{n-2}(az)\,\mathrm{d}z\,,\quad n > 0$$

$$\int \sec^n(az)\,\mathrm{d}z = \frac{\sin(az)\sec^{n-1}(az)}{a(n-1)} + \frac{n-2}{n-1}\int \sec^{n-2}(az)\,\mathrm{d}z\,,\quad n > 1$$

$$\int \tan^n(az)\,\mathrm{d}z = \frac{1}{a(n-1)}\tan^{n-1}(az) - \int \tan^{n-2}(az)\,\mathrm{d}z\,, \quad n \neq 1$$

$$\int \cot^n(az)\,\mathrm{d}z = -\frac{1}{a(n-1)}\cot^{n-1}(az) - \int \cot^{n-2}(az)\,\mathrm{d}z\,, \quad n \neq 1$$

$$\int \frac{\cos^n(az)}{\sin^m(az)}\,\mathrm{d}z = -\frac{\cos^{n-1}(az)}{a(m-1)\sin^{m-1}(az)} - \frac{n-1}{m-1}\int \frac{\cos^{n-2}(az)}{\sin^{m-2}(az)}\,\mathrm{d}z\,, \quad m \neq 1$$

$$\int \frac{\cos^n(az)}{\sin(az)}\,\mathrm{d}z = -\frac{\cos^{n-1}(az)}{a(n-1)} - \int \frac{\cos^{n-2}(az)}{\sin(az)}\,\mathrm{d}z\,, \quad n \neq 1$$

$$\int \frac{\sin^n(az)}{\cos^m(az)}\,\mathrm{d}z = \frac{\sin^{n-1}(az)}{a(m-1)\cos^{m-1}(az)} - \frac{n-1}{m-1}\int \frac{\sin^{n-2}(az)}{\cos^{m-2}(az)}\,\mathrm{d}z\,, \quad m \neq 1$$

$$\int \frac{\sin^n(az)}{\cos(az)}\,\mathrm{d}z = -\frac{\sin^{n-1}(az)}{a(n-1)} + \int \frac{\sin^{n-2}(az)}{\cos(az)}\,\mathrm{d}z\,, \quad n \neq 1$$

$$\int z\sin(az)\,\mathrm{d}z = \frac{1}{a^2}\sin(az) - \frac{z}{a}\cos(az) \qquad \int z\cos(az)\,\mathrm{d}z = \frac{1}{a^2}\cos(az) + \frac{z}{a}\sin(az)$$

$$\int \frac{\sin(az)}{z}\,\mathrm{d}z = \sum_{n=0}^{\infty}(-1)^n\frac{(az)^{2n+1}}{(2n+1)\cdot(2n+1)!} \qquad \int \frac{\cos(az)}{z}\,\mathrm{d}z = \ln|az| + \sum_{n=1}^{\infty}(-1)^n\frac{(az)^{2n}}{(2n)\cdot(2n)!}$$

$$\int \sin(az)\sin(bz)\,\mathrm{d}z = \frac{\sin((b-a)z)}{2(b-a)} - \frac{\sin((b+a)z)}{2(b+a)}\,, \quad |a| \neq |b| \qquad \int \csc(az)\sec(az)\,\mathrm{d}z = \frac{1}{a}\ln|\tan(az)|$$

$$\int \cos(az)\cos(bz)\,\mathrm{d}z = \frac{\sin((b-a)z)}{2(b-a)} + \frac{\sin((b+a)z)}{2(b+a)}\,, \quad |a| \neq |b| \qquad \int \sec(az)\tan(az)\,\mathrm{d}z = \frac{1}{a}\sec(az)$$

$$\int \sin(az)\cos(bz)\,\mathrm{d}z = -\frac{\cos((a-b)z)}{2(a-b)} - \frac{\cos((b+a)z)}{2(b+a)}\,, \quad |a| \neq |b| \qquad \int \csc(az)\cot(az)\,\mathrm{d}z = -\frac{1}{a}\csc(az)$$

$$\int \sin(az)\tan(az)\,\mathrm{d}z = \frac{1}{a}(\ln|\sec(az)+\tan(az)| - \sin(az))$$

$$\int_{-c}^{c}\sin(z)\,\mathrm{d}z = 0 \qquad \int_0^{\pi}\sin(mz)\sin(nz)\,\mathrm{d}z = 0\,, \quad m \neq n$$

$$\int_{-c}^{c}\cos(z)\,\mathrm{d}z = 2\int_0^{c}\cos(z)\,\mathrm{d}z = 2\sin(c) \qquad \int_0^{\pi}\cos(mz)\cos(nz)\,\mathrm{d}z = 0\,, \quad m \neq n$$

$$\int_{-c}^{c}\tan(z)\,\mathrm{d}z = 0 \qquad \int_0^{2\pi}\sin^{2m+1}(z)\cos^{2n+1}(z)\,\mathrm{d}z = 0\,, \quad n, m \in \mathbb{Z}$$

1.14.4 Inverse Trigonometric Functions

$$\int \arcsin(az)\,\mathrm{d}z = z\arcsin(az) + \frac{\sqrt{1-a^2z^2}}{a} \qquad \int \arccos(az)\,\mathrm{d}z = z\arccos(az) - \frac{\sqrt{1-a^2z^2}}{a}$$

$$\int \operatorname{arccsc}(az)\,\mathrm{d}z = z\operatorname{arccsc}(az) + \frac{1}{a}\operatorname{artanh}\left(\sqrt{1-\frac{1}{a^2z^2}}\right) \qquad \int \operatorname{arcsec}(az)\,\mathrm{d}z = z\operatorname{arcsec}(az) - \frac{1}{a}\operatorname{arcosh}|az|$$

$$\int \arctan(az)\,\mathrm{d}z = z\arctan(az) - \frac{1}{2a}\ln(a^2z^2+1) \qquad \int \operatorname{arccot}(az)\,\mathrm{d}z = z\operatorname{arccot}(az) + \frac{1}{2a}\ln(a^2z^2+1)$$

$$\int \arcsin^n(az)\,\mathrm{d}z = z\arcsin^n(az) + \frac{n\sqrt{1-a^2z^2}\arcsin^{n-1}(az)}{a} - n(n-1)\int \arcsin^{n-2}(az)\mathrm{d}z$$

$$\int \arccos^n(az)\,\mathrm{d}z = z\arccos^n(az) - \frac{n\sqrt{1-a^2z^2}\arccos^{n-1}(az)}{a} - n(n-1)\int \arccos^{n-2}(az)\mathrm{d}z$$

1.14.5 Hyperbolic Functions

$$\int \sinh(az)\,\mathrm{d}z = \frac{1}{a}\cosh(az) \qquad \int \cosh(az)\,\mathrm{d}z = \frac{1}{a}\sinh(az) \qquad \int \tanh(az)\,\mathrm{d}z = \frac{1}{a}\ln(\cosh(az))$$

$$\int \sinh^2(az)\,\mathrm{d}z = \frac{1}{4a}\sinh(2az) - \frac{z}{2} \qquad \int \cosh^2(az)\,\mathrm{d}z = \frac{1}{4a}\sinh(2az) + \frac{z}{2} \qquad \int \tanh^2(az)\,\mathrm{d}z = z - \frac{1}{a}\tanh(az)$$

$$\int \operatorname{csch}(az)\,\mathrm{d}z = \frac{1}{a}\ln\left|\tanh\left(\frac{ax}{2}\right)\right| \qquad \int \operatorname{sech}(az)\,\mathrm{d}z = \frac{2}{a}\arctan(e^{az}) \qquad \int \coth(az)\,\mathrm{d}z = \frac{1}{a}\ln|\sinh(az)|\ ,$$

$$\int \sinh^n(az)\,\mathrm{d}z = \frac{1}{an}(\sinh^{n-1}(az))\,(\cosh(az)) - \frac{n-1}{n}\int \sinh^{n-2}(az)\,,\mathrm{d}z \quad n>0$$

$$\int \operatorname{csch}^n(az)\,\mathrm{d}z = \frac{\cosh(az)\operatorname{csch}^{n-1}(az)}{a(n-1)} - \frac{n-2}{n-1}\int \operatorname{csch}^{n-2}(az)\,\mathrm{d}z\,, \quad n\neq 1$$

$$\int \cosh^n(az)\,\mathrm{d}z = \frac{1}{an}(\cosh^{n-1}(az))\,(\sinh(az)) + \frac{n-1}{n}\int \cosh^{n-2}(az)\,\mathrm{d}z\,, \quad n>0$$

$$\int \operatorname{sech}^n(az)\,\mathrm{d}z = \frac{\sinh(az)\operatorname{sech}^{n-1}(az)}{a(n-1)} + \frac{n-2}{n-1}\int \operatorname{sech}^{n-2}(az)\,\mathrm{d}z\,, \quad n\neq 1$$

$$\int \tanh^n(az)\,\mathrm{d}z = -\frac{1}{a(n-1)}\tanh^{n-1}(az) + \int \tanh^{n-2}(az)\,\mathrm{d}z\,, \quad n\neq 1$$

$$\int \coth^n(az)\,\mathrm{d}z = -\frac{1}{a(n-1)}\coth^{n-1}(az) + \int \coth^{n-2}(az)\,\mathrm{d}z\,, \quad n\neq 1$$

$$\int \frac{\cosh^n(az)}{\sinh^m(az)}\,\mathrm{d}z = -\frac{\cosh^{n-1}(az)}{a(m-1)\sinh^{m-1}(az)} - \frac{n-1}{m-1}\int \frac{\cosh^{n-2}(az)}{\sinh^{m-2}(az)}\,\mathrm{d}z\,, \quad m \neq 1$$

$$\int \frac{\sinh^m(az)}{\cosh^n(az)}\,\mathrm{d}z = -\frac{\sinh^{m-1}(az)}{a(n-1)\cosh^{n-1}(az)} - \frac{m-1}{n-1}\int \frac{\sinh^{m-2}(az)}{\cosh^{n-2}(az)}\,\mathrm{d}z\,, \quad n \neq 1$$

$$\int z\,\sinh(az)\,\mathrm{d}z = \frac{1}{a}\cosh(az) - \frac{1}{a^2}\sinh(az) \qquad \int z\cosh(az)\,\mathrm{d}z = \frac{1}{a}\sinh(az) - \frac{1}{a^2}\cosh(az)$$

$$\int \sinh(az)\sinh(bz)\,\mathrm{d}z = \frac{1}{a^2-b^2}(a\sinh(bz)\cosh(az) - b\cosh(bz)\sinh(az))\,, \quad a^2 \neq b^2$$

$$\int \cosh(az)\cosh(bz)\,\mathrm{d}z = \frac{1}{a^2-b^2}(a\sinh(az)\cosh(bz) - b\cosh(az)\sinh(bz))\,, \quad a^2 \neq b^2$$

$$\int \cosh(az)\sinh(bz)\,\mathrm{d}z = \frac{1}{a^2-b^2}(a\sinh(az)\sinh(bz) - b\cosh(az)\cosh(bz))\,, \quad a^2 \neq b^2$$

1.14.6 Inverse Hyperbolic Functions

$$\int \operatorname{arsinh}(az)\,\mathrm{d}z = z\operatorname{arsinh}(az) - \frac{\sqrt{a^2z^2+1}}{a} \qquad \int \operatorname{arcsch}(az)\,\mathrm{d}z = z\operatorname{arcsch}(az) + \frac{1}{a}\operatorname{arcoth}\left(\sqrt{\frac{1}{a^2z^2}+1}\right)$$

$$\int \operatorname{arcosh}(az)\,\mathrm{d}z = z\operatorname{arcosh}(az) - \frac{\sqrt{(az+1)\,(az-1)}}{a} \qquad \int \operatorname{arsech}(az)\,\mathrm{d}z = z\operatorname{arsech}(az) - \frac{2}{a}\arctan\left(\sqrt{\frac{1-az}{1+az}}\right)$$

$$\int \operatorname{artanh}(az)\,\mathrm{d}z = z\operatorname{artanh}(az) + \frac{1}{2a}\ln(1-a^2z^2) \qquad \int \operatorname{arcoth}(az)\,\mathrm{d}z = z\operatorname{arcoth}(az) + \frac{1}{2a}\ln(a^2z^2-1)$$

$$\int \operatorname{arsinh}^n(az)\,\mathrm{d}z = z\operatorname{arsinh}^n(az) - \frac{n\sqrt{a^2z^2+1}\operatorname{arsinh}^{n-1}(az)}{a} + n(n-1)\int \operatorname{arsinh}^{n-2}(az)\mathrm{d}z$$

$$\int \operatorname{arcosh}^n(az)\,\mathrm{d}z = z\operatorname{arcosh}^n(az) - \frac{n\sqrt{(az+1)\,(az-1)}\operatorname{arcosh}^{n-1}(az)}{a} + n(n-1)\int \operatorname{arcosh}^{n-2}(az)\mathrm{d}z$$

1.14.7 Exponential Functions

A few special functions are used in this section, namely the error function:

$$\operatorname{erf}(z) = \frac{1}{\sqrt{\pi}}\int_{-z}^{z} e^{-t^2}\mathrm{d}t\,, \tag{29}$$

and the exponential integral:

$$\operatorname{Ei}(z) = -\int_{-z}^{\infty} \frac{e^{-t}}{t}\,\mathrm{d}t\,. \tag{30}$$

$$\int z^n e^{cz}\,\mathrm{d}z = \left(\frac{\partial}{\partial c}\right)^n \frac{e^{cz}}{c}$$

$$\int a^{cz}\,\mathrm{d}z = \frac{a^{cz}}{c\ln(a)}\,,\quad a>0\,;\ a\neq 1$$

$$\int e^{cz}\sin(bz)\,\mathrm{d}z = \frac{e^{cz}}{c^2+b^2}(c\sin(bz)-b\cos(bz))$$

$$\int e^{cz}\cos(bz)\,\mathrm{d}z = \frac{e^{cz}}{c^2+b^2}(c\cos(bz)-b\sin(bz))$$

$$\int e^{cz}\ln(z)\,\mathrm{d}z = \frac{1}{c}(e^{cz}\ln|z| - \mathrm{Ei}(cz))$$

$$\int \frac{1}{\sigma\sqrt{2\pi}} e^{-(\frac{z-\mu}{\sqrt{2}\sigma})^2}\,\mathrm{d}z = \frac{1}{2}\mathrm{erf}\left(\frac{z-\mu}{\sqrt{2}\sigma}\right)$$

$$\int e^{-az^2\pm bz\pm c}\,\mathrm{d}z = \frac{\sqrt{\pi}}{2\sqrt{a}} e^{\frac{b^2}{4a}\pm c}\,\mathrm{erf}\left(\frac{2az\mp b}{2\sqrt{a}}\right)\,,\quad a>0$$

$$\int_0^\infty e^{-z}\ln(z)\,\mathrm{d}z = -\gamma \quad \text{(Euler-Mascheroni Constant)}$$

$$\int_0^\infty \frac{\sin(mz)}{e^{2\pi z}-1}\,\mathrm{d}z = \frac{1}{4}\coth\left(\frac{m}{2}\right) - \frac{1}{2m}$$

1.14.8 Logarithmic Functions

$$\int \log_a(z)\,\mathrm{d}z = \frac{z\ln(z)-z}{\ln(a)}$$

$$\int \log(az)\,\mathrm{d}z = z\ln(az) - z$$

$$\int \log(az+b)\,\mathrm{d}z = \frac{(az+b)\ln(az+b)-(az+b)}{a}$$

$$\int \ln(z^2+a^2)\,\mathrm{d}z = z\ln(z^2+a^2) - 2z + 2\arctan\left(\frac{z}{a}\right)a$$

$$\int \frac{1}{\ln(z)}\,\mathrm{d}z = \ln|\ln(z)| + \ln(z) + \sum_{k=2}^{\infty}\frac{(\ln(z))^k}{k\cdot k!}$$

$$\int \frac{1}{z^n\ln(z)}\,\mathrm{d}z = \ln|\ln(z)| + \sum_{k=1}^{\infty}(-1)^k\frac{(n-1)^k(\ln(z))^k}{k\cdot k!}$$

$$\int (\ln(z))^n\,\mathrm{d}z = z\sum_{k=0}^{n}(-1)^{n-k}(\ln(z))^k\frac{n!}{k!}$$

$$\int z^m\ln(z)\,\mathrm{d}z = z^{m+1}\left(\frac{\ln(z)}{m+1} - \frac{1}{(m+1)^2}\right)\,,\quad m\neq -1$$

$$\int \sin(\ln(z))\,\mathrm{d}z = \frac{z}{2}(\sin(\ln(z)) - \cos(\ln(z)))$$

$$\int \cos(\ln(z))\,\mathrm{d}z = \frac{z}{2}(\sin(\ln(z)) + \cos(\ln(z)))$$

$$\int e^z\left(z\ln(z) - z - \frac{1}{z}\right)\mathrm{d}z = e^z(z\ln(z) - z - \ln(z))$$

$$\int e^{-z}\left(\frac{1}{z} - \ln(z)\right)\mathrm{d}z = e^{-z}\ln(z)$$

$$\int e^z\left(\frac{1}{\ln(z)} - \frac{1}{z\,(\ln(z))^2}\right)\mathrm{d}z = \frac{e^z}{\ln(z)}$$

1.14.9 Special Functions

Integral Representations:

Heaviside Step Function:

$$\theta(z) = \frac{1}{2\pi i}\lim_{\varepsilon\to 0}\int_{-\infty}^{\infty}\frac{e^{itz}}{t-i\varepsilon}\,\mathrm{d}t$$

Dirac Delta Function:

$$\delta(z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{itz}\,\mathrm{d}t$$

Gamma Function:

$$\Gamma(z) = \int_0^{\infty} t^{z-1}e^{-t}\,\mathrm{d}t\,,\quad \mathrm{Re}(z)\notin\mathbb{Z}^-$$

Incomplete Gamma Function:

$$\Gamma(a,z) = \int_z^{\infty} t^{a-1}e^{-t}\,\mathrm{d}t$$

Digamma Function:

$$\psi(z) = \int_0^1 \frac{1-t^{z-1}}{1-t}\,\mathrm{d}t\ -\gamma\,,\quad \mathrm{Re}(z)>0$$

Error Function:

$$\mathrm{erf}(z) = \frac{1}{\sqrt{\pi}}\int_{-z}^{z} e^{-t^2}\mathrm{d}t$$

Polygamma Function:

$$\psi^{\nu}(z) = \frac{1}{\Gamma(-\nu)}\int_0^z \psi(t+1)\,(z-t)^{-(\nu+1)}\,\mathrm{d}t\ -\frac{z^{-(\nu+1)}}{\Gamma(-\nu)}\left(\ln(z)-\psi(-\nu)-\gamma\right),\quad \mathrm{Re}(\nu)<0$$

Beta Function:

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t\,,\quad \mathrm{Re}(a),\ \mathrm{Re}(b)>0$$

Incomplete Beta Function:

$$B_z(a,b) = \int_0^z t^{a-1}(1-t)^{b-1}\,\mathrm{d}t\,,\quad \mathrm{Re}(a),\ \mathrm{Re}(b)>0$$

Polylogarithm:

$$\mathrm{Li}_{\nu}(z) = \frac{z}{\Gamma(\nu)}\int_0^{\infty}\frac{t^{\nu-1}}{e^t-z}\,\mathrm{d}t\,,\quad \mathrm{Re}(\nu)>0$$

Riemann Zeta Function:

$$\zeta(z) = \frac{1}{\Gamma(z)}\int_0^{\infty}\frac{t^{z-1}}{e^t-z}\,\mathrm{d}t\,,\quad \mathrm{Re}(z)>1$$

Cosine Integral:

$$\mathrm{Ci}(z) = -\int_z^{\infty}\frac{\cos(t)}{t}\,\mathrm{d}t$$

Sine Integral:

$$\mathrm{Si}(z) = \int_0^z \frac{\sin(t)}{t}\,\mathrm{d}t$$

Fresnel Sine Integral:

$$S(z) = \int_0^z \sin\left(\frac{\pi t^2}{2}\right)\mathrm{d}t$$

Fresnel Cosine Integral:

$$C(z) = \int_0^z \cos\left(\frac{\pi t^2}{2}\right)\mathrm{d}t$$

Hyperbolic Cosine Integral:

$$\mathrm{Chi}(z) = \int_0^z \frac{\cosh(t)-1}{t}\,\mathrm{d}t\ +\ln(z)+\gamma$$

Hyperbolic Sine Integral:

$$\mathrm{Shi}(z) = \int_0^z \frac{\sinh(t)}{t}\,\mathrm{d}t$$

Exponential Integral:

$$\mathrm{Ei}(z) = -\int_{-z}^{\infty}\frac{e^{-t}}{t}\,\mathrm{d}t$$

Logarithmic Integral:

$$\mathrm{li}(z) = \int_0^z \frac{1}{\ln(t)}\,\mathrm{d}t$$

Poly Exponential Integral:

$$E_{\nu}(z) = z^{\nu-1}\int_z^{\infty} t^{-\nu}e^{-t}\,\mathrm{d}t\,,\quad |\arg(z)|<\pi$$

Hermite Polynomials:

$$H_n(z) = \frac{2^n}{\sqrt{\pi}}\int_{-\infty}^{\infty} e^{-t^2}(z+it)^n\,\mathrm{d}t$$

Complete Elliptic Integral of the 1st Kind:

$$E(z) = \int_0^{\frac{\pi}{2}}\sqrt{1-z\sin^2(t)}\,\mathrm{d}t = \int_0^1\sqrt{\frac{1-z\,t^2}{1-t^2}}\,\mathrm{d}t\,,\quad |\arg(1-z)|<\pi$$

Complete Elliptic Integral of the 2nd Kind:

$$K(z) = \int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{1-z\sin^2(t)}}\,\mathrm{d}t = \int_0^1\frac{1}{\sqrt{(1-z\,t^2)\,(1-t^2)}}\,\mathrm{d}t\,,\quad |\arg(1-z)|<\pi$$

Complete Elliptic Integral of the 3rd Kind:

$$\Pi(n\,|\,m) = \int_0^{\frac{\pi}{2}} \frac{1}{\left(1 - n\sin^2(t)\right)\sqrt{1 - m\sin^2(t)}}\,\mathrm{d}t = \int_0^1 \frac{1}{\left(1 - n\,t^2\right)\sqrt{(1 - m\,t^2)\,(1 - t^2)}}\,\mathrm{d}t$$

Incomplete Elliptic Integral of the 1st Kind:

$$E(z\,|\,m) = \int_0^z \sqrt{1 - m\sin^2(t)}\,\mathrm{d}t = \int_0^{\sin(z)} \sqrt{\frac{1 - m\,t^2}{1 - t^2}}\,\mathrm{d}t\,,\quad -\frac{\pi}{2} \le z \le \frac{\pi}{2}$$

Incomplete Elliptic Integral of the 2nd Kind:

$$F(z\,|\,m) = \int_0^z \frac{1}{\sqrt{1 - m\sin^2(t)}}\,\mathrm{d}t = \int_0^{\sin(z)} \frac{1}{\sqrt{(1 - m\,t^2)\,(1 - t^2)}}\,\mathrm{d}t\,,\quad -\frac{\pi}{2} \le z \le \frac{\pi}{2}$$

Incomplete Elliptic Integral of the 3rd Kind:

$$\Pi(n;\,z\,|\,m) = \int_0^z \frac{1}{\left(1 - n\sin^2(t)\right)\sqrt{1 - m\sin^2(t)}}\,\mathrm{d}t = \int_0^{\sin(z)} \frac{1}{\left(1 - n\,t^2\right)\sqrt{(1 - m\,t^2)\,(1 - t^2)}}\,\mathrm{d}t\,,\quad -\frac{\pi}{2} \le z \le \frac{\pi}{2}$$

Bessel Function of the 1st Kind:

$$J_\nu(z) = 2\left(\frac{z}{2}\right)^\nu \int_0^1 \frac{(1 - t^2)^{\nu - \frac{1}{2}}}{\sqrt{\pi}\,\Gamma(\nu + \frac{1}{2})}\cos(zt)\,\mathrm{d}t\,,\quad \mathrm{Re}(\nu) > -\frac{1}{2}$$

Bessel Function of the 2nd Kind:

$$Y_\nu(z) = \frac{2^{1+\nu} z^{-\nu}}{\sqrt{\pi}\,\Gamma(\frac{1}{2} - \nu)} \int_1^\infty \frac{\cos(zt)}{(t^2 - 1)^{\nu + \frac{1}{2}}}\,\mathrm{d}t\,,\quad |\mathrm{Re}(\nu)| < \frac{1}{2},\quad z \ne 0$$

Modified Bessel Function of the 1st Kind:

$$I_\nu(z) = 2\left(\frac{z}{2}\right)^\nu \int_0^1 \frac{(1 - t^2)^{\nu - \frac{1}{2}}}{\sqrt{\pi}\,\Gamma(\nu + \frac{1}{2})}\cosh(zt)\,\mathrm{d}t\,,\quad \mathrm{Re}(\nu) > -\frac{1}{2}$$

Modified Bessel Function of the 2nd Kind:

$$K_\nu(z) = \int_0^\infty e^{-z\cosh(t)}\cosh(\nu t)\,\mathrm{d}t\,,\quad \mathrm{Re(z)} > 0$$

Spherical Bessel Function of the 1st Kind:

$$j_\nu(z) = \int_0^1 \frac{\cos(zt)}{\Gamma(\nu + 1)}\left(\frac{z}{2(t^2 - 1)}\right)^\nu \mathrm{d}t\,,\quad \mathrm{Re}(\nu) > -1$$

Spherical Bessel Function of the 2nd Kind:

$$y_\nu(z) = \int_1^\infty \frac{\cos(zt)}{\Gamma(-\nu)}\left(\frac{2}{z(t^2 - 1)}\right)^{\nu + 1} \mathrm{d}t\,,\quad |\mathrm{Re}(\nu)| < 1,\quad z \ne 0$$

Laguerre Polynomials:

$$L_n(z) = \frac{1}{2\pi}\int_0^{2\pi} e^{-ze^{it}}\,(1 + e^{-it})^n\,\mathrm{d}t$$

Associated Laguerre Polynomials:

$$L_n^\lambda(z) = \frac{1}{\Gamma(n + 1)} z^{-\frac{\lambda}{2}} e^z \int_0^\infty e^{-t} t^{n + \frac{\lambda}{2}}\,J_\lambda(2\sqrt{tz})\,\mathrm{d}t\,,\quad \mathrm{Re}(n + \lambda) > -1$$

Legendre Polynomials of the 1st Kind:

$$P_n(z) = \frac{1}{\pi}\int_0^\pi \left(z - \sqrt{z^2 - 1}\,\cos(t)\right)^n \mathrm{d}t = \frac{2^n}{\pi}\int_{-\infty}^\infty \frac{(it + z)^n}{(t^2 + 1)^{n+1}}\,\mathrm{d}t$$

Legendre Polynomials of the 2nd Kind:

$$Q_n(z) = \frac{1}{2^{\nu+1}\nu!}\int_{-1}^1 \frac{1}{z - t}\frac{\partial^\nu (t^2 - 1)^\nu}{\partial t^\nu}\,\mathrm{d}t$$

Associated Legendre Polynomials of the 1st Kind:

$$P_n^{\mu}(z) = \frac{1}{\Gamma(-\mu)} \left(1 - z^2\right)^{\frac{\mu}{2}} \int_z^1 P_n(t)\,(t-z)^{-(\mu+1)}\,\mathrm{d}t\,, \quad \mathrm{Re}(\mu) < 0$$

Associated Legendre Polynomials of the 2nd Kind:

$$Q_n^{\mu}(z) = \frac{\Gamma(n+1)}{2\,\Gamma(n-\mu+1)} \int_0^{\infty} \left(e^{-\frac{\mu i \pi}{2}} \left(z + \sqrt{z^2-1}\cosh(t)\right)^{-(n+1)} + e^{\frac{\mu i \pi}{2}} \left(z - \sqrt{z^2-1}\cosh(t)\right)^{-(n+1)}\right) \cosh(\mu t)\,\mathrm{d}t$$

$$-1 < z < 1\,;\ \mathrm{Re}(n+\mu) > -1\,;\ \mathrm{Re}(n-\mu) > -1$$

Spherical Harmonics:

$$Y_n^m(\theta,\phi) = \frac{1}{4\pi^{3/2} i^{3m} n!} \sqrt{(2n+1)\,(n-m)!\,(n+m)!} \int_0^{2\pi} (\cos(\theta) + i\cos(t-\phi)\sin(\theta))^n e^{imt}\,\mathrm{d}t$$

Chebyshev Polynomials of the 1st Kind:

$$T_n(z) = \frac{1}{2} \int_0^{\pi} \left(z + \sqrt{z^2-1}\cos(t)\right)^{n-1} \left(z + (1+n)\sqrt{z^2-1}\cos(t)\right) \sin(t)\,\mathrm{d}t$$

Chebyshev Polynomials of the 2nd Kind:

$$U_n(z) = \frac{n+1}{2} \int_0^{\pi} \left(z + \sqrt{z^2-1}\cos(t)\right)^{n} \sin(t)\,\mathrm{d}t\,, \quad z \notin (-\infty, -1)$$

Hermite Functions:

$$H_\nu(z) = \frac{2^{\nu+1}}{\sqrt{\pi}} e^{z^2} \int_0^{\infty} e^{-t^2} t^{\nu} \cos\left(2zt - \frac{\pi\nu}{2}\right)\,\mathrm{d}t\,, \quad \mathrm{Re}(\nu) > -1$$

Laguerre Functions:

$$L_\nu(z) = -\frac{\sin(\nu\pi)}{\pi} \int_0^1 e^{-t}(1-t)^{\nu}\,\mathrm{d}t\,, \quad -1 < \mathrm{Re}(\nu) < 0$$

Associated Laguerre Functions:

$$L_\nu^{\lambda}(z) = -\int_0^1 \frac{\sin(\nu\pi)(1-t)^{\nu+\lambda}}{\pi e^t t^{\nu+1}}\,\mathrm{d}t\,, \quad \begin{array}{c}\mathrm{Re}(\nu+\lambda) > -1\,,\\ \mathrm{Re}(\nu) < 0\end{array}$$

Legendre Functions of the 1st Kind:

$$P_\nu(z) = \frac{1}{\pi} \int_0^{\pi} \left(z + \sqrt{z^2-1}\,\cos(t)\right)^{-(\nu+1)}\,\mathrm{d}t\,, \quad \mathrm{Re}(z) > 0$$

Associated Legendre Functions of the 1st Kind:

$$P_\nu^{\mu}(z) = \int_z^1 \frac{P_\nu(t)\left(1-z^2\right)^{\frac{\mu}{2}}}{\Gamma(-\mu)(t-z)^{\mu+1}}\,\mathrm{d}t\,, \quad \mathrm{Re}(\mu) < 0$$

Legendre Functions of the 2nd Kind:

$$Q_\nu(z) = \int_0^{\infty} \left(z + \sqrt{z^2-1}\cosh(t)\right)^{-(1+\nu)}\,\mathrm{d}t + \frac{i\pi}{2} P_\nu(z)\,, \quad \mathrm{Re}(z) > 1\,;\ \mathrm{Re}(\nu) > -1$$

Associated Legendre Functions of the 2nd Kind:

$$Q_\nu^\mu(z) = \frac{\Gamma(\nu+1)}{2\,\Gamma(\nu-\mu+1)} \int_0^\infty \left(e^{-\frac{\mu i \pi}{2}} \left(z + \sqrt{z^2-1}\cosh(t)\right)^{-(\nu+1)} + e^{\frac{\mu i \pi}{2}} \left(z - \sqrt{z^2-1}\cosh(t)\right)^{-(\nu+1)} \right) \cosh(\mu t)\, \mathrm{d}t$$

$$-1 < z < 1\ ;\ \mathrm{Re}(\mu+\nu) > -1\ ;\ \mathrm{Re}(\nu-\mu) > -1$$

Chebyshev Functions of the 1st Kind:

$$T_\nu(z) = \frac{1}{2} \int_0^\pi \left(z + \sqrt{z^2-1}\cos(t)\right)^{\nu-1} \left(z + (1+\nu)\sqrt{z^2-1}\cos(t)\right) \sin(t)\, \mathrm{d}t$$

Chebyshev Functions of the 2nd Kind:

$$U_\nu(z) = \frac{\nu+1}{2} \int_0^\pi \left(z + \sqrt{z^2-1}\cos(t)\right)^{\nu} \sin(t)\, \mathrm{d}t\ ,\quad z \notin (-\infty, -1)$$

Hypergeometric Functions:

$${}_pF_q(a_1,\ldots,a_p;b_1,\ldots,b_q;z) = \frac{\Gamma(b_q)}{\Gamma(a_p)\,\Gamma(b_q-a_p)} \int_0^1 t^{a_p-1}(1-t)^{b_q-a_p-1}\, {}_{p-1}F_{q-1}(a_1,\ldots,a_{p-1};b_1,\ldots,b_{q-1};zt)$$

$\mathrm{Re}(b_q) > \mathrm{Re}(a_p) > 0$ for $p \le q$; $\mathrm{Re}(b_q) > \mathrm{Re}(a_p) > 0$ for $p = q+1$ and $|\arg(1-z)| < \pi$

Integrals:

Integrals involving Dirac Delta and Heaviside Step Functions:

$$\int \theta(z)\, \mathrm{d}z = z\,\theta(z) \qquad \int f(z)\,\delta(z-a)\, \mathrm{d}z = f(a) \qquad \int f(z)\,\delta^{(n)}(z-a)\, \mathrm{d}z = (-1)^n f^{(n)}(a)$$

$$\int \delta(z)\, \mathrm{d}z = \theta(z) \qquad \int f(z)\,\delta'(z-a)\, \mathrm{d}z = -f'(a)$$

Integrals involving Sine and Cosine, and Fresnel Sine and Cosine Integral Functions:

$$\int \mathrm{Ci}(b+az)\, \mathrm{d}z = \frac{(b+az)\,\mathrm{Ci}(b+az) - \sin(b+az)}{a} \qquad \int_0^\infty t^{\alpha-1}\,\mathrm{Si}(t)\, \mathrm{d}t = -\frac{\Gamma(\alpha)}{\alpha}\sin\left(\frac{\pi\alpha}{2}\right)\ ,\quad -1 < \mathrm{Re}(\alpha) < 0$$

$$\int \mathrm{Si}(b+az)\, \mathrm{d}z = \frac{(b+az)\,\mathrm{Si}(b+az) + \cos(b+az)}{a} \qquad \int_0^\infty e^{-tz}\,\mathrm{Ci}(t)\, \mathrm{d}t = -\frac{\ln(z^2+1)}{2z}\ ,\quad \mathrm{Re}(z) > 0$$

$$\int z^{\alpha-1}\mathrm{Ci}(az)\, \mathrm{d}z = \frac{z^\alpha\left(\Gamma(\alpha,\,-iaz)\,(-iaz)^{-\alpha} + 2\,\mathrm{Ci}(az) + (iaz)^{-\alpha}\,\Gamma(\alpha,\,iaz)\right)}{2\alpha}$$

$$\int z^{\alpha-1}\mathrm{Si}(az)\, \mathrm{d}z = \frac{z^\alpha\left(-i\Gamma(\alpha,\,-iaz)\,(-iaz)^{-\alpha} + 2\,\mathrm{Si}(az) + i(iaz)^{-\alpha}\,\Gamma(\alpha,\,iaz)\right)}{2\alpha}$$

Integrals involving Hyperbolic Sine and Cosine Integral Functions:

$$\int \mathrm{Shi}(b+az)\,\mathrm{d}z = \frac{(b+az)\,\mathrm{Shi}(b+az)+\cosh(b+az)}{a} \qquad \int_0^\infty e^{-tz}\,\mathrm{Chi}(t)\,\mathrm{d}t = -\frac{\ln(z^2-1)}{2z}\,,\ \ \mathrm{Re}(z)>1$$

$$\int \mathrm{Chi}(b+az)\,\mathrm{d}z = \frac{(b+az)\,\mathrm{Chi}(b+az)-\sinh(b+az)}{a}$$

$$\int z^{\alpha-1}\mathrm{Chi}(az)\,\mathrm{d}z = \frac{z^\alpha(\Gamma(\alpha,\,-az)\,(-az)^{-\alpha}+2\,\mathrm{Chi}(az)+(az)^{-\alpha}\,\Gamma(\alpha,\,az))}{2\alpha}$$

$$\int z^{\alpha-1}\mathrm{Shi}(az)\,\mathrm{d}z = \frac{z^\alpha(\Gamma(\alpha,\,-az)\,(-az)^{-\alpha}+2\,\mathrm{Shi}(az)-(az)^{-\alpha}\,\Gamma(\alpha,\,az))}{2\alpha}$$

$$\int e^{bz}\mathrm{Shi}(az)\,\mathrm{d}z = \frac{\mathrm{Ei}((b-a)z)-\mathrm{Ei}\,((a+b)z)+2e^{bz}\,\mathrm{Shi}(az)}{2b}$$

Integrals involving Exponential and Logarithmic Integral Functions:

$$\int \mathrm{Ei}(b+az)\,\mathrm{d}z = \frac{(b+az)\,\mathrm{Ei}(b+az)-e^{b+az}}{a} \qquad \int z^{\alpha-1}\mathrm{Ei}(az)\,\mathrm{d}z = \frac{z^\alpha(\Gamma(\alpha,\,-az)\,(-az)^{-\alpha}+\alpha\,\mathrm{Ei}(az))}{\alpha}$$

$$\int e^{bz}\,\mathrm{Ei}(az)\,\mathrm{d}z = \frac{e^{bz}\,\mathrm{Ei}(az)-\mathrm{Ei}(z(a+b))}{b} \qquad \int_0^\infty e^{-tz}\mathrm{Ei}(t)\,\mathrm{d}t = -\frac{\ln(z-1)}{z}\,,\ \ \mathrm{Re}(z)>1$$

$$\int \ln(bz)\,\mathrm{Ei}(az)\,\mathrm{d}z = \frac{\mathrm{Ei}(az)\,(1+az\ln(bz)-az)-e^{az}\,(\ln(bz)-1)}{a}$$

$$\int \mathrm{li}(az)\,\mathrm{d}z = z\,\mathrm{li}(az)-\frac{\mathrm{Ei}(2\ln(az))}{a} \qquad \int z^{\alpha-1}\,\mathrm{li}(az)\,\mathrm{d}z = \frac{(az)^\alpha\,\mathrm{li}(az)-\mathrm{Ei}(\alpha+1)\ln(az)}{a^\alpha\,\alpha}$$

$$\int \frac{\mathrm{li}(z)}{z}\,\mathrm{d}z = \ln(z)\,\mathrm{li}(z)-z \qquad \int_0^\infty t^{\alpha-1}\,\mathrm{li}(t)\,\mathrm{d}t = \frac{\ln(-\alpha-1)}{\alpha}\,,\ \ \mathrm{Re}(\alpha)<-1$$

Integrals involving Poly Exponential, Polylogarithm, and Zeta Functions:

$$\int E_\nu(az)\,\mathrm{d}z = -\frac{E_{\nu+1}(az)}{a} \qquad \int z^{\alpha-1}E_\nu(az)\,\mathrm{d}z = (E_\nu(az)-E_{1-\alpha}(az))\frac{z^\alpha}{\alpha+\nu-1}$$

$$\int \mathrm{Li}_\nu(z)\,\mathrm{d}z = \sum_{k=1}^\infty \frac{z^{k+1}}{(k+1)\,k^\nu}\,,\ \ |z|<1 \qquad \int_0^z \frac{1}{t}\mathrm{Li}_\nu(t)\,\mathrm{d}t = \mathrm{Li}_{\nu+1}(z)$$

$$\int \zeta(z)\,\mathrm{d}z = z-\sum_{k=2}^\infty \frac{k^{-z}}{\ln(k)}\,,\ \ \mathrm{Re}(z)>1$$

Integrals involving Elliptic Integral Functions:

$$\int E(az)\,\mathrm{d}z = \frac{2}{3a}((az+1)\,E(az)+(az-1)\,K(az)) \qquad \int K(az)\,\mathrm{d}z = \frac{2}{a}(E(az)+(az-1)\,K(az))$$

Integrals involving Gamma, Beta, and Error Functions:

$$\int \operatorname{erf}(b+az)\,\mathrm{d}z = \frac{1}{a\sqrt{\pi}}\,e^{-(b+az)^2} + \frac{(b+az)\operatorname{erf}(b+az)}{a} \qquad \int z^{\alpha-1}e^{bz^2}\operatorname{erf}(az)\,\mathrm{d}z = \frac{\sqrt{\pi}[\operatorname{erf}(az)]^2}{4a}$$

$$\int z^{\alpha-1}\operatorname{erf}(az)\,\mathrm{d}z = \frac{1}{\alpha\, a^{\alpha}\sqrt{\pi}}\,\Gamma\left(\frac{\alpha+1}{2},\, a^2z^2\right) + \frac{z^{\alpha}\operatorname{erf}(az)}{\alpha}$$

$$\int e^{-a^2z^2}\operatorname{erf}(az)\,\mathrm{d}z = -\frac{a}{\sqrt{\pi}(-b)^{\frac{\alpha+1}{2}}}\sum_{k=0}^{\infty}\frac{b^{-k}a^{2k}}{(2k+1)\,k!}\Gamma\left(\frac{\alpha+1}{2}+k,\, -bz^2\right)$$

Integrals involving Bessel-Type Functions:

$$\int_0^{\infty} J_\nu(z)\,\mathrm{d}z = 1\,, \quad \operatorname{Re}(\nu) > 0 \qquad \int z^{\nu+1}J_\nu(z)\,\mathrm{d}z = \frac{e^{-iaz}z^{\nu+1}}{2\nu+1}(iJ_{\nu+1}(az) + J_\nu(z))$$

$$\int z^{\nu}e^{-iaz}J_\nu(az)\,\mathrm{d}z = z^{\nu+1}J_{\nu+1}(z) \qquad \int z^{\nu+1}I_\nu(z)\,\mathrm{d}z = z^{\nu+1}I_{\nu+1}(z)$$

Integrals involving Hermite and Laguerre Functions:

$$\int H_\nu(z)\,\mathrm{d}z = \frac{1}{2\,(\nu+1)}H_{\nu+1}(z) \qquad \int_{-\infty}^{\infty} e^{-t^2}\,H_m(t)\,H_n(t)\,\mathrm{d}t = 2^n n!\sqrt{\pi}\delta_{n,m}$$

$$\int L_\nu(z)\,\mathrm{d}z = L_\nu(z) - L_{\nu+1}(z) \qquad \int_0^{\infty} e^{-t}\,L_m(t)\,L_n(t)\,\mathrm{d}t = \delta_{n,m}\,, \quad n,m\in\mathbb{N}$$

$$\int L_\nu^{\lambda}(z)\,\mathrm{d}z = -L_{\nu+1}^{\lambda-1}(z) \qquad \int_0^{\infty} e^{-t}\,t^{\lambda}\,L_m^{\lambda}(t)\,L_n^{\lambda}(t)\,\mathrm{d}t = \frac{\Gamma(\lambda+n+1)}{n!}\delta_{n,m}\,, \quad n,m\in\mathbb{N}\,;\ \operatorname{Re}(\lambda) > -1$$

Integrals involving Legendre Functions:

$$\int P_\nu(z)\,\mathrm{d}z = \frac{P_{\nu+1}(z) - P_{\nu-1}(z)}{2\nu+1} \qquad \int_{-1}^{1} P_n(z)\,P_l(z)\,\mathrm{d}z = \frac{2\,\delta_{n,l}}{(2n+1)}$$

$$\int_{-1}^{1} P_n^m(z)\,P_l^m(z)\,\mathrm{d}z = \frac{2\,(m+n)!\,\delta_{n,l}}{(2n+1)\,(n-m)!}\,, \quad n,l,m\in\mathbb{N}$$

Integrals involving Chebyshev Functions:

$$\int T_\nu(az)\,\mathrm{d}z = \frac{1}{2a}\left(\frac{T_{\nu-1}(az)}{1-\nu} + \frac{T_{\nu+1}(az)}{1+\nu}\right)\,, \quad \nu\neq 1 \qquad \int_{-1}^{1}\frac{T_m(z)\,T_n(z)}{\sqrt{1-z^2}}\,\mathrm{d}z = \frac{\pi}{2}\delta_{m,n}\,, \quad m^2+n^2\neq 0$$

$$\int U_\nu(az)\,\mathrm{d}z = \frac{1}{a(\nu+1)}T_{\nu+1}(az) \qquad \int_{-1}^{1} U_m(z)\,U_n(z)\sqrt{1-z^2}\,\mathrm{d}z = \frac{\pi}{2}\delta_{m,n}$$

Integrals involving Spherical Harmonics Functions:

$$\int Y_\lambda^\mu(\theta,\phi)\,\mathrm{d}\phi = -\frac{i}{\mu}Y_\lambda^\mu(\theta,\phi) \qquad \int_0^\pi \csc(\theta)\,Y_\lambda^{\mu_1}(\theta,\phi)\,\bar{Y}_\lambda^{\mu_2}(\theta,\phi)\,\mathrm{d}\theta = \frac{2\lambda+1}{4\pi\mu_1}\delta_{\mu_1,\mu_2}\,,\quad \lambda,\mu_1,\mu_2\in\mathbb{N}\,;\ \mu_1,\mu_2\le\lambda$$

$$\int_0^\pi \sin(\theta)\,Y_{\lambda_1}^\mu(\theta,\phi)\,Y_{\lambda_2}^\mu(\theta,\phi)\,\mathrm{d}\theta = \frac{e^{2i\mu\phi}}{2\pi}\delta_{\lambda_1,\lambda_2}\,,\quad \mu,\lambda_1,\lambda_2\in\mathbb{N}$$

$$\int_0^{2\pi}\int_0^\pi \sin(\theta)\,Y_{\lambda_1}^{\mu_1}(\theta,\phi)\,\bar{Y}_{\lambda_2}^{\mu_2}(\theta,\phi)\,\mathrm{d}\theta\,\mathrm{d}\phi = \delta_{\mu_1,\mu_2}\delta_{\lambda_1,\lambda_2}\,,\quad \lambda_1,\lambda_2,\mu_1,\mu_2\in\mathbb{Z}$$

Integrals involving Hypergeometric Functions:

$$\int {}_pF_q(a_1,\ldots,a_p;b_1,\ldots,b_q;z)\,\mathrm{d}z = \frac{\prod_{j=1}^q(b_j-1)}{\prod_{j=1}^p(a_j-1)}\,{}_pF_q(a_1-1,\ldots,a_p-1;b_1-1,\ldots,b_q-1;z)$$

1.14.10 A Master Formula

$$M_4(a;r,s) = \int_0^\infty \left(\frac{z^2}{z^4+2az^2+1}\right)^r \frac{z^2+1}{z^2(z^s+1)}\,\mathrm{d}z = \frac{B(r-\frac{1}{2},\frac{1}{2})}{2^{r+1/2}(a+1)^{r-1/2}}\,, \tag{31}$$

where $B(p,q)$ is Euler's beta function given by:

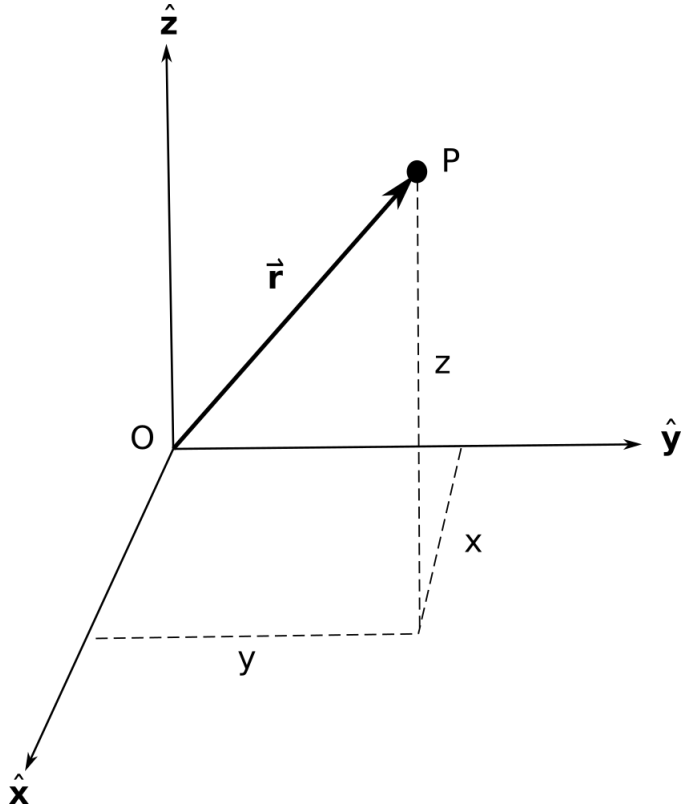
$$B(p,q) = \int_0^1 z^{p-1}(1-z)^{q-1}\,\mathrm{d}z = \frac{\Gamma(p)\,\Gamma(q)}{\Gamma(p+q)} \tag{32}$$

Through the use of transformations, derivatives, changes of variables, and variation of the parameters we can derive many different integrals from this master formula.

1.15 Coordinate Systems

There are numerous coordinate systems that can be used to frame a problem in an illuminating way. In this section we will list some more frequently used coordinate systems and leave the general discussion for section 23. We define the vector field $\boldsymbol{A}\in\mathbb{R}^3$ and the scalar field $f\in\mathbb{R}$.

1.15.1 Cartesian Coordinates



The tuple (x, y, z) describes a point in Cartesian space, where $-\infty < x, y, z < \infty$.

Position Vector: $\vec{\boldsymbol{r}} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}$

Basis Vectors: $\hat{\boldsymbol{x}} = \dfrac{\partial \vec{\boldsymbol{r}}}{\partial x}\,,\ \hat{\boldsymbol{y}} = \dfrac{\partial \vec{\boldsymbol{r}}}{\partial y}\,,\ \hat{\boldsymbol{z}} = \dfrac{\partial \vec{\boldsymbol{r}}}{\partial z}$

Infinitesimal Displacement: $\vec{\boldsymbol{dr}} = \mathrm{d}x\hat{\boldsymbol{x}} + \mathrm{d}y\hat{\boldsymbol{y}} + \mathrm{d}z\hat{\boldsymbol{z}}$

Volume Element: $\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$

Gradient: $\boldsymbol{\nabla} f(x, y, z) = \dfrac{\partial f}{\partial x}\hat{\boldsymbol{x}} + \dfrac{\partial f}{\partial y}\hat{\boldsymbol{y}} + \dfrac{\partial f}{\partial z}\hat{\boldsymbol{z}}$

Total Derivative: $\mathrm{d}f(x, y, z) = \boldsymbol{\nabla} f \cdot \vec{\mathbf{d}}\boldsymbol{r} = \dfrac{\partial f}{\partial x}\mathrm{d}x + \dfrac{\partial f}{\partial y}\mathrm{d}y + \dfrac{\partial f}{\partial z}\mathrm{d}z$

Divergence: $\boldsymbol{\nabla} \cdot \boldsymbol{A} = \dfrac{\partial A_x}{\partial x} + \dfrac{\partial A_y}{\partial y} + \dfrac{\partial A_z}{\partial z}$

Curl: $\boldsymbol{\nabla} \times \boldsymbol{A} = \left(\dfrac{\partial A_z}{\partial y} - \dfrac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\dfrac{\partial A_x}{\partial z} - \dfrac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\dfrac{\partial A_y}{\partial x} - \dfrac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}}$

Laplacian: $\boldsymbol{\nabla}^2 f(x, y, z) = \dfrac{\partial^2 f}{\partial x^2} + \dfrac{\partial^2 f}{\partial y^2} + \dfrac{\partial^2 f}{\partial z^2}$

1.15.2 Cylindrical Coordinates

From Cartesian coordinates, the transformations:

$$\begin{cases} x = r\cos(\phi) \\ y = r\sin(\phi) \\ z = z \end{cases} \quad \Rightarrow \quad \begin{cases} r = \sqrt{x^2+y^2}\,, & 0 < r < \infty \\ \phi = \arctan\left(\frac{y}{x}\right)\,, & 0 < \phi < 2\pi \\ z = z\,, & -\infty < z < \infty \end{cases}$$

yield the cylindrical coordinate system.

Position Vector: $\boldsymbol{r} = r\cos(\phi)\hat{\boldsymbol{x}} + r\sin(\phi)\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}} = r\hat{\boldsymbol{r}} + z\hat{\boldsymbol{z}}$

Basis Vectors:

$$\hat{\boldsymbol{r}} = \frac{\partial \boldsymbol{r}}{\partial r} = \cos(\phi)\hat{\boldsymbol{x}} + \sin(\phi)\hat{\boldsymbol{y}}$$

$$\hat{\boldsymbol{\phi}} = \frac{1}{r}\frac{\partial \boldsymbol{r}}{\partial \phi} = -\sin(\phi)\hat{\boldsymbol{x}} + \cos(\phi)\hat{\boldsymbol{y}}$$

$$\hat{\boldsymbol{z}} = \frac{\partial \boldsymbol{r}}{\partial z} = \hat{\boldsymbol{z}}$$

Infinitesimal Displacement: $\mathbf{d}\boldsymbol{r} = \mathrm{d}r\hat{\boldsymbol{r}} + r\mathrm{d}\phi\hat{\boldsymbol{\phi}} + \mathrm{d}z\hat{\boldsymbol{z}}$

Volume Element: $\mathrm{d}V = \mathrm{d}r\,(r\mathrm{d}\phi)\,\mathrm{d}z$

Gradient: $$\boldsymbol{\nabla} f(r,\phi,z) = \frac{\partial f}{\partial r}\hat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z}\hat{\boldsymbol{z}}$$

Total Derivative: $$\mathrm{d}f(r,\phi,z) = \boldsymbol{\nabla} f \cdot \mathbf{d}\boldsymbol{r} = \frac{\partial f}{\partial r}\mathrm{d}r + \frac{\partial f}{\partial \phi}\mathrm{d}\phi + \frac{\partial f}{\partial z}\mathrm{d}z$$

Divergence: $$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{\partial A_r}{\partial r} + \frac{1}{r}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

Curl: $$\boldsymbol{\nabla}\times\boldsymbol{A} = \left(\frac{1}{r}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right)\hat{\boldsymbol{r}} + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right)\hat{\boldsymbol{\phi}} + \frac{1}{r}\left(\frac{\partial}{\partial r}(rA_\phi) - \frac{\partial A_r}{\partial \phi}\right)\hat{\boldsymbol{z}}$$

Laplacian: $$\boldsymbol{\nabla}^2 f(r,\phi,z) = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$$

The Cartesian basis vectors in terms of the cylindrical basis vectors are:

$\hat{\boldsymbol{x}} = \cos(\phi)\hat{\boldsymbol{r}} - \sin(\phi)\hat{\boldsymbol{\phi}}$
$\hat{\boldsymbol{y}} = \sin(\phi)\hat{\boldsymbol{r}} + \cos(\phi)\hat{\boldsymbol{\phi}}$
$\hat{\boldsymbol{z}} = \hat{\boldsymbol{z}}$

1.15.3 Spherical Coordinates

From Cartesian coordinates, the transformations:

$$\begin{cases} x = r\sin(\theta)\cos(\phi) \\ y = r\sin(\theta)\sin(\phi) \\ z = r\cos(\theta) \end{cases} \quad \Rightarrow \quad \begin{cases} r = \sqrt{x^2+y^2+z^2}\,, & 0 < r < \infty \\ \theta = \arctan\left(\dfrac{\sqrt{x^2+y^2}}{z}\right), & 0 < \theta < \pi \\ \phi = \arctan\left(\dfrac{y}{x}\right), & 0 < \phi < 2\pi \end{cases}$$

yield the spherical coordinate system.

Position Vector: $\boldsymbol{r} = r\sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + r\sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + r\cos(\theta)\hat{\boldsymbol{z}} = r\hat{\boldsymbol{r}}$

Basis Vectors:

$$\hat{\boldsymbol{r}} = \frac{\partial \boldsymbol{r}}{\partial r} = \sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + \sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + \cos(\theta)\hat{\boldsymbol{z}}$$

$$\hat{\boldsymbol{\theta}} = \frac{1}{r}\frac{\partial \boldsymbol{r}}{\partial \theta} = \cos(\theta)\cos(\phi)\hat{\boldsymbol{x}} + \cos(\theta)\sin(\phi)\hat{\boldsymbol{y}} - \sin(\theta)\hat{\boldsymbol{z}}$$

$$\hat{\boldsymbol{\phi}} = \frac{1}{r\sin(\theta)}\frac{\partial \boldsymbol{r}}{\partial \phi} = -\sin(\phi)\hat{\boldsymbol{x}} + \cos(\phi)\hat{\boldsymbol{y}}$$

Infinitesimal Displacement: $\mathbf{d}\boldsymbol{r} = \mathrm{d}r\hat{\boldsymbol{r}} + r\mathrm{d}\theta\hat{\boldsymbol{\theta}} + r\sin(\theta)\mathrm{d}\phi\hat{\boldsymbol{\phi}}$

Volume Element: $\mathrm{d}V = \mathrm{d}r\,(r\mathrm{d}\theta)\,(r\sin(\theta)\,\mathrm{d}\phi)$

Gradient:

$$\boldsymbol{\nabla} f(r,\theta,\phi) = \frac{\partial f}{\partial r}\hat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin(\theta)}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}}$$

Total Derivative:

$$\mathrm{d}f(r,\theta,\phi) = \boldsymbol{\nabla} f \cdot \mathbf{d}\boldsymbol{r} = \frac{\partial f}{\partial r}\mathrm{d}r + \frac{\partial f}{\partial \theta}\mathrm{d}\theta + \frac{\partial f}{\partial \phi}\mathrm{d}\phi$$

Divergence:

$$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin(\theta)}\frac{\partial}{\partial \theta}(\sin(\theta)A_\theta) + \frac{1}{r\sin(\theta)}\frac{\partial A_\phi}{\partial \phi}$$

Curl:

$$\boldsymbol{\nabla}\times\boldsymbol{A} = \frac{1}{r\sin(\theta)}\left(\frac{\partial}{\partial \theta}(\sin(\theta)A_\phi) - \frac{\partial A_\theta}{\partial \phi}\right)\hat{\boldsymbol{r}} + \frac{1}{r}\left(\frac{1}{\sin(\theta)}\frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r}(rA_\phi)\right)\hat{\boldsymbol{\theta}} + \frac{1}{r}\left(\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial \theta}\right)\hat{\boldsymbol{\phi}}$$

Laplacian:

$$\boldsymbol{\nabla}^2 f(r,\theta,\phi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin(\theta)}\frac{\partial}{\partial \theta}\left(\sin(\theta)\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2(\theta)}\frac{\partial^2 f}{\partial \phi^2}$$

The Cartesian basis vectors in terms of the spherical basis vectors are:

$\hat{\boldsymbol{x}} = \sin(\theta)\cos(\phi)\hat{\boldsymbol{r}} + \cos(\theta)\cos(\phi)\hat{\boldsymbol{\theta}} - \sin(\phi)\hat{\boldsymbol{\phi}}$

$\hat{\boldsymbol{y}} = \sin(\theta)\sin(\phi)\hat{\boldsymbol{r}} + \cos(\theta)\sin(\phi)\hat{\boldsymbol{\theta}} + \cos(\phi)\hat{\boldsymbol{\phi}}$

$\hat{\boldsymbol{z}} = \cos(\theta)\hat{\boldsymbol{r}} - \sin(\theta)\hat{\boldsymbol{\theta}}$

1.16 Geometric Quantities

1.16.1 2-Dimensional Relations

In two dimensions we can speak about the area of a two-dimensional figure or the surface area of a three-dimensional figure. The former can be thought of as the integral:

$$A = \int_a^b f(x)\,\mathrm{d}x\,, \tag{33}$$

which represents the area enclosed by the function $f(x)$ and the interval $a < x < b$. The area between the graphs of two functions $f(x)$ and $g(x)$ is given by:

$$A = \int_a^b f(x)\,\mathrm{d}x - \int_a^b g(x)\,\mathrm{d}x = \int_a^b f(x) - g(x)\,\mathrm{d}x\,, \tag{34}$$

where it is understood that $f(x) > g(x)$ over the interval $a < x < b$ so that $A \geq 0$. We can also compute the area enclosed by a parametric curve $\boldsymbol{u}(t) = (x(t), y(t))$ with the line integral:

$$A = \frac{1}{2}\oint_{t_0}^{t_1} (x\dot{y} - y\dot{x})\,\mathrm{d}t = \frac{1}{2}\oint_{t_0}^{t_1} (\boldsymbol{u} \times \dot{\boldsymbol{u}})_z\,\mathrm{d}t\,, \tag{35}$$

where $(\cdot)_z$ indicates the z-component of the cross-product.

The surface area of a three-dimensional figure is given in terms of the parametric surface $\boldsymbol{r} = \boldsymbol{r}(u, v)$, where $\boldsymbol{r}$ is a continuously differentiable vector function of $(u, v) \in D \subset \mathbb{R}^2$:

$$A = \iint_D \left(\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}\right)\mathrm{d}u\mathrm{d}v\,. \tag{36}$$

Some common two-dimensional figures are discussed below.

Triangle:

The three main special types of triangles according to side length are the equilateral (3 equal length sides), isosceles (2 equal length sides), and scalene (All different length sides) triangles. By internal angle they are the right (1 90° angle), obtuse (1 angle > 90°), and acute (All angles < 90°) -angled triangles.

The Sine, Cosine, and Tangent laws can be used to compute the side lengths of any Euclidean triangle, provided at least three of the angles and/or side lengths are known. The perimeter of a triangle is given by $p = b_1 + b_2 + b_3$. The area of a triangle is given by a few different formulae, each of which has certain advantages depending on the given information. They are:

$$A = \frac{1}{2}\,\text{base} \times \text{height}\,,$$

(Useful when the height perpendicular to the base is easily found)

$$A = \frac{1}{2}\,b_1\,b_2\sin(\theta_3) = \frac{1}{2}\,b_2\,b_3\sin(\theta_1) = \frac{1}{2}\,b_3\,b_1\sin(\theta_2)\,,$$

(Useful when given two side lengths and one internal angle)

$$A = \frac{b_2^2\sin(\theta_1)\sin(\theta_1+\theta_2)}{2\sin(\theta_2)} = \frac{b_1^2\sin(\theta_2)\sin(\theta_3)}{2\sin(\theta_2+\theta_3)}\,,$$

(Useful when given one side length and two internal angles)

$$A = \sqrt{s\,(s-a)\,(s-b)\,(s-c)}\,,$$

where $s = p/2$ is the semiperimeter of the triangle. (Useful when given three side lengths and no internal angles)

The last method involves the use of Cartesian coordinates and matrices. Given three general vertices $A = (x_A, y_A)$, $B = (x_B, y_B)$, and $C = (x_C, y_C)$, the area of the triangle ABC is given by:

$$\text{Area} = \frac{1}{2}\left\|\begin{matrix} x_A & x_B & x_C \\ y_A & y_B & y_C \\ 1 & 1 & 1 \end{matrix}\right\| = \frac{1}{2}\left|(x_A - x_C)\,(y_B - y_A) - (x_A - x_B)\,(y_C - y_A)\right|.$$

Polygons:

Polygons are typically classified by the number of sides n they have, their convexity, and/or their symmetry. A general polygon classified under number of sides is denoted as an n-gon (The other two classes can branch off from this one). A convex polygon is one with all its interior angles $< 180°$, and a concave polygon is a polygon where at least one interior angle is $\geq 180°$ and the boundary does not cross itself (simple). In terms of symmetry, an equiangular polygon is one where all corner angles are equal, a cyclic polygon is one where all corners lie on a single circle, an equilateral polygon is one where all edges are the same length, a tangential polygon is one where all sides are tangent to an inscribed circle, and a regular polygon is one which is both cyclic and equilateral or equilateral and equiangular.

The sum of the interior angles in a simple n-gon is $(n-2) \times 180°$, while the sum of the interior angles in a convex n-gon is $\left(1 - \frac{2}{n}\right) \times 180°$.

The area of a simple polygon with vertices (in order) $(x_0, y_0), (x_1, y_1), \ldots, (x_{n-1}, y_{n-1})$ is given by the equation:

$$A = \frac{1}{2}\sum_{i=0}^{n-1}(x_i\,y_{i+1} - x_{i+1}\,y_i)\,, \tag{37}$$

where $x_n = x_0$ and $y_n = y_0$.

If the polygon is regular then the area can be given by the following:

$$A = \frac{1}{2}\,nsa = \frac{1}{2}\,pa = \frac{1}{4}\,ns^2\cot\left(\frac{\pi}{n}\right) = na^2\tan\left(\frac{\pi}{n}\right) = \frac{1}{2}\,nR^2\sin\left(\frac{2\pi}{n}\right)\,, \tag{38}$$

where s is the side length, p is the perimeter, R is the circumradius (the distance from the centre of the polygon to one of its vertices), and a is the apothem (the distance from the centre of the polygon to the midpoint of its side length). R and a are related through the expression

$$R = \frac{s}{2\sin\left(\frac{\pi}{n}\right)} = \frac{a}{\cos\left(\frac{\pi}{n}\right)}\,. \tag{39}$$

For $n = 3$ we have the familiar triangle (discussed above), for $n = 4$ we have the quadrilateral, which has the important subcategories of parallelogram and trapezoid, for $n = 5$ we have the pentagon as a well known construction, for $n = 6$ we have the hexagon as a well known construction, and so on for increasing number of sides. The areas for some of these shapes are given below:

Parallelogram: $A = bh$

Trapezoid: $A = \dfrac{(a+b)\,h}{2} = \dfrac{a+b}{4\,|b-a|}\sqrt{(-a+b+c+d)\,(a-b+c+d)\,(a-b+c-d)\,(a-b-c+d)}$

Pentagon: $A = \dfrac{5}{4}\,s^2 \cot\left(\dfrac{\pi}{5}\right)$

Hexagon: $A = \dfrac{3\sqrt{3}}{2}\,s^2$

Circle:

The equation describing a circle is: $(x-x_0)^2 + (y-y_0)^2 = a^2$ or in parametric form is: $(x,y) = (a\cos(t), a\sin(t))$ for $0 \le t \le 2\pi$, and note that the eccentricity of a circle is equal to 0.

The area of a circle is:

$$A = \int_{-a}^{a} \sqrt{a^2 - x^2}\,\mathrm{d}x = \pi a^2\,, \tag{40}$$

and the circumference is:

$$C = \int_0^{\pi} 2r\,\mathrm{d}\theta = 2\pi r\,. \tag{41}$$

Ellipse:

The equation describing an ellipse is: $\dfrac{(x-x_0)^2}{a^2} + \dfrac{(y-y_0)^2}{b^2} = 1$ or in parametric form is: $(x,y) = (a\cos(t), b\sin(t))$ for $0 \le t \le 2\pi$, and the eccentricity $e = \sqrt{1 - \dfrac{b^2}{a^2}}$ gives a measure of how ovular an ellipse is.

The area of an ellipse is:

$$A = \int_{-a}^{a} 2b\sqrt{1 - \frac{x^2}{a^2}}\,\mathrm{d}x = \pi a b\,, \tag{42}$$

and the circumference is:

$$C = \int_0^{\pi/2} 4a\sqrt{1 - e^2 sin(\theta)^2}\,\mathrm{d}\theta = 4aE(e) = 2\pi a\left[1 - \sum_{n=1}^{\infty}\left(\frac{(2n-1)!!}{(2n)!!}\right)^2 \frac{e^{2n}}{2n-1}\right], \tag{43}$$

where $E(k)$ is the complete elliptic integral of the second kind.

Hyperbola:

The equation describing a hyperbola is: $\dfrac{(x-x_0)^2}{a^2} - \dfrac{(y-y_0)^2}{b^2} = 1$ or in parametric form is: $(x,y) = (\pm a\cosh(t), b\sinh(t))$ for $t \in \mathbb{R}$, and the eccentricity of a hyperbola is $e = \sqrt{1 + \dfrac{b^2}{a^2}}$.

The area under a hyperbola is:

$$A = b\int_{x_0}^{a} \sqrt{\frac{x^2}{a^2} - 1}\,\mathrm{d}x = \frac{b}{2a}\left(x\sqrt{x^2 - a^2} + a^2 \ln\left|x + \sqrt{x^2 - a^2}\right|\right)\Bigg|_{x_0}^{a}\,, \tag{44}$$

and the arc length is:

$$s = b\int_{\mathrm{arcosh}(\frac{x_1}{a})}^{\mathrm{arcosh}(\frac{x_2}{a})} \sqrt{1 + \left(1 + \frac{a^2}{b^2}\right)\sinh^2(v)}\;\mathrm{d}v = -ib\left[E\left(iz\,\middle|\,1 + \frac{a^2}{b^2}\right)\right]_{\mathrm{arcosh}(\frac{x_1}{a})}^{\mathrm{arcosh}(\frac{x_2}{a})}\,. \tag{45}$$

Super Shapes:

A superellipse is a generalization of many two-dimensional shapes and is given by the equation:

$$\left|\frac{(x-x_0)}{a}\right|^n + \left|\frac{(y-y_0)}{b}\right|^n = 1\,, \tag{46}$$

where $n, a, b \in \mathbb{R}^+$. The superellipse has extreme points $(\pm a, 0)$ and $(0, \pm b)$, and has four pseudo-vertices at $(\pm 2^{-1/n}a, \pm 2^{-1/n}b)$. The superellipse can also be described by the parametric equations:

$$x(t) = |\cos(t)|^{2/n} \cdot a\,\mathrm{sgn}(\cos(t)) \quad \text{and} \quad y(t) = |\sin(t)|^{2/n} \cdot b\,\mathrm{sgn}(\sin(t))\,, \tag{47}$$

where $0 \le t \le 2\pi$.

The area inside a superellipse can be calculated using the formula:

$$A = 4\,a\,b\,\frac{\left(\Gamma\left(1+\frac{1}{n}\right)\right)^2}{\Gamma\left(1+\frac{2}{n}\right)}\,. \tag{48}$$

For the case when $n = 1$ the curve is a parallelogram with corners $(\pm a, 0)$ and $(0, \pm b)$. Some common parallelograms include the square (four sides and angles of equal size), rectangle (four angles of equal size), and rhombus (four sides of equal size). The area of any parallelogram is simply $A = \text{Base} \times \text{Height}$. For the case when $n = 2$ the curve is an ordinary ellipse and when $a = b$, reduces to a circle.

There also exists a superhyperbola, which is similar to the superellipse but describes hyperbolic geometry instead. It is given by the equation:

$$\left|\frac{(x-x_0)}{a}\right|^n - \left|\frac{(y-y_0)}{b}\right|^n = 1\,, \tag{49}$$

where $n, a, b \in \mathbb{R}^+$. The superhyperbola can also be described by the parametric equations:

$$x(t) = |\sec(t)|^{2/n} \cdot a\,\mathrm{sgn}(\sec(t)) \quad \text{and} \quad y(t) = |\tan(t)|^{2/n} \cdot b\,\mathrm{sgn}(\tan(t))\,, \tag{50}$$

where $-\frac{\pi}{2} \le t \le \frac{\pi}{2}$.

The Super Formula is a generalization of the superellipse, and is described by the equation:

$$r(\phi) = \left(\left|\frac{1}{a}\cos\left(\frac{x\phi}{4}\right)\right|^{n_2} + \left|\frac{1}{b}\sin\left(\frac{y\phi}{4}\right)\right|^{n_3}\right)^{-1/n_1}\,. \tag{51}$$

1.16.2 3-Dimensional Relations

In three dimensions we can speak of the volume of a figure, which can be thought of as the integral over a region $D \subseteqq \mathbb{R}^3$:

$$V = \iiint_D \det\frac{\partial(u,v,w)}{\partial(x,y,z)}\,\mathrm{d}\boldsymbol{r}\,, \tag{52}$$

where $\det\frac{\partial(u,v,w)}{\partial(x,y,z)}$ is the determinant of the 3×3 Jacobian matrix of partial derivatives.

Certain shapes can be generated by rotating a plane curve around some straight line that lies in the same plane (axis of revolution). These types of shapes are known as solids of revolution and have a volume given by the equation:

$$V = \int_0^\pi \int_a^b [f(x)]^2\,\mathrm{d}x\,\mathrm{d}\theta = \int_a^b \pi[f(x)]^2\,\mathrm{d}x\,, \tag{53}$$

if the area between the curve $f(x)$ and the axis is being revolved, or by the equation:

$$V = \int_a^b \pi \left| [f(x)]^2 - [g(x)]^2 \right| \, \mathrm{d}x \,, \tag{54}$$

if the area between the curves $f(x)$ and $g(x)$ and the lines $x = a$ and $x = b$ about the x-axis is being revolved. The above method is used when the integration takes place over an axis that is parallel to the axis of revolution (called method of discs). If the integration should be taken over an axis that is perpendicular to the axis of rotation (called method of shells), then the volume is given by the equation:

$$V = \int_0^{2\pi} \int_a^b x \, |f(x)| \, \mathrm{d}x \, \mathrm{d}\phi = \int_a^b 2\pi x \, |f(x)| \, \mathrm{d}x \,, \tag{55}$$

if the area between the curve $f(x)$ and the axis is being revolved, or by the equation:

$$V = \int_0^{2\pi} \int_a^b x \, |f(x) - g(x)| \, \mathrm{d}x \, \mathrm{d}\phi = \int_a^b 2\pi x \, |f(x) - g(x)| \, \mathrm{d}x \,. \tag{56}$$

If the curve is given in parametric form $(x(t), y(t))$ on an interval $[a, b]$, then the volumes of the solids generated by revolving the curve around the x and y axes are:

$$V_x = \int_a^b \pi y^2 \frac{\mathrm{d}x}{\mathrm{d}t} \, \mathrm{d}t \quad \text{and} \quad V_y = \int_a^b \pi x^2 \frac{\mathrm{d}y}{\mathrm{d}t} \, \mathrm{d}t \,.$$

The areas of the surfaces, under the same circumstances as above, are given by:

$$A_x = \int_a^b 2\pi y \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2} \, \mathrm{d}t \quad \text{and} \quad A_y = \int_a^b 2\pi x \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2} \, \mathrm{d}t \,.$$

Some figures may not be closed, in which case they will only have a volume if they are bounded by an additional surface. Examples include hyperboloids and parabaloids.

Super Shapes:

The superellipse and super formula can both be extended to three-dimensional space. Some of the three-dimensional cousins of the superellipses are the superquadrics, superellipsoids, and supertoroids. The superquadrics are given by the equation:

$$\left| \frac{(x - x_0)}{a} \right|^r + \left| \frac{(y - y_0)}{b} \right|^s + \left| \frac{(z - z_0)}{c} \right|^t = 1 \,, \tag{57}$$

where $r, s, t, a, b, c \in \mathbb{R}^+$. If r, s, t are allowed to be negative real numbers then the resulting shapes are sometimes called super-hyperboloids, whose boundaries extend to infinity. Superquadrics can also be defined parametrically, using the surface parameters u and v, as:

$$x(u, v) = a \, \mathrm{sgn}(\cos(v)) \, |\cos(v)|^{2/r} \, \mathrm{sgn}(\cos(u)) \, |\cos(u)|^{2/r}$$

$$y(u, v) = b \, \mathrm{sgn}(\cos(v)) \, |\cos(v)|^{2/s} \, \mathrm{sgn}(\sin(u)) \, |\sin(u)|^{2/s}$$

$$z(u, v) = c \, \mathrm{sgn}(\sin(v)) \, |\sin(v)|^{2/t}$$

where $-\frac{\pi}{2} \leq v \leq \frac{\pi}{2}$ and $-\pi \leq u \leq \pi$.

For the case when $r = s = t = 1$ and $a = b = c$ the shape is a regular octahedron, for the case when $r = s = t = 2$ the shape is an ellipsoid and when $a = b = c$ the shape reduces to a sphere.

The superellipsoids are given by the inequality:

$$\left(\left|\frac{(x-x_0)}{a}\right|^r + \left|\frac{(y-y_0)}{b}\right|^r\right)^{t/r} + \left|\frac{(z-z_0)}{c}\right|^t \leq 1\,, \tag{58}$$

where $r, t, a, b, c \in \mathbb{R}^+$. Superellipsoids can also be defined parametrically, again using the surface parameters u and v, as:

$$x(u,v) = a\,\mathrm{sgn}(\cos(v))\,|\cos(v)|^{2/t}\,\mathrm{sgn}(\cos(u))\,|\cos(u)|^{2/r}$$

$$y(u,v) = b\,\mathrm{sgn}(\cos(v))\,|\cos(v)|^{2/t}\,\mathrm{sgn}(\sin(u))\,|\sin(u)|^{2/r}$$

$$z(u,v) = c\,\mathrm{sgn}(\sin(v))\,|\sin(v)|^{2/t}$$

where $-\frac{\pi}{2} \leq v \leq \frac{\pi}{2}$ and $-\pi \leq u \leq \pi$.

The volume inside a superellipsoid can be calculated using the formula:

$$V = \frac{8\,a\,b\,c}{3\,r\,t}\,\frac{\Gamma\left(\frac{1}{r}\right)\Gamma\left(\frac{1}{r}\right)\Gamma\left(\frac{2}{t}\right)\Gamma\left(\frac{1}{t}\right)}{\Gamma\left(\frac{2}{r}\right)\Gamma\left(\frac{3}{t}\right)}\,. \tag{59}$$

For the case when $r, t > 2$ the shape is a cuboid; for the case when $r = 2$ and $t > 2$ the shape is a cylindroid; for the case when $t = 2$ and $r > 2$ the shapes resemble pillows; for the case when either $r < 1$ or $t < 1$ the shape is pinched; for the case when $r = t = 2$ the shape is an ellipsoid, and furthermore if $a = b = c$ then the shape is a sphere.

The supertoroids are given by the equation:

$$\left(\left(\left|\frac{(x-x_0)}{a}\right|^r + \left|\frac{(y-y_0)}{b}\right|^r\right)^{1/r} - \frac{R}{\sqrt{a^2+b^2}}\right)^t + \left|\frac{(z-z_0)}{c}\right|^t = 1\,, \tag{60}$$

where $r, t, a, b, c, R \in \mathbb{R}^+$ and R is the radius of the torus. Supertoroids can also be defined parametrically, again using the surface parameters u and v, as:

$$x(u,v) = a\left(\frac{R}{\sqrt{a^2+b^2}} + \mathrm{sgn}(\cos(v))\,|\cos(v)|^{2/t}\right)\mathrm{sgn}(\cos(u))\,|\cos(u)|^{2/r}$$

$$y(u,v) = b\left(\frac{R}{\sqrt{a^2+b^2}} + \mathrm{sgn}(\cos(v))\,|\cos(v)|^{2/t}\right)\mathrm{sgn}(\sin(u))\,|\sin(u)|^{2/r}$$

$$z(u,v) = c\,\mathrm{sgn}(\sin(v))\,|\sin(v)|^{2/t}$$

where $-\pi \leq v \leq \pi$ and $-\pi \leq u \leq \pi$.

The volume inside a supertoroid can be calculated using the formula:

$$V = \frac{8\,a\,b\,c\,\alpha}{r\,t}\,\frac{\Gamma\left(\frac{1}{r}\right)\Gamma\left(\frac{1}{r}\right)\Gamma\left(\frac{1}{t}\right)\Gamma\left(\frac{1}{t}\right)}{\Gamma\left(\frac{2}{r}\right)\Gamma\left(\frac{2}{t}\right)}\,, \tag{61}$$

where $\alpha > 1$ is the hole diameter of the supertoroid.

For the case when $r, t > 2$ the shape is a rounded square toroid; for the case when $t = 2$ and $r > 2$ the shapes resemble piping; for the case when either $r < 1$ or $t < 1$ the shape is a pinched toroid; for the case when $r = t = 2$ the shape is a toroid, and furthermore if $a = b = c$ then the shape is a torus.

Superhyperboloids exist in two separate categories: those that are smoothly defined over the whole domain (one piece) and those that are disjointly defined over the whole domain (two pieces). The superhyperboloids of one piece are given by the equation:

$$\left(\left|\frac{(x-x_0)}{a}\right|^r + \left|\frac{(y-y_0)}{b}\right|^r\right)^{t/r} - \left|\frac{(z-z_0)}{c}\right|^t = 1\,, \tag{62}$$

where $r, t, a, b, c \in \mathbb{R}^+$. Superhyperboloids of one piece can also be defined parametrically, again using the surface parameters u and v, as:

$$x(u,v) = a\,\mathrm{sgn}(\sec(v))\,|\sec(v)|^{2/t}\,\mathrm{sgn}(\cos(u))\,|\cos(u)|^{2/r}$$

$$y(u,v) = b\,\mathrm{sgn}(\sec(v))\,|\sec(v)|^{2/t}\,\mathrm{sgn}(\sin(u))\,|\sin(u)|^{2/r}$$

$$z(u,v) = c\,\mathrm{sgn}(\tan(v))\,|\tan(v)|^{2/t}$$

where $-\dfrac{\pi}{2} \le v \le \dfrac{\pi}{2}$ and $-\pi \le u \le \pi$.

For the case when $r = t = 2$ the shape is a hyperboloid of one piece.

The superhyperboloids of two pieces are given by the equation:

$$\left(\left|\frac{(x-x_0)}{a}\right|^r - \left|\frac{(y-y_0)}{b}\right|^r\right)^{t/r} - \left|\frac{(z-z_0)}{c}\right|^t = 1\,, \tag{63}$$

where $r, t, a, b, c \in \mathbb{R}^+$. Superhyperboloids of two pieces can also be defined parametrically, again using the surface parameters u and v, as:

$$x(u,v) = a\,\mathrm{sgn}(\sec(v))\,|\sec(v)|^{2/t}\,\mathrm{sgn}(\sec(u))\,|\sec(u)|^{2/r} \tag{64}$$

$$y(u,v) = b\,\mathrm{sgn}(\sec(v))\,|\sec(v)|^{2/t}\,\mathrm{sgn}(\tan(u))\,|\tan(u)|^{2/r} \tag{65}$$

$$z(u,v) = c\,\mathrm{sgn}(\tan(v))\,|\tan(v)|^{2/t} \tag{66}$$

where $-\dfrac{\pi}{2} \le v \le \dfrac{\pi}{2}$, $-\frac{\pi}{2} \le u \le \frac{\pi}{2}$ (for the first piece), and $\frac{\pi}{2} \le u \le \frac{3\pi}{2}$ (for the second piece).

For the case when $r = t = 2$ the shape is a hyperboloid of two pieces.

The spherical product of super formulae is the three-dimensional analogue of the two-dimensional formula given previously. This spherical product can be defined parametrically, using the parameters θ and ϕ, as:

$$x(\theta,\phi) = \left(\left|\frac{1}{a}\cos\left(\frac{x_1\theta}{4}\right)\right|^{n_2} + \left|\frac{1}{b}\sin\left(\frac{y_1\theta}{4}\right)\right|^{n_3}\right)^{-1/n_1}\cos(\theta)\left(\left|\frac{1}{a}\cos\left(\frac{x_2\phi}{4}\right)\right|^{n_2} + \left|\frac{1}{b}\sin\left(\frac{y_2\phi}{4}\right)\right|^{n_3}\right)^{-1/n_1}\cos(\phi)$$

$$y(\theta,\phi) = \left(\left|\frac{1}{a}\cos\left(\frac{x_1\theta}{4}\right)\right|^{n_2} + \left|\frac{1}{b}\sin\left(\frac{y_1\theta}{4}\right)\right|^{n_3}\right)^{-1/n_1}\sin(\theta)\left(\left|\frac{1}{a}\cos\left(\frac{x_2\phi}{4}\right)\right|^{n_2} + \left|\frac{1}{b}\sin\left(\frac{y_2\phi}{4}\right)\right|^{n_3}\right)^{-1/n_1}\cos(\phi)$$

$$z(\theta,\phi) = \left(\left|\frac{1}{a}\cos\left(\frac{x_2\phi}{4}\right)\right|^{n_2} + \left|\frac{1}{b}\sin\left(\frac{y_2\phi}{4}\right)\right|^{n_3}\right)^{-1/n_1}\sin(\phi)$$

where $-\dfrac{\pi}{2} \le \theta \le \dfrac{\pi}{2}$ and $-\pi \le \phi \le \pi$.

Ellipsoid:

The equation describing an ellipsoid is given by:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \,, \tag{67}$$

where $a, b, c \in \mathbb{R}^+$. We can also describe the ellipsoid parametrically, using the parameters $0 \leq \phi < 2\pi$ and $0 < \theta < \pi$, with the equations:

$$\begin{aligned} x &= a \sin(\theta) \cos(\phi) \\ y &= b \sin(\theta) \sin(\phi) \\ z &= c \cos(\theta) \end{aligned}$$

The ellipsoid is a special case of the superellipsoids and superquadrics (when the exponents are all equal). Further classification of the ellipsoid results in three distinct shapes: when $a = b > c$ we have an oblate spheroid (bulge at equator), when $a = b < c$ we have a prolate spheroid (bulge at poles), and when $a = b = c$ we have a sphere.

The volume of an ellipsoid can be calculated using the formula:

$$V = \int_0^{2\pi} \int_0^1 \int_0^{\pi} abcr^2 \sin(\theta) \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{4\pi}{3} abc \,, \tag{68}$$

where we recover the volume of a sphere when $a = b = c$.

The surface area of an ellipsoid can be calculated using the formula:

$$S = 2\pi c^2 + \frac{2\pi ab}{\sin(\varphi)} \Big(E(\varphi, k) \sin^2(\varphi) + F(\varphi, k) \cos^2(\varphi) \Big) \,, \tag{69}$$

where $k^2 = \dfrac{a^2(b^2 - c^2)}{b^2(a^2 - c^2)}$, $\cos(\varphi) = \dfrac{c}{a}$, $E(\varphi, k)$ is the incomplete elliptic integral of the second kind, and $F(\varphi, k)$ is the incomplete elliptic integral of the first kind.

Parallelepiped:

A parallelepiped is the three-dimensional construction of a figure with six faces, each of which is a parallelogram. Some specific cases of the parallelepiped are the cube (all faces are squares), rectangular cuboid (all faces are rectangles), square cuboid (two faces are squares and four faces are rectangles), rhombohedron (all faces are rhombuses), and rhombic prism (two faces are rhombuses and four faces are rectangles).

The volume of a parallelepiped can be calculated by considering vectors or through geometric properties. If we consider the vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ that generate the parallelepiped (see figure), then the area of the base of the parallelepiped is given by $B = |\boldsymbol{a}| \cdot |\boldsymbol{b}| \cdot \sin(\gamma) = |\boldsymbol{a} \times \boldsymbol{b}|$ and the height is given by $h = |\boldsymbol{c}| \cdot |\cos(\theta)|$, so that the volume is:

$$V = B \cdot h = (|\boldsymbol{a}| \, |\boldsymbol{b}| \sin(\gamma)) \cdot (|\boldsymbol{c}| \, |\cos(\theta)|) = |\boldsymbol{a} \times \boldsymbol{b}| \, |\boldsymbol{c}| \, |\cos(\theta)| = |\, (\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c} \,| \,. \tag{70}$$

Alternatively, if we consider the triple product of $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$, then the volume is given by:

$$V = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \,. \tag{71}$$

If we instead only consider the three side lengths a, b, c and the corresponding angles α, β, γ, then the volume is given by:

$$V = \begin{vmatrix} \boldsymbol{a}\cdot\boldsymbol{a} & \boldsymbol{a}\cdot\boldsymbol{b} & \boldsymbol{a}\cdot\boldsymbol{c} \\ \boldsymbol{b}\cdot\boldsymbol{a} & \boldsymbol{b}\cdot\boldsymbol{b} & \boldsymbol{b}\cdot\boldsymbol{c} \\ \boldsymbol{c}\cdot\boldsymbol{a} & \boldsymbol{c}\cdot\boldsymbol{b} & \boldsymbol{c}\cdot\boldsymbol{c} \end{vmatrix} = abc\sqrt{1 + 2\cos(\alpha)\cos(\beta)\cos(\gamma) - \cos^2(\alpha) - \cos^2(\beta) - \cos^2(\gamma)}\,. \tag{72}$$

If $a = b = c$ and $\alpha = \beta = \gamma = \pi/2$ then we get the volume of a cube $V = a^3$; if only $\alpha = \beta = \gamma = \pi/2$ then we get the volume of a cuboid $V = abc$.

The surface area of a parallelepiped can be calculated using the formula:

$$S = 2\cdot(|\boldsymbol{a}\times\boldsymbol{b}| + |\boldsymbol{a}\times\boldsymbol{c}| + |\boldsymbol{b}\times\boldsymbol{c}|) = 2\,(ab\sin(\gamma) + bc\sin(\alpha) + ca\sin(\beta))\,. \tag{73}$$

If $a = b = c$ and $\alpha = \beta = \gamma = \pi/2$ then we get the surface area of a cube $S = 6a^2$; if only $\alpha = \beta = \gamma = \pi/2$ then we get the surface area of a cuboid $S = 2(ab + bc + ca)$.

Pyramid:

Pyramids consist of an n-gon base connected to a point (called the apex) from its vertices. Some specific cases of pyramids are the square pyramid $n = 4$ (square base), tetrahedron $n = 3$ (triangular base), rectangular pyramid $n = 4$ (rectangular base), pentagonal pyramid $n = 5$ (pentagonal base), and hexagonal pyramid $n = 6$ (hexagonal base).

The volume of a pyramid can be calculated from the area of the base B and the height from base to apex h using the formula:

$$V = \int_0^h \frac{B}{h^2}(h-y)^2\,\mathrm{d}y = \frac{1}{3}B\cdot h = \frac{n}{12}hs^2\cot\left(\frac{\pi}{n}\right)\,, \tag{74}$$

where the last equality holds if the base is a regular n-gon and n is the number of side lengths s of the polygon face. For the above mentioned shapes corresponding to $n = 3$, the volume is $V = 4hs^2/\sqrt{3} = s^3/6\sqrt{2}$; for $n = 4$ it is $V = hs^2/3$ (square pyramid) or $V = hlw/3$ (rectangular pyramid); for $n = 5$ it is $V = 5hs^2\cot(\pi/5)/12$, and for $n = 6$ it is $V = \sqrt{3}hs^2/2$.

The surface area of a pyramid with base perimeter P and slant height $L = \sqrt{h^2 + r^2}$ can be calculated using the formula:

$$S = B + \frac{PL}{2}\,. \tag{75}$$

For the above mentioned shapes corresponding to $n = 3$, the surface area is $S = \sqrt{3}s^2$; for $n = 4$ it is $S = s^2 + 2sL$ (square pyramid) or $S = lw(1 + 2L)$ (rectangular pyramid); for $n = 5$ it is $S = 5s^2\cot(\pi/5)/4 + 5sL/2$, and for $n = 6$ it is $S = 3s(\sqrt{3}s/2 + L)$.

Cone:

A cone can be thought of as the limit as n (number of polygon sides) goes to infinity of a pyramid. Some specific cones include the circular and elliptic cones.

For right cones (base is directly perpendicular to height), the positive portion of the equation:

$$z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}\,, \tag{76}$$

describes an elliptic cone in Cartesian coordinates, and a circular cone when $a = b$ (note that these cones are not bounded and hence have an infinite volume and surface area).

The volume of any conic solid is simply $V = \frac{1}{3}A_B h$, where A_B is the area of the base and h is the height. For a right circular cone, the volume is

$$V = \int_0^{\arctan(\frac{R}{h})} \int_0^{2\pi} \int_0^{h\sec(\theta)} r^2 \sin(\theta)\, \mathrm{d}r\, \mathrm{d}\theta\, \mathrm{d}\phi = \frac{1}{3}\pi r^2 h\,,$$

(using spherical coordinates).

The surface area of a right circular cone is $S = \pi r(r + \sqrt{r^2 + h^2})$.

Prism:

Prisms consist of two n-gons, one as the base and the other separated by parallelograms, which connect the sides of the two n-gons. Some specific cases of prisms are the cube $n = 4$ (a translated square), triangular prism $n = 3$ (a translated triangle), rectangular prism $n = 4$ (a translated rectangle), pentagonal prism $n = 5$ (a translated pentagon), and hexagonal prism $n = 6$ (a translated hexagon).

The volume of a prism can be calculated from the area of the base B and the height h using the formula:

$$V = B \cdot h = \frac{n}{4}\, hs^2 \cot\left(\frac{\pi}{n}\right)\,, \tag{77}$$

where the last equality holds if the base is a regular n-gon and n is the number of side lengths s of the polygon faces. For the above mentioned shapes corresponding to $n = 3$, the volume is $V = \sqrt{3}hs^2/4$; for $n = 5$ it is $V = 5hs^2 \cot(\pi/5)/4$, and for $n = 6$ it is $V = 9hs^2/\sqrt{12}$.

The surface area of a right prism with base perimeter P can be calculated using the formula:

$$S = 2B + Ph = \frac{n}{2}\, s^2 \cot\left(\frac{\pi}{n}\right) + nsh\,, \tag{78}$$

where the last equality holds if the base is a regular n-gon. For the above mentioned shapes corresponding to $n = 3$, the surface area is $S = \sqrt{3}hs^2/2 + 3sh$; for $n = 5$ it is $S = 5hs^2 \cot(\pi/5)/2 + 5sh$, and for $n = 6$ it is $S = 3\sqrt{3}hs^2 + 6sh$.

Cylinder:

A cylinder can be thought of as the limit as n (number of polygon sides) goes to infinity of a an n-gonal prism. Some specific cylinders include the circular and elliptic cylinders.

For right cylinders (base is directly perpendicular to height), the positive portion of the equation:

$$1 = \frac{x^2}{a^2} + \frac{y^2}{b^2}\,, \tag{79}$$

describes an elliptic cylinder in Cartesian coordinates, and a circular cylinder when $a = b$ (note that these cylinders are not bounded and hence have an infinite volume and surface area).

The volume of any cylindrical solid is simply $V = A_B h$, where A_B is the area of the base and h is the height. For any solid circular cylinder, the volume is

$$V = \int_0^h \int_0^{2\pi} \int_0^r r\, \mathrm{d}r\, \mathrm{d}z\, \mathrm{d}\phi = \pi r^2 h\,,$$

(using cylindrical coordinates). We can also think of the solid circular cylinder volume as an integral of the area of the circle over the height of the cylinder (think solids of revolution). The volume of the solid elliptical cylinder is best visualized this way, having volume:

$$V = \int_0^h A(x)\, \mathrm{d}x = \int_0^h \pi ab\, \mathrm{d}x = \pi abh\,.$$

The surface area of a circular cylinder is $S = e \times p + 2B$, where e is the length of an element, p is the perimeter of a right section of the cylinder, and B is the area of the base of the cylinder. For a solid right circular cylinder, the surface area is $S = 2\pi r(r + h)$. If the right circular cylinder is open (i.e. missing the top and bottom elements), then the surface area is just $S = 2\pi rh$.

Toroid:

A toroid is a doughnut like solid with a hole in the middle that is formed by revolving a shape around an axis of revolution. When that shape is a circle, the toroid is called a torus.

The volume and surface area of a toroid can be calculated using the equations for the solid and surface of revolution methods described earlier for a suitable area. The square toroid has a volume $V = 2\pi RA$, where A is the square section area and R is the radius of the overall shape, and has a surface area $S = 2\pi RC$, where C is the overall circumference.

The equation describing the torus is given by:

$$\left(\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}} - \frac{R}{\sqrt{a^2 + b^2}}\right)^2 + \frac{z^2}{c^2} = 1\,, \tag{80}$$

where $a, b, c, R \in \mathbb{R}^+$ and R is the major radius of the torus. We can also describe the torus parametrically, using the parameters $0 \leq \phi < 2\pi$ and $0 < \theta < \pi$, with the equations:

$$x(\theta, \phi) = a\left(\frac{R}{\sqrt{a^2 + b^2}} + \cos(\theta)\right)\cos(\phi)$$

$$y(\theta, \phi) = b\left(\frac{R}{\sqrt{a^2 + b^2}} + \cos(\theta)\right)\sin(\phi)$$

$$z(\theta, \phi) = c\sin(\theta)$$

The torus is a special case of the supertoroids (all exponents equal to 2). When $a = b = c$, the volume of the torus can be calculated using the formula:

$$V = 2\int_0^{2\pi}\int_{R\sin(\theta)-\sqrt{a^2-R^2\cos^2(\theta)}}^{R\sin(\theta)+\sqrt{a^2-R^2\cos^2(\theta)}}\int_{\arccos(\frac{a}{R})}^{\frac{\pi}{2}} r^2\sin(\theta)\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi = (\pi a^2)\,(2\pi R) = 2\pi^2 Ra^2\,, \tag{81}$$

(using spherical coordinates).

The surface area of a torus for $a = b = c$ can be calculated using the formula:

$$S = 2\pi\int_0^{2\pi}\int_{\arccos(\frac{a}{R})}^{\frac{\pi}{2}} R^2\sin(\theta)\,d\theta\,d\phi = \int_{R-a}^{R+a}\frac{2\pi ax}{\sqrt{r^2 - (x-R)^2}}\,\mathrm{d}x = (2\pi a)\,(2\pi R) = 4\pi^2 Ra\,. \tag{82}$$

Hyperboloid:

There are two types of hyperboloids, namely hyperboloids of one sheet and of two sheets. Hyperboloids of one sheet are connected surfaces with negative Gaussian curvature everywhere, implying that the intersection of the hyperboloid with its tangent plane at a point consists of two branches of curves with distinct tangents at the point. The equation describing a one sheet hyperboloid is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1\,, \tag{83}$$

where $a, b, c \in \mathbb{R}^+$. We can also describe the one sheet hyperboloid parametrically, using the parameters $0 \leq \theta < 2\pi$ and $-\infty < v < \infty$, with the equations:

$$x = a\cosh(v)\cos(\theta)$$
$$y = b\cosh(v)\sin(\theta)$$
$$z = c\sinh(v)$$

Hyperboloids of two sheets are surfaces with two connected components, which have positive Gaussian curvature everywhere, implying that the intersection of the hyperboloid with its tangent plane at a point happens only at this point. The equation describing a two sheet hyperboloid is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1\,, \tag{84}$$

where $a, b, c \in \mathbb{R}^+$. We can also describe the two sheet hyperboloid parametrically, using the parameters $0 \leq \theta < 2\pi$ and $0 \leq v < \infty$, with the equations:

$$x = a\sinh(v)\cos(\theta)$$
$$y = b\sinh(v)\sin(\theta)$$
$$z = \pm c\cosh(v)$$

(Neither the volume nor the surface area of the two hyperboloids can be calculated unless they are bounded by intersection with other surfaces)

We can further extend the notion of volume to an integral over a region $D \subseteqq \mathbb{R}^n$:

$$V = \int \cdots \int_D \det \frac{\partial(\boldsymbol{r'})}{\partial(\boldsymbol{r})}\, \mathrm{d}\boldsymbol{r}\,, \tag{85}$$

where $\det \frac{\partial(\vec{r'})}{\partial(\vec{r})}$ is the determinant of the $n \times n$ Jacobian matrix of partial derivatives.

1.17 Vector Identities

The cross product between vectors $\boldsymbol{A}, \boldsymbol{B}$ is defined as

$$\boldsymbol{A} \times \boldsymbol{B} = |\boldsymbol{A}|\ |\boldsymbol{B}| \sin(\theta_{\boldsymbol{A}\,\boldsymbol{B}})\boldsymbol{e}_{\perp \boldsymbol{A}\,\boldsymbol{B}} \tag{86}$$

and the dot product as

$$\boldsymbol{A} \cdot \boldsymbol{B} = |\boldsymbol{A}|\ |\boldsymbol{B}| \cos(\theta_{\boldsymbol{A}\,\boldsymbol{B}})\,. \tag{87}$$

The following is a list of various algebraic properties of vectors, with some involving the dot $(\cdot)$ and cross $(\times)$ products (For a more general discussion of vectors see sections 2-38). For vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D} \in \mathbb{R}^3$ and scalar $c \in \mathbb{R}$:

Commutativity of Addition:

$\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$

Commutativity of Dot Product:

$\boldsymbol{A} \cdot \boldsymbol{B} = \boldsymbol{B} \cdot \boldsymbol{A}$

Anti-Commutativity of Cross Product:

$\boldsymbol{A} \times \boldsymbol{B} = -\boldsymbol{B} \times \boldsymbol{A}$

Distributivity of Scalar Multiplication over Addition:

$c(\boldsymbol{A} + \boldsymbol{B}) = c\boldsymbol{A} + c\boldsymbol{B}$

Distributivity of Dot Product over Addition:

$$(\boldsymbol{A}+\boldsymbol{B})\cdot\boldsymbol{C}=\boldsymbol{A}\cdot\boldsymbol{C}+\boldsymbol{B}\cdot\boldsymbol{C}$$

Distributivity of Cross Product over Addition:

$$(\boldsymbol{A}+\boldsymbol{B})\times\boldsymbol{C}=\boldsymbol{A}\times\boldsymbol{C}+\boldsymbol{B}\times\boldsymbol{C}$$

Scalar Triple Product:

$$\boldsymbol{A}\cdot(\boldsymbol{B}\times\boldsymbol{C})=\boldsymbol{B}\cdot(\boldsymbol{C}\times\boldsymbol{A})=\boldsymbol{C}\cdot(\boldsymbol{A}\times\boldsymbol{B})$$

$$=|\boldsymbol{ABC}|=\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$$

Vector Triple Product:

$$\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})=(\boldsymbol{A}\cdot\boldsymbol{C})\boldsymbol{B}-(\boldsymbol{A}\cdot\boldsymbol{B})\boldsymbol{C}$$

$$(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{C}=(\boldsymbol{A}\cdot\boldsymbol{C})\boldsymbol{B}-(\boldsymbol{B}\cdot\boldsymbol{C})\boldsymbol{A}$$

Jacobi Identity:

$$\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})=(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{C}+\boldsymbol{B}\times(\boldsymbol{A}\times\boldsymbol{C})$$

$$\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})+\boldsymbol{C}\times(\boldsymbol{A}\times\boldsymbol{B})+\boldsymbol{B}\times(\boldsymbol{C}\times\boldsymbol{A})=0$$

CBinet-Cauchy Identity in 3-D:

$$(\boldsymbol{A}\times\boldsymbol{B})\cdot(\boldsymbol{C}\times\boldsymbol{D})=(\boldsymbol{A}\cdot\boldsymbol{C})(\boldsymbol{B}\cdot\boldsymbol{D})-(\boldsymbol{B}\cdot\boldsymbol{C})(\boldsymbol{A}\cdot\boldsymbol{D})$$

Lagrange's Identity in 3-D:

$$|\boldsymbol{A}\times\boldsymbol{B}|^2=(\boldsymbol{A}\cdot\boldsymbol{A})(\boldsymbol{B}\cdot\boldsymbol{B})-(\boldsymbol{A}\cdot\boldsymbol{B})^2$$

Also note that in three dimensions, we can express a vector $\boldsymbol{D}$ in terms of a basis $\{\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}\}$ as:

$$\boldsymbol{D}=\frac{\boldsymbol{D}\cdot(\boldsymbol{B}\times\boldsymbol{C})}{|\boldsymbol{ABC}|}\boldsymbol{A}+\frac{\boldsymbol{D}\cdot(\boldsymbol{C}\times\boldsymbol{A})}{|\boldsymbol{ABC}|}\boldsymbol{B}+\frac{\boldsymbol{D}\cdot(\boldsymbol{A}\times\boldsymbol{B})}{|\boldsymbol{ABC}|}\boldsymbol{C} \tag{88}$$

1.18 Vector Calculus Identities

In the succeeding section where the definitions of the vector calculus operators will be given, we will be using tensor index notation and it should be understood that we employ the Einstein summation convention as well.

The gradient of a function of n variables $f(\boldsymbol{x})$ (denoted as a scalar field), is the vector field:

$$\boldsymbol{\nabla}f(\boldsymbol{x})=\frac{\partial f}{\partial x_i}\boldsymbol{e}_i \tag{89}$$

The divergence of an n-dimensional vector field $\boldsymbol{F}$ is the scalar function:

$$\boldsymbol{\nabla}\cdot\boldsymbol{F}=\frac{\partial F_i}{\partial x_i} \tag{90}$$

The curl of an n-dimensional vector field $\boldsymbol{F}$ is the vector field:

$$\boldsymbol{\nabla}\times\boldsymbol{F}=\varepsilon^{ijk}\boldsymbol{e}_i\frac{\partial F_k}{\partial x^j}\,, \tag{91}$$

where ε^{ijk} is the Levi-Civita symbol; equal to 0 for $i=j, j=k, i=k$, equal to 1 for an even permutation of $\{i,j,k\}$, and equal to -1 for an odd permutation of $\{i,j,k\}$. The Laplacians of a function of n variables $f(\boldsymbol{x})$ and an nL-dimensional vector field $\boldsymbol{F}$ are, respectively:

$$\boldsymbol{\nabla}^2 f(\boldsymbol{x})=(\boldsymbol{\nabla}\cdot\boldsymbol{\nabla})f(\boldsymbol{x})=\frac{\partial^2 f}{\partial x_i\partial x^i}\quad\text{and}\quad\boldsymbol{\nabla}^2\boldsymbol{F}=(\boldsymbol{\nabla}\cdot\boldsymbol{\nabla})\boldsymbol{F} \tag{92}$$

(Note that both the divergence of the divergence of a vector field and the curl of the divergence of a vector field are undefined)

Now a list of some vector calculus identities using the proceeding definitions, for scalar fields f, g and vector fields $\boldsymbol{F}, \boldsymbol{G}$:

Gradient:

$$\nabla(f+g) = \nabla f + \nabla g$$

$$\nabla(f\boldsymbol{F}) = \nabla f \otimes \boldsymbol{F} + f\nabla \boldsymbol{F}$$

$$\nabla(fg) = g\nabla f + f\nabla g$$

$$\nabla(\boldsymbol{F}\cdot\boldsymbol{G}) = (\boldsymbol{F}\cdot\nabla)\boldsymbol{G} + (\boldsymbol{G}\cdot\nabla)\boldsymbol{F} + \boldsymbol{F}\times(\nabla\times\boldsymbol{G}) + \boldsymbol{G}\times(\nabla\times\boldsymbol{F})$$

Divergence:

$$\nabla\cdot(\boldsymbol{F}+\boldsymbol{G}) = \nabla\cdot\boldsymbol{F} + \nabla\cdot\boldsymbol{G}$$

$$\nabla\cdot(\boldsymbol{F}\times\boldsymbol{G}) = (\nabla\times\boldsymbol{F})\cdot\boldsymbol{G} - (\nabla\times\boldsymbol{G})\cdot\boldsymbol{F}$$

$$\nabla\cdot(f\boldsymbol{F}) = f\nabla\cdot\boldsymbol{F} + \boldsymbol{F}\cdot\nabla f$$

$$\nabla\cdot(\nabla\times\boldsymbol{F}) = 0$$

Curl:

$$\nabla\times(\boldsymbol{F}+\boldsymbol{G}) = \nabla\times\boldsymbol{F} + \nabla\times\boldsymbol{G}$$

$$\nabla\times(f\nabla g) = \nabla f\times\nabla g$$

$$\nabla\times(f\boldsymbol{F}) = f(\nabla\times\boldsymbol{F}) + \nabla f\times\boldsymbol{F}$$

$$\nabla\times(\boldsymbol{F}\times\boldsymbol{G}) = \boldsymbol{F}(\nabla\cdot\boldsymbol{G}) - \boldsymbol{G}(\nabla\cdot\boldsymbol{F}) + (\boldsymbol{F}\cdot\nabla)\boldsymbol{F}) - (\boldsymbol{F}\cdot\nabla)\boldsymbol{G}$$

$$\nabla\times(\nabla f) = 0$$

Laplacian:

$$\nabla\cdot(\nabla f) = \nabla^2 f$$

$$\nabla^2(fg) = g\nabla^2 f + 2(\nabla g)\cdot(\nabla f) + (\nabla^2 g)f$$

$$\nabla^2\boldsymbol{F} = \nabla(\nabla\cdot\boldsymbol{F}) - \nabla\times(\nabla\times\boldsymbol{F})$$

$$\nabla^2(f\boldsymbol{F}) = \boldsymbol{F}\nabla^2 f + 2(\nabla f\cdot\nabla)\boldsymbol{F} + f(\nabla^2\boldsymbol{F})$$

$$\nabla\cdot(g\nabla f) = g\nabla^2 f + \nabla g\cdot\nabla f$$

$$\nabla^2(\boldsymbol{F}\cdot\boldsymbol{G}) = \boldsymbol{F}\cdot\nabla^2\boldsymbol{G} - \boldsymbol{G}\cdot\nabla^2\boldsymbol{F} + 2\nabla\cdot((\boldsymbol{G}\cdot\nabla)\boldsymbol{F} + \boldsymbol{G}\times(\nabla\times\boldsymbol{F}))$$

Third Derivatives:

$$\nabla^2(\nabla f) = \nabla(\nabla\cdot(\nabla f)) = \nabla(\nabla^2 f)$$

$$\nabla^2(\nabla\times\boldsymbol{F}) = -\nabla\times(\nabla\times(\nabla\times\boldsymbol{F})) = \nabla\times(\nabla^2\boldsymbol{F})$$

$$\nabla^2(\nabla\cdot\boldsymbol{F}) = \nabla\cdot(\nabla(\nabla\cdot\boldsymbol{F})) = \nabla\cdot(\nabla^2\boldsymbol{F})$$

In terms of integrals, we have the following identities, where V is a three-dimensional volume, S is a two-dimensional boundary surface of V, C is a one-dimensional boundary of S,f, g are scalar fields, $\boldsymbol{F}$ is a vector field, and $\partial f/\partial n$ is the directional derivative of a scalar field:

Divergence Theorem:

$$\oiint_S \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{S} = \iiint_V (\nabla\cdot\boldsymbol{F})\,\mathrm{d}V$$

$$\oiint_S f\mathrm{d}\boldsymbol{S} = \iiint_V \nabla f\,\mathrm{d}V$$

$$\oiint_S \boldsymbol{F}\times\mathrm{d}\boldsymbol{S} = -\iiint_V (\nabla\times\boldsymbol{F})\,\mathrm{d}V$$

Stokes' Theorem:

$$\oint_C \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{\ell} = \iint_S (\nabla\times\boldsymbol{F})\cdot\mathrm{d}\boldsymbol{S}$$

$$\oint_C f\,\mathrm{d}\boldsymbol{\ell} = -\iint_S \nabla f\times\mathrm{d}\boldsymbol{S}$$

Green's First Identity:

$$\oiint_S f\nabla g\cdot\mathrm{d}\boldsymbol{S} = \iiint_V (f\nabla^2 g + \nabla g\cdot\nabla f)\,\mathrm{d}V$$

Green's Second Identity:

$$\oiint_S (f\boldsymbol{\nabla} g - g\boldsymbol{\nabla} f)\cdot \mathrm{d}\boldsymbol{S} = \oiint_S \left(f\frac{\partial g}{\partial n} - g\frac{\partial f}{\partial n}\right)\,\mathrm{d}S = \iiint_V \left(f\boldsymbol{\nabla}^2 g - g\boldsymbol{\nabla}^2 f\right)\,\mathrm{d}V$$

$$\iiint_V \boldsymbol{F}\cdot\boldsymbol{\nabla} f\,\mathrm{d}V = \oiint_S f\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{S} - \iiint_V f\boldsymbol{\nabla}\cdot\boldsymbol{F}\,\mathrm{d}V$$

$$\iiint_V f\boldsymbol{\nabla}\cdot\boldsymbol{F}\,\mathrm{d}V = \iint_S f\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{S} - \iiint_V \boldsymbol{\nabla} f\cdot\boldsymbol{F}\,\mathrm{d}V$$

1.19 Matrix Calculus Identities

1.20 Commutator Identities

2 Set Theory

2.1 Axioms of Set Theory

2.1.1 Axioms of Zermelo-Fraenkel

1. **Axiom of Extensionality:** If X and Y have the same elements, then $X = Y$.
2. **Axiom of Pairing:** For any a and b there exists a set $\{a,\ b\}$ that contains exactly a and b.
3. **Axiom Schema of Separation:** If P is a property (with parameter p), then for any X and p there exists a set $Y = \{u \in X\ :\ P(u,p)\}$ that contains all those $u \in X$ that have property P.
4. **Axiom of Union:** For any X there exists a set $Y = \cup X$, the union of all elements of X.
5. **Axiom of Power Set:** For any X there exists a set $Y = P(X)$, the set of all subsets of X
6. **Axiom of Infinity:** There exists an infinite set.
7. **Axiom Schema of Replacement:** If a class F is a function, then for any X there exists a set $Y = F(X) = \{F(x)\ :\ x \in X\}$.
8. **Axiom of Regularity:** Every non-empty set has an $\in$-minimal element.
9. **Axiom of Choice:** Every Family of non-empty sets has a choice function. (Axioms 1.1-1.8 constitute Zermelo-Fraenkel axiomatic set theory ZF; the addition of axiom 1.9 makes it ZFC - ZF with the Axiom of Choice)

2.1.2 Formulae

The formulae of set theory are built up from the atomic formulae

$$x \in y \quad \text{and} \quad x = y\,,$$

by means of connectives

$$\varphi \wedge \psi, \qquad \varphi \vee \phi, \qquad \neg\varphi, \qquad \varphi \to \psi, \qquad \varphi \leftrightarrow \psi$$

(conjunction, disjunction, negation, implication, equivalence), and quantifiers

$$\forall\varphi \quad \text{and} \quad \exists\,\varphi$$

(for all, there exists).

2.1.3 Classes

If $\varphi(x, p_1, \ldots, p_n)$ is a formula, we call

$$C = \{x \,:\, \varphi(x, p_1, \ldots, p_n)\}$$

a class. Members of the class C are all those sets x that satisfy $\varphi(x, p_1, \ldots, p_n)$:

$$x \in C \quad \text{if and only if} \quad \varphi(x, p_1, \ldots, p_n).$$

Two classes are considered equal if they have the same elements: If

$$C = \{x \,:\, \varphi(x, p_1, \ldots, p_n)\} \quad \text{and} \quad D = \{x \,:\, \psi(x, q_1, \ldots, q_m)\},$$

then $C = D$ if and only if for all x

$$\varphi(x, p_1, \ldots, p_n) \leftrightarrow \psi(x, q_1, \ldots, q_m).$$

Inclusion of classes is defined as (C is a subclass of D)

$$C \subset D \quad \text{if and only if for all } x,\ x \in C \text{ implies } x \in D,$$

as well as the following operations on classes:

$$\begin{aligned}
C \cap D &= \{x \,:\, x \in C \text{ and } x \in D\},\\
C \cup D &= \{x \,:\, x \in C \text{ or } x \in D\},\\
C - D &= \{x \,:\, x \in C \text{ and } x \notin D\},\\
\cup C &= \{x \,:\, x \in S \text{ for some } S \in C\} = \cup\{S \,:\, S \in C\}.
\end{aligned}$$

3 Calculus

3.1 Taylor Series

The Taylor series for a function f of one variable, about the point $x = x_0$, is given by the sum:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}\,(x - x_0)^n\,, \tag{93}$$

where $f^{(n)}(x_0)$ is the n-th derivative of f evaluated at the point $x = x_0$. It is often used as a method to approximate continuous functions. For example, a linear approximation for the function $f(x)$ is just the Taylor series truncated after the first two terms in the series:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)\,. \tag{94}$$

With all approximations, some form of error is accrued, which for the Taylor series is represented by the expression:

$$R_k(x) = \frac{f^{(k+1)}(a)}{(k+1)!}\,(x - x_0)^{k+1}\,, \tag{95}$$

where $x_0 < a < x$.

The Taylor series also generalizes to functions of more than one variable:

$$\begin{aligned}
f(x_1,\ldots,x_d) &= \sum_{n_1=0}^{\infty}\cdots\sum_{n_d=0}^{\infty}\frac{(x_1-a_1)^{n_1}\cdots(x_d-a_d)^{n_d}}{n_1!\cdots n_d!}\left(\frac{\partial^{n_1+\cdots+n_d} f(a_1,\ldots,a_d)}{\partial x_1^{n_1}\cdots\partial x_d^{n_d}}\right)\\
&= \sum_{|\alpha|\le p}\frac{(\boldsymbol{x}-\boldsymbol{a})^{\alpha}}{\alpha!}\,\frac{\partial^{|\alpha|} f(\boldsymbol{a})}{\partial \boldsymbol{x}^{\alpha}} + \sum_{|\alpha|=p}(\boldsymbol{x}-\boldsymbol{a})^{\alpha}\,h_{\alpha}(\boldsymbol{x})\\
&= f(a_1,\ldots,a_d) + \sum_{j=1}^{d}\frac{\partial f(a_1,\ldots,a_d)}{\partial x_j}(x_j-a_j) + \frac{1}{2!}\sum_{j=1}^{d}\sum_{k=1}^{d}\frac{\partial^2 f(a_1,\ldots,a_d)}{\partial x_j\,\partial x_k}(x_j-a_j)\,(x_k-a_k)\\
&\quad + \frac{1}{3!}\sum_{j=1}^{d}\sum_{k=1}^{d}\sum_{l=1}^{d}\frac{\partial^3 f(a_1,\ldots,a_d)}{\partial x_j\,\partial x_k\,\partial x_l}(x_j-a_j)\,(x_k-a_k)\,(x_l-a_l) + \cdots\,,
\end{aligned}$$

where $\alpha = (\alpha_1,\ldots,\alpha_d)$, $|\alpha| = \alpha_1 + \cdots + \alpha_d$, $\alpha! = \alpha_1!\cdots\alpha_d!$, $\boldsymbol{x}^{\alpha} = x_1^{\alpha_1}\cdots x_d^{\alpha_d}$, and $\lim_{\boldsymbol{x}\to\boldsymbol{a}} h_{\alpha}(\boldsymbol{x}) = 0$.

If f is $p+1$ times continuously differentiable in a closed ball $B = \{\boldsymbol{y}\in\mathbb{R}^d : ||\boldsymbol{a}-\boldsymbol{y}|| \le r\}$ for some $r > 0$, then the Taylor series can be written as:

$$f(x_1,\ldots,x_d) = \sum_{|\alpha|\le p}\frac{(\boldsymbol{x}-\boldsymbol{a})^{\alpha}}{\alpha!}\,\frac{\partial^{|\alpha|} f(\boldsymbol{a})}{\partial \boldsymbol{x}^{\alpha}} + \sum_{|\beta|=p+1}(\boldsymbol{x}-\boldsymbol{a})^{\beta}\,R_{\beta}(\boldsymbol{x})\,,$$

and an exact formula for the error term can be given by:

$$R_{\beta}(\boldsymbol{x}) = \frac{|\beta|}{\beta!}\int_0^1 (1-t)^{|\beta|-1}\,\frac{\partial^{|\beta|} f(\boldsymbol{a}+t(\boldsymbol{x}-\boldsymbol{a}))}{\partial \boldsymbol{x}^{\beta}}\,\mathrm{d}t\,.$$

4 Advanced Calculus

5 Differential Equations

6 Dynamical Systems and Chaos

7 Partial Differential Equations

8 Real Analysis

8.1 The Real Number System

8.1.1 Algebraic Axioms of Real Numbers

1. There is a binary operation called addition and denoted + such that if x and y are real numbers, $x + y$ is also a real number.

2. Addition is associative, i.e.

$$(x + y) + z = x + (y + z) \quad \text{for all } x, y, z \in \mathbb{R}.$$

3. Addition is commutative, i.e.

$$x + y = y + x \quad \text{for all } x, y \in \mathbb{R}.$$

4. A unique additive identity exists. There exists a real number denoted 0 which satisfies

$$x + 0 = x = 0 + x \quad \text{for all } x \in \mathbb{R}.$$

5. A unique additive inverse exists. For each $x \in \mathbb{R}$, there exists $y \in mathbbR$ such that

$$x + y = 0 = y + x$$

6. There is a binary operation called multiplication and denoted such that if x and y are real numbers, then $x \cdot y$ (or xy) is a real number.

7. Multiplication is associative, i.e.

$$(xy)z = x(yz) \quad \text{for all } x, y, z \in \mathbb{R}.$$

8. Multiplication is commutative, i.e.

$$xy = yx \quad \text{for all } x, y \in \mathbb{R}.$$

9. A unique multiplicative identity exists. There exists a real number, different from 0, denoted 1 which satisfies

$$x \cdot 1 = x = 1 \cdot x \quad \text{for all } x \in \mathbb{R}.$$

10. Unique multiplicative inverses exist for non-zero real numbers. For any $x \in \mathbb{R}$ with $x \neq 0$, there exists $y \in \mathbb{R}$ such that

$$xy = 1 = yx$$

11. Multiplication distributes over addition, i.e.

$$x(y+z) = xy + xz \qquad (y+z)x = yx + zx \qquad \text{for all } x, y, z \in \mathbb{R}.$$

(Any system satisfying axioms 1-11 is called a field)

12. There is a subset P of $\mathbb{R}$ called the positive real numbers satisfying
(i) If x and y are in P, then $x + y$ and xy are in P
(ii) If x is in $\mathbb{R}$, exactly one of the following statements is true:

$$x \in P \quad \text{or} \quad x = 0 \quad \text{or} \quad -x \in P$$

13. A non-empty subset X of real numbers, for which there exists a real number a such that $x \leq a$ for all $x \in X$, has a least upper bound.

9 Complex Analysis

10 Abstract Algebra

11 Topology

12 Algebraic Topology

13 Differential Topology

14 Linear Algebra

15 Multi-linear Algebra

16 Functional Analysis

We start with some definitions regarding normed spaces.

Normed Space: A normed space X is a vector space with a norm defined on it. In particular, a Banach space is a complete normed space.

Norm: A real or complex valued function on a vector space X whose value at an $x \in X$ is denoted by $||x||$, and which has the properties

(I) $||x|| \geq 0$

(II) $\quad ||x|| = 0 \quad \Longleftrightarrow \quad x = 0$

(III) $\quad ||\alpha x|| = |\alpha|||x||$

(IV) $\quad ||x + y|| \leq ||x|| + ||y||\,,$

for arbitrary vectors $x, y \in X$ and any scalar α.

A norm on X defines a metric d on X which is given by

$$d(x, y) = ||x - y|| \qquad x, y \in X$$

and is called the *metric induced by the norm*. The metric d satisfies

(a) $\quad d(x + a, y + a) = d(x, y)$

(b) $\quad d(\alpha x, \alpha y) = |\alpha| d(x, y)$

for all $x, y, a \in X$ and every scalar α.

Notice that property (IV) of the norm implies

$$\big|||y|| - ||x||\big| \leq ||y - x||\,,$$

which implies that the norm is continuous. A very common norm is the one defined for the Banach space L^p :

$$||x|| = \left(\sum_{j=1}^{\infty} |\xi_j|^p\right)^{1/p}\,,$$

which induces the metric

$$d(x, y) = ||x - y|| = \left(\sum_{j=1}^{\infty} |\xi_j - \eta_j|^p\right)^{1/p}\,.$$

Another important norm is the one defined for the space of continuous functions with continuous n-th derivatives $C^n[a, b]$:

$$||y||_n = \sum_{i=0}^{n} \max_{a \leq x \leq b} |y^{(i)}(x)|\,.$$

stuff

Linear Functional: A linear operator F in a vector space X over a field K, which maps every element in the domain of F $\mathscr{D}(F) \subseteq X$ to the field K, i.e. the map

$$F : \mathscr{D}(F) \longrightarrow K\,,$$

where $K = \mathbb{R}$ if X is real and $K = \mathbb{C}$ if X is complex. Let $F[f]$ be a functional on X for each $f \in X$, then a linear functional has the following properties

(I) $\quad F[\alpha f] = \alpha F[f] \qquad \forall f \in X$ and any scalar α

(II) $\quad F[f_1 + f_2] = F[f_1] + F[f_2] \qquad \forall f_1, f_2 \in X\,.$

Bounded Linear Functional: A bounded linear operator F with range in the scalar field of the normed space X in which the domain $\mathscr{D}(F)$ lies. Thus there exists a real number c such that for all $x \in \mathscr{D}(F)$,

$$|F(x)| \leq c||x|| \,.$$

Furthermore, the norm of F is

$$||F|| = \sup_{\substack{x \in \mathscr{D}(F) \\ x \neq 0}} \frac{|F(x)|}{||x||} \,.$$

Continuity and Boundedness Theorem: A linear functional F with domain $\mathscr{D}(F)$ in a normed space is continuous if and only if F is bounded.

more stuff

16.1 Calculus of Variations

Increment of a Functional: Let $h, f \in X$ and $\epsilon > 0$ be a small positive constant, then an increment of the functional F is

$$\Delta F[h] = F[f + \epsilon h] - F[f] \,,$$

where $f = f(x)$ and $h = h(x)$.

Now suppose that

$$\Delta F[h] = \phi[h] + \epsilon||h|| + O(\epsilon^2) \,,$$

where $\phi[h]$ is a linear functional and $\epsilon \to 0$ as $||h|| \to 0$. Then $F[f]$ is said to be *differentiable*, and the principle linear part of $\Delta F[h]$, i.e. $\phi[h]$, is called the variation of $F[f]$ and is denoted by $\delta F[h]$. It can be shown that $\delta F[h]$ of a differentiable function $F[f]$ is unique.

Theorem: A necessary condition for the differentiable functional F to have an extremum for $f = f_0$ is that $\delta F[h] = 0$ for $f = f_0$ and all admissible $h \in X$.

The extrema can further be classified into strong and weak extrema: A functional $F[f]$ is said to have a strong extremum if for $f = f_0$ there exists $\epsilon > 0$ such that $F[f] - F[f_0]$ has the same sign for all $f \in \mathscr{D}(F)$ which satisfy $||f - f_0||_0 < \epsilon$ (here $||x - y||_0$ is the norm on $C^0[a, b]$), while a weak extremum follows the same definition except the norm $||f - f_0||_0$ is replaced with $||f - f_0||_1$ (the norm on $C^1[a, b]$). Also note that two functionals are said to be equivalent if they have the same extremals.

Now that a necessary condition for an extremum has been given, the simplest variational problem is finding the weak extremum of the functional

$$F[f] = \int_a^b G(x, f, f') \, \mathrm{d}x \,,$$

for which $G(x, y, z)$ has continuous first and second partial derivatives with respect to all of its arguments, and the functions $f(x)$ are continuously differentiable for $a \leq x \leq b$ with boundary conditions $f(a) = A$ and $f(b) = B$.

Proceed by first calculating the first variation of F and setting it equal to zero like so

$$\delta F = \int_a^b \left(\frac{\mathrm{d}G}{\mathrm{d}f} h(x) + \frac{\mathrm{d}G}{\mathrm{d}f'} h'(x) \right) \mathrm{d}x = \int_a^b \left(\frac{\mathrm{d}G}{\mathrm{d}f} - \frac{\mathrm{d}}{\mathrm{d}x} \frac{\mathrm{d}G}{\mathrm{d}f'} \right) h(x) \, \mathrm{d}x = 0 \,,$$

so clearly the equation

$$\frac{\mathrm{d}G}{\mathrm{d}f} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\mathrm{d}G}{\mathrm{d}f'} = 0$$

must be satisfied; it is known as the *Euler-Lagrange equation*. The E-L equation can be generalized to

$$\frac{\mathrm{d}G}{\mathrm{d}f_\alpha} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\mathrm{d}G}{\mathrm{d}f'_\alpha} = 0$$

for $F[f_1, f_2, \ldots, f_n]$ and f_i only depending on x, or to

$$\frac{\partial G}{\partial f_\alpha} - \nabla_\alpha \frac{\partial G}{\partial f'_\alpha} = 0\,,$$

for $F[f_1, f_2, \ldots, f_n]$ and G depending on variables $x_1, \ldots, x_n$. If there are constraints on the extremal curves in the form

$$\int_a^b M_j(x; f_1, \ldots, f_n; f'_1, \ldots, f'_n)\,\mathrm{d}x = \ell_j \quad (j = 1, \ldots, k)\,,$$

where the ℓ_j are constants, then the E-L equations become

$$\frac{\partial}{\partial f_i}\left(G + \sum_{j=1}^{k} \lambda_j M_j\right) - \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\partial}{\partial f'_i}\left(G + \sum_{j=1}^{k} \lambda_j M_j\right)\right] = 0 \quad (i = 1, \ldots, n)\,,$$

where the constants $\{\lambda_1, \ldots, \lambda_k\}$ are called *Lagrange Multipliers*.

For a functional depending on higher-order derivatives, i.e.

$$F[f] = \int_a^b G(\boldsymbol{x}, f, f', \ldots, f^{(n)})\,\mathrm{d}\boldsymbol{x}\,,$$

the function f must satisfy the *Euler-Poisson Equations*:

$$\sum_{i=0}^{m} \sum_{n_1 \le \ldots \le n_i} (-1)^i \frac{\partial^i}{\partial x_{n_1} \cdots \partial x_{n_i}} \left(\frac{\partial G}{\partial f_j^{(n_1)} \cdots \partial f_j^{(n_i)}}\right) = 0\,.$$

Functional Derivative: The functional derivative is defined through the relation

$$\int \frac{\delta F}{\delta f(x)} h(x)\,\mathrm{d}x = \delta F = \lim_{\epsilon \to 0} \frac{F[f + \epsilon h] - F[f]}{\epsilon} = \left[\frac{\mathrm{d}F[f + \epsilon h]}{\mathrm{d}\epsilon}\right]_{\epsilon=0}\,,$$

where h is an arbitrary function and ϵh is the variation of f; it has the properties

(1) $$\frac{\delta(c_1 F_1 + c_2 F_2)[f]}{\delta f(x)} = c_1 \frac{\delta F_1[f]}{\delta f(x)} + c_2 \frac{\delta F_2[f]}{\delta f(x)} \qquad \text{(Linearity)}$$

(2) $$\frac{\delta(F_1 F_2)[f]}{\delta f(x)} = \frac{\delta F_1[f]}{\delta f(x)} F_2[f] + F_1[f] \frac{\delta F_2[f]}{\delta f(x)} \qquad \text{(Product Rule)}$$

(3) $$\frac{\delta(F_1\big[F_2[f]\big])}{\delta f(z)} = \int \left(\left.\frac{\delta F_1\big[F_2\big]}{\delta F_2(x)}\right|_{F_2 = F_2[f]} \cdot \frac{\delta F_2[f](x)}{\delta f(z)}\right)\mathrm{d}x \qquad \text{(Chain Rule)}\,.$$

The above definition can be generalized to define the nth-*order functional derivative*:

$$\left.\frac{\mathrm{d}^n F[\boldsymbol{f}+\epsilon\boldsymbol{h}]}{\mathrm{d}\epsilon^n}\right|_{\epsilon=0} = \int \mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\cdots\mathrm{d}\boldsymbol{x}_n \frac{\delta^n F[\boldsymbol{f}]}{\delta f(\boldsymbol{x}_1)\delta f(\boldsymbol{x}_2)\cdots\delta f(\boldsymbol{x}_n)} h(\boldsymbol{x}_1)h(\boldsymbol{x}_2)\cdots h(\boldsymbol{x}_n)\,,$$

where the functional $F[\boldsymbol{f}+\epsilon\boldsymbol{h}]$ is given by the formula

$$F[\boldsymbol{f}+\epsilon\boldsymbol{h}] = \sum_{n=0}^{N}\frac{\epsilon^n}{n!}\int \mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\cdots\mathrm{d}\boldsymbol{x}_n \frac{\delta^n F[\boldsymbol{f}]}{\delta f(\boldsymbol{x}_1)\delta f(\boldsymbol{x}_2)\cdots\delta f(\boldsymbol{x}_n)} h(\boldsymbol{x}_1)h(\boldsymbol{x}_2)\cdots h(\boldsymbol{x}_n) + O(\epsilon^{N+1})\,.$$

ex. The functional derivative of the functional $F[f] = f(x)$ is found through first finding $F[f+\epsilon h] = f(x)+\epsilon h(x)$, then taking the derivative with respect to ϵ:

$$\frac{\mathrm{d}F[f+\epsilon h]}{\mathrm{d}\epsilon} = h(x) \quad \underset{\epsilon=0}{\longrightarrow} \quad \left.\frac{\mathrm{d}F[f+\epsilon h]}{\mathrm{d}\epsilon}\right|_{\epsilon=0} = h(x) = \int \frac{\delta F[f]}{\delta f(x')}h(x')\,\mathrm{d}x' = \int \delta(x-x')h(x')\,\mathrm{d}x'$$

thus the functional derivative is

$$\frac{\delta F[f]}{\delta f(x')} = \delta(x-x')\,.$$

With the notion of functional derivatives at hand, we can then define the functional Taylor series of an arbitrary functional $F[f]$, centred about some multivariate function $\bar{f}$, (provided it's functional Taylor series exists) as:

$$F[\boldsymbol{f}] = \sum_{n=0}^{\infty}\frac{1}{n!}\prod_{i=0}^{n}\int \mathrm{d}\boldsymbol{x}_i \left.\frac{\delta^i F[\boldsymbol{f}]}{\delta f(\boldsymbol{x}_i)}\right|_{\boldsymbol{f}=\bar{\boldsymbol{f}}} (\boldsymbol{f}(\boldsymbol{x}_i) - \bar{\boldsymbol{f}}(\boldsymbol{x}_i))\,,$$

or for F depending on a number of multivariate functions $\{\boldsymbol{f}_k\}$ as:

$$F[\{\boldsymbol{f}_k\}] = \prod_{k=1}^{N}\sum_{n_k=0}^{\infty}\frac{1}{n_k!}\prod_{i_k=0}^{n_k}\int \mathrm{d}\boldsymbol{x}_{i_k} \left.\frac{\delta^{i_k} F[\{\boldsymbol{f}_k\}]}{\delta f_k(\boldsymbol{x}_{i_k})}\right|_{\boldsymbol{f}_k=\bar{\boldsymbol{f}}_k} (\boldsymbol{f}_k(\boldsymbol{x}_{i_k}) - \bar{\boldsymbol{f}}_k(\boldsymbol{x}_{i_k}))\,.$$

Noether's Theorem: If the functional

$$F[f] = \int_{x_0}^{x_1} G(x,f,f')\,\mathrm{d}x$$

is invariant under the family of transformations

$$x^* = \Phi(x,f,f';\epsilon)$$
$$f_i^* = \Psi(x,f,f';\epsilon)$$

for arbitrary x_0 and x_1, then

$$\sum_{i=1}^{n}\frac{\mathrm{d}G}{\mathrm{d}f_i'}\left[\frac{\partial\Psi_i(x,f,f';\epsilon)}{\partial\epsilon}\right]_{\epsilon=0} + \left(G - \sum_{i=1}^{n}\frac{\mathrm{d}G}{\mathrm{d}f_i'}f_i'\right)\left[\frac{\partial\Phi(x,f,f';\epsilon)}{\partial\epsilon}\right]_{\epsilon=0} = \text{constant}\,.$$

In other words, every one-parameter family of transformations leaving $F[f]$ invariant leads to a first integral of its system of Euler equations.

Bilinear Functional: A linear functional $B[x,y]$ where $x,y\in X$ such that B is a functional of y for fixed x and a functional of x for fixed y. It has the following properties:

(1) $\qquad B[x+y,z] = B[x,z]+B[y,z] \qquad$ and $\qquad B[x,y+z] = B[x,y]+B[x,z]$

(2) $\qquad B[\alpha x, y] = \alpha B[x, y] \qquad$ and $\qquad B[x, \alpha y] = \alpha B[x, y]$,

$\forall x, y, z \in X$ and $\alpha \in \mathbb{R}$. If $y = x$ then $B[x, x]$ is called a quadratic functional, and is positive-definite if $B[x, x] > 0$ for every non-zero element x.

A bilinear functional is called a *bilinear form* if it is defined on a finite-dimensional space. Every bilinear form can be represented as

$$B[x, y] = \sum_{i=1}^{n} \sum_{k=1}^{n} b_{ik} \hat{e}_x^i \hat{e}_y^k ,$$

where $\{\hat{e}_x^1, \hat{e}_x^2, \ldots, \hat{e}_x^n\}$ and $\{\hat{e}_y^1, \hat{e}_y^2, \ldots, \hat{e}_y^n\}$ are the components of the vectors x and y relative to some basis. Note that a quadratic form can be defined by setting $x = y$ and $\hat{e}_x = \hat{e}_y$.

ex.

$B[x, y] = \int_a^b \alpha(t)\, x(t)\, y(t)\, \mathrm{d}t$ is a bilinear functional and $A[x] = B[x, x] = \int_a^b \alpha(t)\, x^2(t)\, \mathrm{d}t$ is a quadratic functional (provided $\alpha(t) > 0 \quad \forall t \in [a, b]$) .

Second Variation: A functional $F[f]$ is twice differentiable if its increment can be written in the form

$$\Delta F[h] = \phi_1[h] + \phi_2[h] + \epsilon ||h||^2 + O(\epsilon^2) \qquad (\epsilon \to 0 \text{ as } ||h||^2 \to 0) ,$$

where $\phi_1[h] = \delta F[h]$ is the first variation, and $\phi_2[h]$ is a quadratic functional called the second variation of $F[f]$ and denoted by $\delta^2 F[h]$.

Strongly Positive: A quadratic functional $\phi_2[h]$ is said to be strongly positive if there exists $k > 0$ such that $\phi_2[h] \geq k||h||^2 \quad \forall h$.

Sufficient Condition for a Minimum: The functional $F[f]$ has a minimum at $f = f_0$ if its first variation vanishes there, i.e. $\delta F[h] = 0$ for $f = f_0$, and its second variation $\delta^2 F[h]$ is strongly positive at $f = f_0$.

There are some special theorems related to functionals of the form

$$F[f] = \int_a^b G(x, f, f')\, \mathrm{d}x ,$$

which are stated in the following section.

Legendre's Theorem: A necessary condition for the functional

$$F[f] = \int_a^b G(x, f, f')\, \mathrm{d}x \qquad f(a) = A \text{ and } f(b) = B ,$$

to have a minimum for the curve $f = f(x)$ is that the inequality

$$\frac{\mathrm{d}^2 G}{\mathrm{d}f'^2} \geq 0$$

be satisfied at every point of the curve.

Conjugate Points: The point $a^* \neq a$ is said to be conjugate to the point a if the equation

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{2}\frac{\partial^2 G}{\partial f'^2} h'(x)\right] + \frac{1}{2} h(x) \left[\frac{\partial^2 G}{\partial f^2} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial^2 G}{\partial f \partial f'}\right)\right] = 0$$

has a solution which vanishes for $x = a$ and $x = a^*$ but is not identically zero.

Jacobi's Theorem: If the extremal $f = f(x)$ corresponds to a minimum of the functional

$$F[f] = \int_a^b G(x, f, f')\,\mathrm{d}x\,,$$

and if

$$\frac{\mathrm{d}^2 G}{\mathrm{d}f'^2} > 0$$

along this extremal, then the open interval (a, b) contains no points conjugate to a.

This leads us to a result called *Jacobi's equation*:

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\partial^2 G}{\partial f'^2} h'(x)\right] + h(x)\left[\frac{\partial^2 G}{\partial f^2} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial^2 G}{\partial f \partial f'}\right)\right] = 0\,,$$

which is the variational equation of the Euler-Lagrange equation. It can be interpreted as the differential equation satisfied by the difference between neighbouring extremals, to within infinitesimal terms of the first-order.

The above results together with the requirement that the Euler-Lagrange equation be satisfied, must be satisfied simultaneously as well as independently to guarantee a weak minimum for the functional $F[f]$, so we summarize them with a theorem:

Theorem: Suppose that for some admissible curve $f = f(x)$, the functional

$$F[f] = \int_a^b G(x, f, f')\,\mathrm{d}x \qquad f(a) = A \text{ and } f(b) = B\,,$$

satisfies the following conditions:

1. The curve $f = f(x)$ is an extremal, i.e. satisfies the Euler-Lagrange equation $\frac{\partial G}{\partial f} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial G}{\partial f'} = 0$
2. Along the curve $f = f(x)$, $\frac{\partial^2 G}{\partial f'^2} > 0$ (Strengthened Legendre Condition)
3. The interval $[a, b]$ contains no points conjugate to the point a (Strengthened Jacobi Condition),

then $F[f]$ has a weak minimum for $f = f(x)$.

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